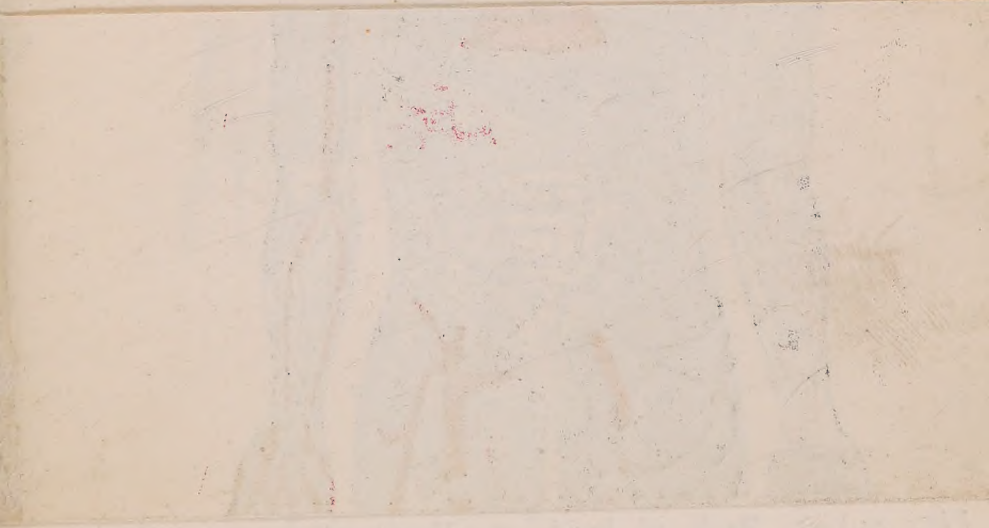
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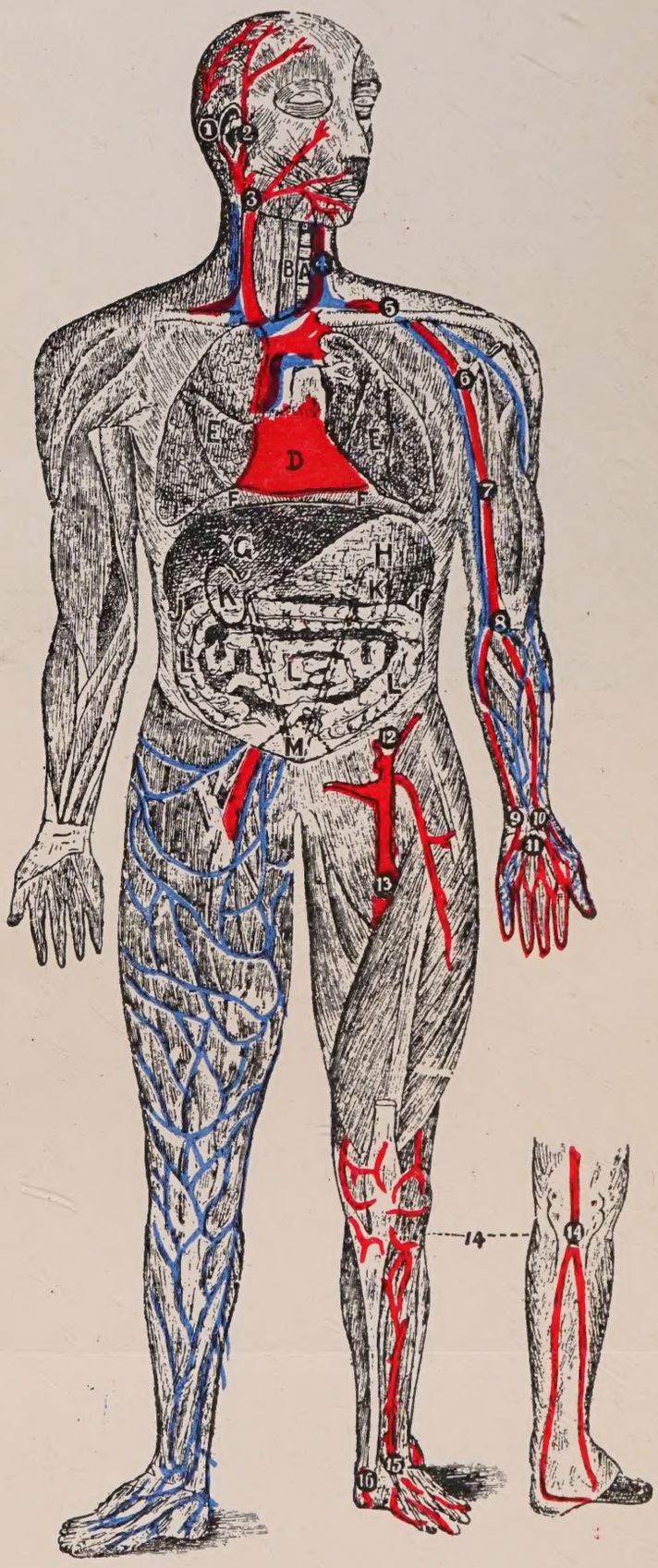

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Muscle, fibres and Tendons clearly shown by dissection. Position of main ARTERIES shown in Red.

VEINS—Network of blue lines. Position of organs in neck, chest and abdomen, shown clearly by dissection.



The Letters indicate organs.—A. Wind Pipe. B. Food Pipe. C. The Aorta. D. The Heart. E. The Lungs. F. Midriff. G. Liver. H. Stomach. I. Spleen. J. Gall-bladder. K. Kidneys with pipe through which urine passes from kidneys to bladder (position shown by dotted lines). L. Bowels. M. Bladder for urine.

The numbers, 1 to 16 are placed on the position of the "pressure points." (*See page 126 and Chapter XVII and Fig. 44.*)—1. Back of head. 2. Front of ear. 3. Corner (angle) of lower jaw. 4. Neck. 5. Behind collar bone. 6. Armpit. 7. Middle of Arm. 8. Bend of Elbow. 9. Ulner or inner side of the wrist. 10. Radial or outer side of the wrist. 11. Palm of hand. 12. Middle of groin. 13. Middle of thigh. 14. Behind knee joint. 15. Middle of ankle. 16. Behind inner protuberance of ankle.

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A MANUAL

OF

FIRST AID FOR INDIA.

BY

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"the improvement of health,
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the mitigation of suffering

IN

Peace and War."

که طاعت بجز خدمت خلق نیست

به تسبیح و سجاده و دلق نیست ـ

"Verily in the sight of God devotion is naught save the Service of His Creation : Not the telling of beads, the prayer carpet, and the patched garment of the Hermit."

PREFACE.

The first edition of this brochure was published in 1925 and brought to the notice of Sir Fredrick Whyte, K.C.S.I., then President of the Indian Council of St. John Ambulance Association. He had the book scrutinised by his Medical Sub-Committee and it was recommended for adoption as the official text book for India. In the interest of the cause of First Aid work in India I agreed to the suggestion to make over the copyright to the Indian Council but the authorities at St. John's Gate in London ruled that the official text-book for England should also be adopted for India and replace the present one written by Colonel Blackham.

On being released from my obligation in this way, I have taken upon myself the task of revising and seeing the book through the press not in any spirit of opposition, but as a supplement to the

official text-book, as I am convinced that the introduction of an Indian colour is necessary to suit Indian conditions and habits of the people. For instance the First Aid Treatment of frost bite may be important in Europe but in India we require a detailed knowledge of the treatment of sun-stroke. Similarly cases of dog bites and snake bites are very rare in England, but are unfortunately too common in India. Suicide by drowning so common in England is almost unknown in India, but suicidal hanging and poisoning by opium is very common. Therefore the above subjects and others of special importance in India such as Burns, Wounds, Insensibility and Poisonings, have received special attention.

I have also tried to give reasons and explain the methods in the hope that it will help students to follow the teachings in a rational way and prevent learning by

rote, which is the inevitable result of compressing the subject too much with the object of producing a book of small size.

Further, in order that students may visualise known and familiar objects and thus get a more lasting impression I have put in a larger number of illustrations and altered the diagrams replacing European by Indian figures. But as it is not possible to alter the recognised conventions regarding tying bandages and of lifting and carrying an injured person, I have maintained the standard English methods. The special features mentioned above, have increased the volume of the book but it is more apparent than real as the matter is confined within the limits of the Syllabus of the Indian Council.

I very much appreciate the help received by me from Mr. W. J. Higuet of the East Indian Railway Ambulance Brigade, Jamalpur and Mr. J. Roy who shared with

me the burden of comparing the proof pages.

I am grateful to Lt.-Col. J. Norman Walker, I.M.S., Surgeon to His Excellency the Viceroy and to Colonel Bhola Nauth, C.I.E., I.M.S. (Retd.), General Secretary, Indian Council of St. John Ambulance Association, for their very generous appreciation of my effort and for many helpful suggestions.

I acknowledge with thanks the help I have received from English standard works especially from "First Aid to the Injured" written by the late Sir James Cantlie, K.B.E., F.R.C.S.

I realise how difficult it is to produce a book free from imperfections specially at its first appearance, and I crave the indulgence of the critic for the errors which he may discover.

HASSAN SUHRAWARDY.

Calcutta, January, 1927.

St. John Ambulance Association.

INDIAN COUNCIL.

SYLLABUS OF INSTRUCTION IN FIRST-AID TO THE INJURED.

FIRST LECTURE.

A. Principles of First-Aid.
B. Brief Description of Human Skeleton and Muscles.
C. Fractures—Causes, varieties, signs and symptoms.
D. Treatment of Fractures—General Rules.
E. Triangular Bandage—Its application to Head, Chest, Back, Shoulder, Elbow, Hand, Hip, Knee and Foot and as Arm-slings—Large, Small, and St. John.

SECOND LECTURE.

A. Individual Fractures—Details of treatment: Skull, Lower Jaw, Shoulder-blade, Collar-bone, arm, Fore-arm, Hand, Thigh, Leg, Knee-cap, Foot, Ribs, Pelvis and Spine.
B. Dislocations, Sprains and Strains—Signs, Symptoms, and treatment.
C. Practice—Treatment of Fractures.

THIRD LECTURE.

A. General Description of the Heart and Blood Vessels.
B. Circulation of the Blood.
C. Varieties of Hæmorrhage and the means of arresting hæmorrhage.
D. Situation of main arteries—Pressure points.

E. Compression of arteries by Digital and Instrumental pressure.
F. Treatment of wounds.
G. Practice—Compression of arteries.

FOURTH LECTURE.

A. Internal Hæmorrhage—Hæmorrhage from special regions. Bruises.
B. Burns, Scalds, Frost bite, Bites of animals, Snake bites, Stings, Fish-hook in skin.
C. Foreign bodies in Eye, Nose and Ear.
D. Practice—Treatment of fractures and hæmorrhage (as in First and Second Lectures).

FIFTH LECTURE.

A. The Nervous System.
B. The Organs and Mechanism of Respiration.
C. Artificial Respiration.
D. Hand-Seats.
E. Practice—Artificial Respiration.

SIXTH LECTURE (FOR MALES ONLY).

A. Poisoning.
B. Improvised methods of lifting and carrying sick or injured.
C. Hand-Seats.
D. Stretcher exercise.
E. Conveyance of sick or injured by rail or in country carts.

SIXTH LECTURE (FOR FEMALES ONLY).

A. Poisoning.
B. Hand—Seats—Lifting and carrying of patients.

C. Preparation for the reception of accidents.
D. Preparation of the bed.
E. Removal of clothes.
F. Preparation for surgeon.

NOTE I.—The subject of poisons should be treated in a general manner; the common poisons classified, and only their general symptoms, effects and treatment taught.

NOTE II.—The latter part of each lecture should be devoted to practical work, such as application of bandages and splints, lifting and carrying wounded on stretchers.

NOTE III.—Male classes must pass in that system of stretcher exercise most suitable for the locality.

NOTE IV.—As little time as possible is to be spent on instruction in anatomical and physiological details. Lecturers and examiners are particularly requested to remember that it is "First-Aid" that has to be taught and tested, and not anatomy and physiology.

NOTE V.—When a candidate is under instruction for the voucher the number and date of his previous certificate should be stated in the Column for Remarks in the Examiner's Report Form.

Mixed classes of men and women are on no account permitted.

No Lecturer may examine his own class for Certificates.

Pupils under sixteen years of age can only be examined in the "Junior" Course (Reference No. 4, India).

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The illustrations marked by asterisks are reproduced from Sir James Cantlies "First Aid to the Injured."

CHAPTER I

" First Aid "—Its Definition—Utility and Limitations--Definition of certain terms.

" FIRST AID " is the term applied to a branch of Medicine and Surgery which deals with certain practical and simple methods of treatment, a knowledge and application of which enables lay persons to render prompt and skilled assistance in cases of sudden illness and accidents. It also teaches how to utilise ordinary things, commonly found in one's home or available at the place of accident, for treatment and transport of the sick and the wounded.

In cases of sudden illness, poisoning, and injuries received in accidents, it is of the highest importance to know *what to do, how much to do,* and *what not to do.* The well-meaning helper, through ignor-

ance of the methods of First Aid, has often endangered the life of the person whom he is so anxious to serve.

In the rural areas of our country, where qualified medical aid is either entirely absent, or difficult to obtain quickly, men bitten by a snake, or a mad dog, or wounded by a poisoned arrow, or suffering from opium or arsenic poisoning, have been saved by simple methods adopted promptly by a First Aid man to prevent the absorption of the poison into the general system, and to neutralise the evil effects of the amount that has already gained entrance into the body.

It must, however, be clearly understood that a *First Aider* is neither a doctor nor even a trained nurse and *should never take the responsibility of continued treatment.* The First Aider only renders temporary help in an emergency by simple methods, the value of which lies in its being prompt,

skilled, efficient and based on scientific knowledge. The First Aider's duty and the object of First Aid is—(*a*) To attend to actual emergencies. (*b*) To endeavour to get medical help as soon as possible. (*c*) To make the patient comfortable. (*d*) To relieve pain or suffering. (*e*) To prevent complications and in this way save the life and expedite recovery of the wounded and the sick. The First Aider's duties end with the arrival of the doctor, or of the admittance of his patient into a hospital.

Definitions.

Signs are those which can be seen and observed on the sufferer (patient).

Symptoms are what the patient feels and states or complains of.

History means the circumstances attending the accident or sudden illness.

Dressings: By this is meant pieces of

medicated or plain germ-free cloth of light material, lint, or cotton wool used for applying on wounds, burns and injuries.

Splints are pieces of planks of light timber or metal sheeting, cut in different sizes suitable to fit different parts of the body and used to support and restrain movement in injured limbs and joints.

Note.—In First Aid, by *Splints* are meant any material improvised and used for giving support to the injured limbs and at the same time preventing movement in the joints above and below the injured part.

CHAPTER II

The Principles of First Aid.

The following fifteen points are the guide to the principles of First Aid treatment :—

1. Look for the cause of the injury

or sudden illness and promptly remove the sufferer from the source of danger, or remove the source of danger and the direct cause of the injury or illness.

2. *Stop* any *severe bleeding,* no matter what the other injuries present on the patient may be. Remember that loss of blood must be stopped promptly and efficiently, as on it depends the future chances of life and recovery of the patient.

3. Place the injured part and the sufferer in as comfortable a position as is possible under the circumstances. Remove all pressure from the injured part and keep it in an easy position.

4. See that an injured person gets as much fresh air as possible. Remove all tight clothing and keep back the crowd of inquisitive by-standers and even sympathetic friends, as they prevent free access of air by forming a cordon round the patient.

5. Promptly cover up, all wounds and abrasions in the skin, with dressings or clean cloth, in order to prevent contamination from the outside air and dust, which are full of germs of disease.

6. Keep the patient warm, in all cases of shock, whether due to injuries or fright and other mental or physical causes, as the patient in a state of shock feels very cold. Adopt methods for reducing and overcoming shock. *Vide* page 238.

Note.—Keep a patient cool in cases when he is suffering from a heat stroke, but warm in cases of burn and scalds, in which he is in a state of great shock.

7. Be *resourceful* so that you may use to the best advantage whatever is handy. Usually, accidents do not happen in places fully equipped with medical appliances, splints, dressings, and bandages. The First Aid man must accustom himself to

utilising ordinary household things and improvising easily available material for his use for treatment and transport.

8. Study and utilise the most suitable method and means of transport for different classes of injury and sickness.

9. Accustom yourself to the method of removal of clothes with the minimum amount of discomfort to the patient and do not take them off unless it is really necessary.

10. Get into the habit of giving clear and definite instructions to the patient and his friends or to by-standers, as to what to do, what not to do, and how to assist you.

11. Be discriminating, and decide, in a case of multiple injuries, which of them should be treated *first* and by you, and which could be dealt with later with the help of friends or by-standers, and which should be reserved for treatment by the doctor.

12. Be observant and tactful. Notice the signs and hear the symptoms and learn the history of the case without upsetting the patient by asking thoughtless questions.

13. Be persevering and continue your efforts even when these are not at first successful. For instance, while performing artificial respiration in cases of suffocation by drowning, do not give up your efforts because they are not quickly successful.

14. Be sympathetic and cheer your patient up and prove yourself a messenger of succour and relief to the sick and the injured, irrespective of caste, creed or colour, and unbiased by considerations of the fact that the sufferer is a friend or an enemy.

15. Endeavour to obtain medical assistance as early as possible, either by getting the doctor to visit the patient or by conveying the patient to a well-equipped

hospital or to the surgery of a doctor. When sending for the doctor do not send verbal messages. Remember to write and let him know, in as few words as possible, just what you think the trouble is. This helps him to come adequately prepared to render assistance and to give orders to his assistants for preparing the operation room or the surgery.

Be prompt. Send a man on a bicycle or a gharry or a motor car. Send a phone message, and in distant places send an express wire.

CHAPTER III

Brief Description of the Human Skeleton.

For " First Aid " students an elementary knowledge of the structure of the body and of the functions of the more important organs and the respiratory, nervous and circulatory systems is necessary. The

science concerned with the *functions* of the organs and systems is known as *Physiology*, and that concerned with the *structure* of the body is known as *Anatomy*.

N.B.—"First Aid" students need only have a general working knowledge of elementary anatomy and physiology and need not learn the details nor the technical names of the different bones of the skeleton.

In the description of the human body the man is supposed to be standing erect with his arms hanging down by the side and the palms of the hands open and directed forwards. The middle line runs vertically from the top of the head to a point midway between the feet. Any portion or part of the body nearer this middle line is said to be situated "internally" or to the "inner side" of the portion or part of the body further away from it. This is expressed by saying that

such and such is situated to the inner side of, or lies internally to, such and such. Thus, in the description of the leg bones, we say the shin bone (the *tibia*) is situated internally or to the inner side of the brooch bone (the *fibula*).

The Human Skeleton.

The frame-work of the human body is called the *Skeleton*. It is built up of two hundred distinct bones of different shapes and sizes.

Bones are composed of one-third animal and two-thirds of earthy matter, intimately blended together. This combination gives both strength and a certain amount of spring in the bones. Animal matter which is soft predominates in children and the earthy matter which gives hardness is predominant in the old. Hence in children the bones are not so brittle as in the aged.

Utility of Bones.

The Bones—

(1) Form the frame-work which gives rigidity to the body.

(2) Determine the general contour, shape and proportions of the body.

(3) Serve as points of attachment to the muscles and as levers on which the muscles act in order to move the limbs or the body.

(4) Protect vital organs by providing firm coverings; the skull protects the brain, the bones of the chest protect the lungs and the heart.

(5) Help in forming joints.

The skeleton is divided into *four different regions*:—

(1) The Skull or "the Head," (2) Bones of the body or "the Trunk," (3) Bones of the upper limb, (4) Bones of the lower limb.

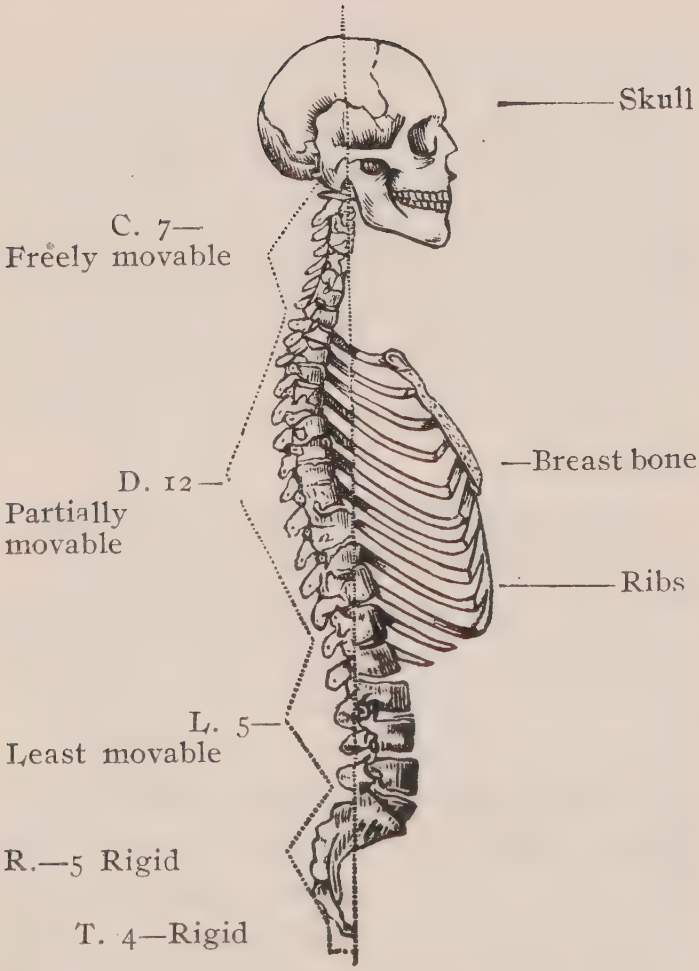


FIG. 1. Skull, Spinal Column, Ribs and Breast-bone.

Skull.

The bones of the head and face form the skull ; they are all knitted together firmly except the lower jaw which is freely movable on a hinge-joint. The skull is supported on the summit of the vertebral or the spinal column. *Fig. 1.*

Its upper region is called the " Vault " or *Culverium.*

Its lower region is called the " Base. "

Its front portion is called the " Face. "

Within the skull (head and face) are contained the brain and the organs and nerves which control the five senses, *viz.,* Sight, Smell, Taste, Touch, Speech.

When bones of the head and face are broken there is thus great danger of damage to special organs of sense and to the brain. Broken fragments may press on the surface of the brain and cause unconsciousness and paralysis. A blow on

the head may, without breakage of a bone, result in unconsciousness on account of the shaking or *Concussion* of the brain.

The lower jaw, the only movable part of the skull, is like a horse-shoe in shape. It frequently gets broken, but this is not as dangerous as fractures of the bones of the skull.

Trunk.

The Bones of the Trunk or Body are :—

(*a*) The Backbone, also called the Spine. (*Vertebral Column*).

(*b*) Breast-bone and Ribs (*Chest*).

(*c*) The Haunch Bones (*Pelvis*).

The Backbone (*Vertebral Column*) is a flexuous and flexible column made up of thirty-three irregularly shaped triangular pieces of bones placed one on top of the other. It forms a long pillar with a central canal, through which passes the prolonga-

tion of the lower and back part of the brain known as the *spinal cord* or *marrow*, which it encloses and protects. (*Vide* Chapter on Nervous System). The Spinal Column gives strength to the skeleton and enables man to maintain an erect position. It supports the skull, and through it is transmitted the weight of the upper part of the body to the lower. Each segment of bone in this column is called a *vertebra* and has a circular hole in it, each segment is placed one on top of the other in such a way that the hole in the body of one fits exactly over that of the other, and in this way the bony channel called the spinal canal is formed. Each segment has also a pad of gristle interposed between it and its fellow above and below. These (*Inter-vertebral discs*) act as buffers and prevent jars and shocks of impact being conveyed to the skull and the brain within it. The discs of gristle also bind the pieces of bones

together firmly but at the same time allow a limited amount of flexibility. The first two vertebræ in the neck are freely movable. The first one is called *Atlas*, on account of its supporting the globe of the head, and the next one is named the *Axis*, as this is the pivot on which the head rotates in its side to side movements. The movement of the spinal column is most marked in the neck (*cervical region*), the movement getting less and less from above downwards, being least in the region of the loin (*lumbar region*). The spine in the regions of the rump and the tail is rigid, the lowest nine pieces of bone of the spinal column joining together firmly in two groups, the upper five of these forming the rump bone (*Sacrum*), and the lowest four constituting the tail bone (*Coccyx*). *See* Fig. 1.

Note.—If the spine gets broken, displaced, or severely jarred, serious consequences—such as

partial or complete paralysis—or death results. The higher the seat of injury, the more dangerous it is, because the parts controlling the action of the heart and breathing are situated in the upper regions. For this reason the man who breaks his neck dies immediately, whereas the man whose spine gets broken in the waist may escape only with paralysis of his lower limbs and may live for some time.

Haunch bones (*Pelvis*) are two thick irregularly shaped flat bones. They meet each other in front but remain separated behind. The *rump bone* fills the gap and forms a complete circle of bone called the *pelvic girdle*. Within the pelvic girdle are placed the urinary bladder and the lower bowels. In the female it also contains the womb and its appendages. On the outer side of each of the bones there is a deep cup-shaped cavity, which is the socket into which the round head of the thigh bone fits and forms the hip joint. The inner surface of the bones is smooth

and forms part of the side wall of the lower part of the abdomen or belly.

Rump bone (*Sacrum*) is a wedge-shaped piece of bone formed by fusion of five bones of the spinal column, situated below the lowest segment in the region of the loin. It forms one rigid mass of bone which, as has been mentioned above, with the two haunch bones completes the pelvic girdle.

Tail bone (*Coccyx*) is a group of four little rudimentary bones of the spine which unite to form one small cone-shaped piece which is connected by a pad of gristle and fibrous bands to the lowest portion of the rump bone.

Chest.

The frame work of the Chest (*Thorax*) consists of :—

(1) The Breast-bone, in front.

(2) Twelve pairs of ribs at the sides.

(3) Backbone behind. (*See* Fig. 1.)

(1) *The Breast-bone* (*Sternum*) is shaped somewhat like a dagger, it forms the middle portion of the front wall of the chest. Its upper ends support the collar bone, and its margins form connections with the gristly front portions of the ribs.

(2) *The Ribs* are elastic arches of bone which form the chief part of the bony frame-work of the chest. There are twelve of them on each side and they span the distance between the twelve segments of the spinal column behind, and the breast-bone in front. The first seven pairs attached to the breast-bone, directly by means of their gristly front ends are termed *true ribs,* while the remaining five pairs are termed *false ribs ;* these are attached indirectly by forming connection with the front portion of the ribs immediately above

them. The last two pairs of the false ribs (11th and 12th pairs) are not attached to the breast-bone and are termed *floating ribs*.

The chest contains the chief organs of circulation, *viz.*, the heart and the great blood vessels and also the organs of respiration, *viz.*, the lungs, the wind pipe and its sub-divisions (*bronchi*). *See* Fig. 43 and the frontispiece.

The chest cavity is separated from the belly (*abdominal cavity*) by a thin dome-shaped flat piece of flesh known as the *midriff* (*diaphragm*). The upper rounded (*convex*) surface of the dome forms the floor of the chest and the hollow (*concave*) lower surface, the roof of the abdomen. *Vide* Fig. in frontispiece.

Bones of Upper Limb. Fig. 2.

These are concerned chiefly with movements connected with holding, eating,

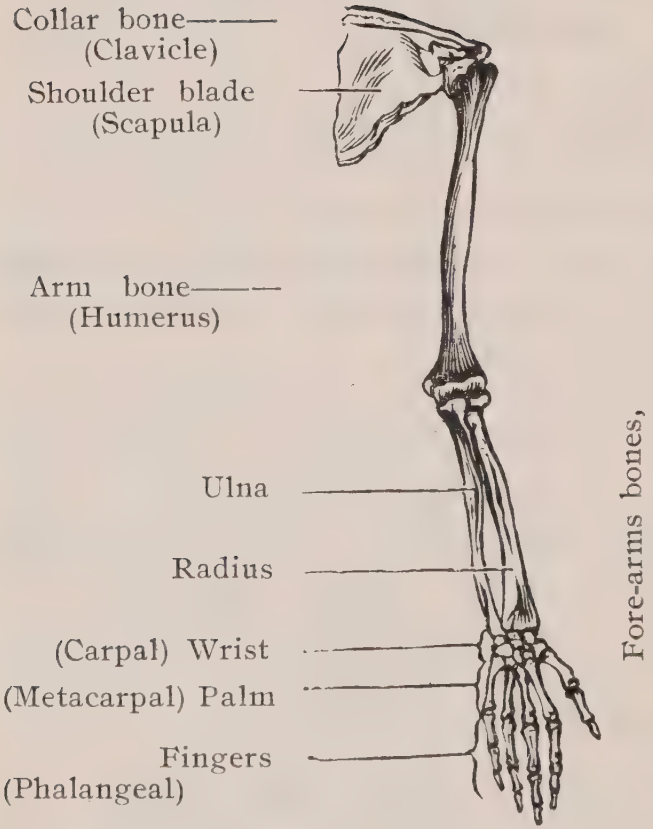


FIG. 2. Bones of the upper limb.

doing work, etc. In shape and number they are the same on the right and left sides, and consist of the following :—

(*a*) *Collar bone* is a small cylindrical bone about 5″ long, shaped like the Italic letter *f* or an open S placed horizontally in front of the chest immediately above the first rib. It forms joints with upper end of breast-bone at one extremity and with the top of shoulder blade bone at the other. *Vide* Fig. 4. This bone is broken more often than any other in the human body and is thus of special interest to the First Aid man.

(*b*) *Shoulder blade bone* is a flat triangular shaped bone lying on the upper part of the back of the chest wall over the ribs ; most of the muscles which move the arms are attached to this bone which is therefore freely movable over the ribs and in consequence is seldom broken except by

a direct violent blow on it. On the upper part of its outer border is a saucer shaped shallow cavity which receives the head of the arm bone and forms the shoulder joint.

(*c*) *Upper Arm Bone.* (One in each limb). This is the longest and largest bone of the upper limb. Its upper end has a rounded head which, with the shallow saucer shaped cavity in the shoulder blade bone, forms the shoulder joint. At its lower end the arm bone has a roller-shaped surface which, with the upper end of the bones of the fore arm, forms the elbow joint.

(*d*) *Forearm bones.* (Two in each limb). The one which extends from the thumb side of the wrist to the thumb side of the elbow (outer side) is called the *Radius*, and the one which extends from the little finger side (inner side) of the wrist to the little finger side of the elbow is called the *Ulna*. The point of the elbow and most of

the elbow joint is formed by the upper end of the ulna. The lower end of the ulna is excluded from the wrist joint proper. The radius is the smaller of the two bones. Its upper end is small and forms a very small part of the elbow joint. The lower end, which is very much larger, forms the whole of the wrist joint.

(*e*) *Wrist bones* consist of eight small irregularly shaped bones arranged in two rows, beautifully adapted to facilitate the delicate movements of the wrist.

(*f*) *Palm bones* are a row of five long bones attached at one end to the wrist bones and at the other to the finger bones. These form the frame work of the palm of the hand and form the prominence of knuckles.

(*g*) *Finger bones* are fourteen in number arranged in a row of three for each finger, and two for the thumb.

Bones of Lower Limb. Fig. 3.

These are concerned mostly with movements connected with walking, kicking, running, etc. In shape and number they are the same on the right and left sides and consist of the following:—

(*a*) *Thigh bone;* this is the longest, largest and strongest bone in the skeleton and forms the hip joint above by its ball-shaped upper end with the cup-shaped cavity on the outer surface of the haunch bone. By its lower end it forms the knee joint with the upper end of the shin bone.

(*b*) *Knee cap* is a small triangular flat bone situated in front of the knee joint. It protects the front of the joint and increases the power (leverage) of the muscles of the front of the thigh by making them act at a greater angle.

(*c*) *Leg bones* are two in number:—

Shin Bone (*Tibia*). It is the second

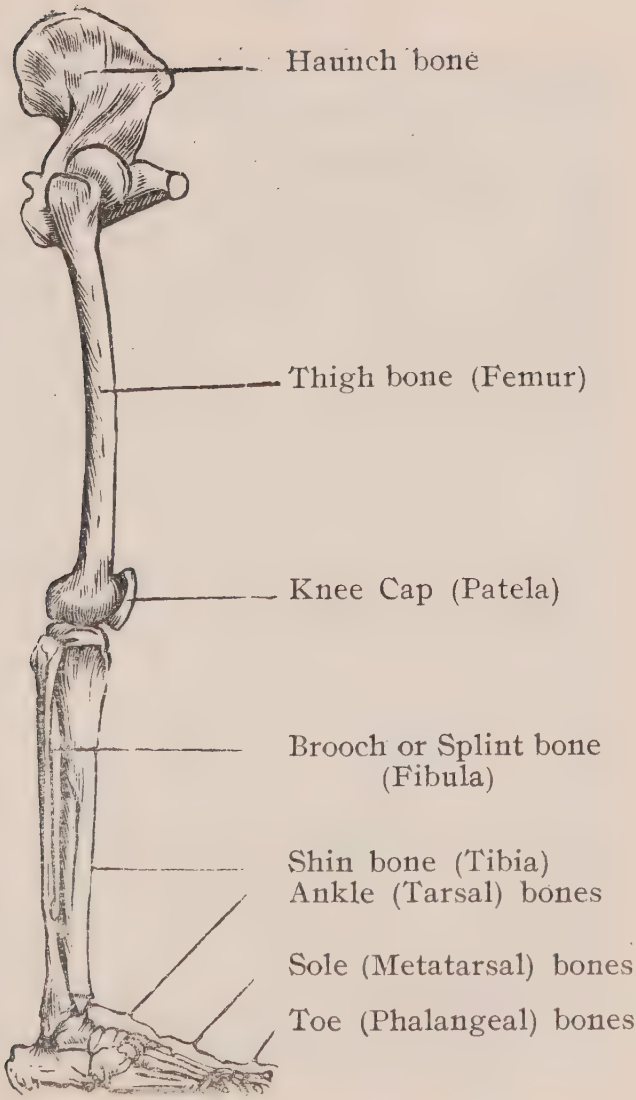


FIG. 3. Bones of the lower limb.

largest bone in the body and is situated at the front and inner side of the leg. The upper end supports the lower end of the thigh bone and with it forms the knee joint. The lower end, which is smaller than the upper, enters into the formation of the ankle joint. Its prolongation downwards forms the inner support of the ankle (Inner malleolus). Most of the inner surface of this bone can be felt under the skin and is called the shin.

Brooch bone (*Fibula*) is the smaller of the two leg bones and in proportion to its length the most slender of all the long bones in the body. Its upper and lower ends are fixed by ligaments to the outer side of its fellow, the shin bone, which receives support from it, hence its name of *splint bone*. The upper end is not included in the knee joint. The lower end enters into the formation of the ankle joint. Its prolongation downwards forms the

outer support for the ankle (External malleolus).

(*d*) *Ankle bones* (*Tarsal bones*) are seven irregular bones arranged in three rows well adapted to facilitate the movements of the ankle and help to form the arch of the foot. Only one bone (Astragalus or Talus) enters into the formation of the ankle joint, the rest go to form the arch which supports the ankle.

(*e*) The sole and instep of the foot is formed of five bones similarly to those in the palm of the hand.

(*f*) The toes, like the fingers, are formed of fourteen little bones, two for the big toe and three for each of the other four.

CHAPTER IV

Joints, Ligaments, Muscles and Tendons.

By *Joints* are meant those places in the body where movements in certain direc-

tions can be performed. In the elbow, the knee and the lower jaw, movement in one direction only like that in a hinge is allowed. In a ball and socket joint, like

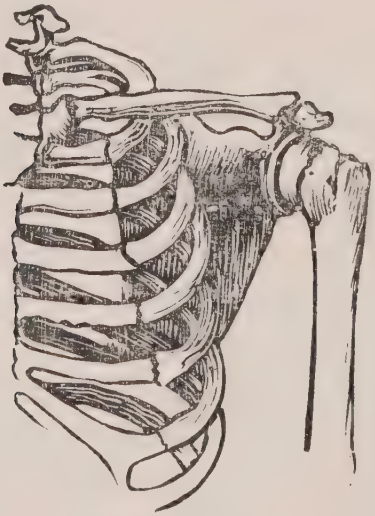


FIG. 4. Shoulder Joint. Collar bone.

the shoulder and the hip joints, movements in many directions are allowed. In the wrist and ankle joints, a more limited range of movement is allowed as compared to a

ball and socket joint, but it is more complex than those in the elbow and the knee joints. In each joint parts of two or more adjacent bones work on each other, but are held together in position by ligaments. *Ligaments* are strong non-yielding bands of tough fibrous tissues which surround the joints and hold the different working surfaces of the bones together. The working surfaces of the bones in the joint are lined by a thin coating of gristle to reduce friction, and the surface of the ligaments within the joints forms a soft flexible sack called the *Capsule* of the joint and has a lining of a delicate membrane which secretes a glairy substance like the white of an egg, called *Synovia* or *joint-oil*, which lubricates the joints.

Muscles.

Muscle is the same thing as that which we call flesh and meat, and is made up of

bundles of red fibres which have the power of contraction and relaxation. The contraction and relaxation of muscles under the influence of the will enable us to perform different movements of the body and to exert power.

Tendons or *Sinews* are the whitish tough fibrous cords into which the fleshy muscles get gathered together before they get attachment to bones.

There are two kinds of muscles, (1) the voluntary and (2) the involuntary.

(1) The *voluntary* muscles are those which can be made to contract or relax under the influence of one's will. These muscles extend from one bone of the skeleton to another with a joint usually intervening between their attachments. Their contraction and relaxation cause movements of the limb and the body. For instance the muscle in the front of the arm (*biceps*) extends from the top of the

shoulder blade bone to which it is attached above, to the upper portion of the bone of the forearm to which it is attached below, the elbow joint intervening. This muscle, when contracted and relaxed at will, bends the forearm backwards and forwards. *See* Fig. in Frontispiece.

(2) The *involuntary* muscles are those which cannot be relaxed or contracted under the influence of one's will. These muscles continue to perform their work even during sleep and insensibility. Contraction of the heart, and movement of the muscles of the stomach and bowels in the process of digestion, are examples of the action of involuntary muscles.

CHAPTER V

Abdomen and its Contents ; their importance in First Aid. (See Figure in Frontispiece.)

The *abdomen* is another name for what is commonly called belly. The *midriff* separates it from the cavity of the chest, *vide* page 21. Some of the most important organs are placed within the abdomen. The First Aid man should know their position in the healthy and in those suffering from common Indian diseases like malaria, dysentery and chronic dyspepsia which alter their normal position of security afforded by nature and make them liable to get easily injured. (1) The *Stomach* lies on the left side under cover of the ribs, excepting for a little distance behind the hollow part just below

the breast bone called the "pit of the stomach." In the dyspeptic, the stomach gets dilated and reaches several inches below the left ribs. (2) The *Liver* lies in the upper part of the abdomen on the right side protected under the right lower ribs and almost entirely covered by them. (3) The *Spleen* lies beneath the ribs at the upper part of the left side of the abdomen between the left side (*fundus*) of the stomach and the midriff. The liver and spleen cannot be felt in the healthy. They often get injured as a result of knocks and blows and kicks on the side (ribs) with serious results. They are often enlarged in the poorer malaria-stricken classes of Indians and can easily be felt beyond the covering of the left lower ribs ; when thus enlarged they are liable to get injured easily, resulting in collapse and death. (4) The bowels (*Intestines*) occupy the greater part of the cavity of the abdomen.

(5) The sweet bread (*Pancreas*) is securely placed behind the stomach with coils of bowels lying in front and is not of much importance to the First Aid man. (6) The kidney lies securely in the loins at the back but occasionally gets injured in crushing injuries. (7) The *urinary bladder,* when full, reaches several inches above the front of the brim of the pelvis and can be easily injured by a kick or a fall.

A kick or a blow on the abdomen without injuring any of the organs may cause serious shock and collapse on account of the disturbance of functions controlled by the sympathetic system of nerves which freely ramify in the bowels and other abdominal contents. The First Aid man may save a patient by promptly overcoming the shock.

CHAPTER VI

Triangular Bandage—its utility and advantages.

Triangular Bandage.

It is used for—

(1) Keeping a dressing in position.
(2) Securing splints in position.
(3) Supporting an injured arm as sling.

Triangular bandage is *most suitable* for First Aid work because—

(1) It can be easily made from any piece of cloth, a yard or thirty-eight inches square by folding it into two from corner to corner and cutting it across. A handkerchief can provide a triangular bandage though it may not always be of the standard size, *i.e.*, 38″ inches square.

(2) Its application is easy and requires very little practice and experience.

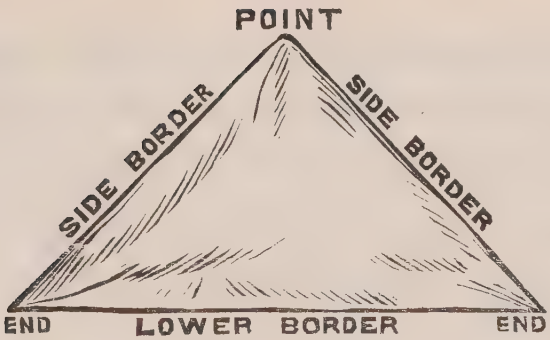
(3) It can be quickly applied and can be utilised in one form or another on any part of the body.

(4) In its application there is not the same likelihood of pressing on blood vessels and stopping the circulation of blood in the limb (*strangulation*) as in an unskilfully applied roller bandage.

The triangular bandage is *used in four different ways, viz.*, (1) Open, (2) Broad, (3) Medium, and (4) Narrow.

Of the three borders of the bandage the base of the triangle or the longest border is called the *lower border* and the two others, the *side borders*. The corner opposite the base or the lower border is called the *point* and the remaining corners the *ends* of the bandage.

The open bandage does not require any description, it is the whole triangular cloth



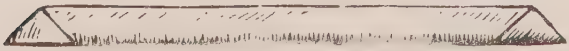
Open Bandage.



Broad Bandage 1-Fold (Incomplete).



Broad Bandage 2-Fold (Complete).



Narrow Bandage (4-Fold).



Dotted Lines show the folds of Medium Bandage.

FIG. 5.

unfolded and spread to its full extent. The *broad* fold bandage is made by bringing the point down to the lower border or base of the triangle and then folding it lengthwise into two. The *narrow* fold bandage is made by folding the broad fold bandage once and thus consists of four folds. The *medium* fold bandage is made by bringing the point down to the centre of the base as in the broad fold bandage but then folding it into three folds. *Vide* Fig. 5.

Note.—When not in use the bandage should be put away folded in such a way that it could be easily brought out ready for use. The following is the standard method:—Fold the triangular bandage into a narrow fold bandage, bring the two free ends to the centre, the two ends now formed should then be turned to the centre, then double the bandage upon itself, reducing it into a small packet.

In tying the ends of bandages fix them

by a *surgeon's knot,* also known as a *reef knot,* as it does not slip and is undone easily. *Granny knots* are hard to undo but liable to slip.

To tie a reef knot:—Hold one end of the bandage in the left hand and the other



Reef knot. Granny knot.

FIG. 6.

end in the right hand. Pass the left end in front of the right hand end and twist one end around the other in the ordinary method of tying a knot ; then pass the left hand behind the right and finish the knot.

Arm Slings.

Three varieties of arm slings are used :—

(1) St. John Arm Sling.
(2) Large Arm Sling.
(3) Small Arm Sling.

Method of Applying St. John Arm Sling.

(1) Stand in front of patient. (2) Lay an open triangular bandage across the chest over the injured limb placing one end on uninjured shoulder and the point overhanging the elbow on injured side. [Fig. 7 (1)]. (3) Pass the lower end of the open triangular bandage under injured limb and carry it across the back and tie it off loosely on top of sound shoulder. [Fig. 7 (2)]. (4) Bring the point folded over the elbow of the injured limb and secure it by a safety pin. [Fig. 7 (3)]. (5) In order to support and raise elbow

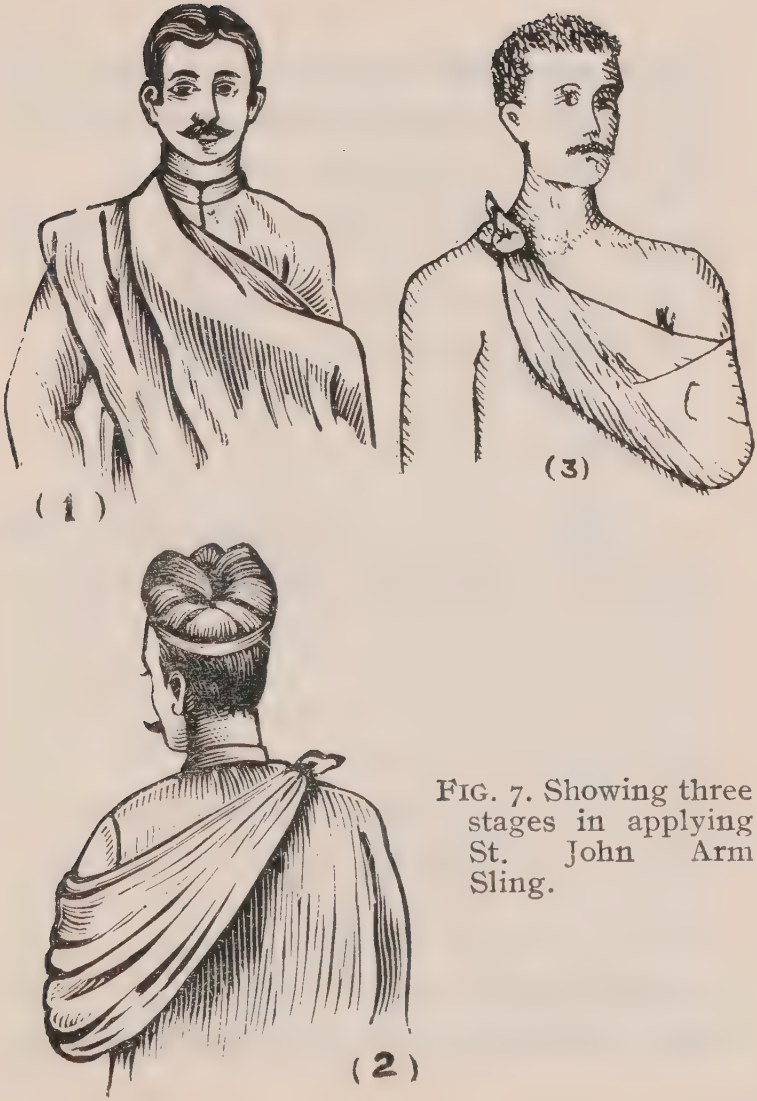


FIG. 7. Showing three stages in applying St. John Arm Sling.

into proper position, undo the knot on the collar bone, readjust the sling and tie off the knot as in Fig. 7 (3).

St. John Arm Sling is the sling to be used for choice when it is not advisable to have any forward movement of the arm as in the case of a broken collar bone.

Method of Applying Large Arm Sling.

(1) Stand in front of patient and hold point of an open triangular bandage in one hand and one of its ends in the other. (2) Place the bandage on the front of body so that the point lies between body and elbow of injured side. (Fig. 8A). (3) Carry the end over the sound shoulder, draw up the bandage sufficiently to allow it (the end) to go round the back of the neck to the shoulder of injured side. (4) Bend elbow and lay the forearm and hand of injured side on bandage placing

the hand a little higher than the elbow. (5) Now bring the lower end of the bandage over the forearm and carry it up to the

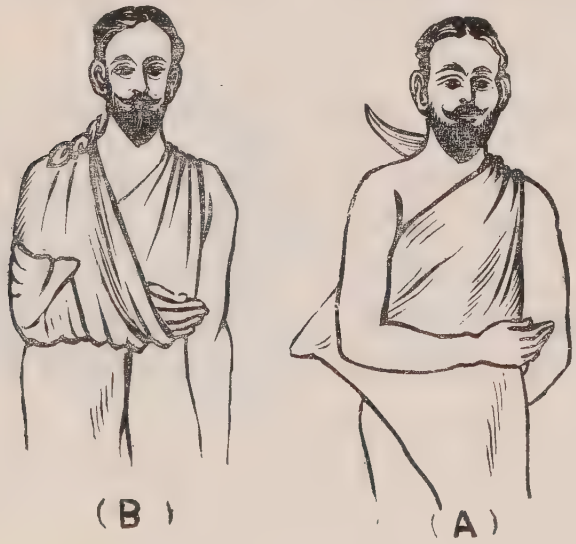


FIG. 8. Large Arm Sling.

end on the injured shoulder where it is tied. (6) Tighten bandage and tie off as in St. John Arm Sling. Bring the point

round to the front of the bandage and secure it there by a safety pin. (Fig. 8-B).

Note.—The large arm sling and St. John arm sling support the weight of the whole arm, and also give rest to the shoulder.

Method of Applying Small Arm Sling.

(1) Make a broad fold bandage. (2) Place one end on the sound shoulder and carry it round the back of the neck to the injured shoulder. (3) Bend elbow on the injured side and place the hand higher than forearm, support the wrist and part of the hand and forearm by the bandage, bring up lower end and tie to the upper end. (Fig. 9).

Note.—Avoid putting the knot on the back of the neck as it will hurt the patient while lying on his back. The hollow in front, below the collar bone is the most suitable place. The top of the shoulder is the next best.

Where a safety pin is employed to secure a

bandage see that the pin bears no weight nor presses into the flesh.



FIG. 9. Small Arm Sling.

Improvised Slings may be used when the real slings are not available. Amongst such may be mentioned passing the hand inside the buttoned up coat, pinning the tail to the front of the coat, and securing the sleeve to the clothing by safety pins.

CHAPTER VII

Application of Triangular Bandage for Retaining Dressings.

(1) *Head.* Fold the lower border or base to form a hem about 2″ wide. Place

the middle of this hemmed border on the forehead of the patient immediately above the eyebrows, make the remaining portion of the triangle cover the top of the head, the point hanging down over the back of



FIG. 10. Head Bandage.

the head. Carry the two ends round the head passing just above the ears towards the centre of the back of the head. Cross the two ends as low down the back of head

as possible, bring forward again and tie off in the centre of the forehead.

Pull the point well down towards the nape of the neck and then turn up and secure on top of head by a safety pin. (Fig. 10).

Note.—If you begin the bandage with the hemmed part well down over the forehead in front and carry it also low down the back of the head, the bandage will not slip.

(2) *Forehead.* The centre of a narrow bandage is placed over the dressings and the bandage carried around the head, above the ears, to the back of the head where it is tied off. In case of hæmorrhage from the forehead or side of the head the ends are crossed at the back of the head and are then brought forward and tied over the centre of the dressings applied over the bleeding part, the object being to apply extra pressure over the bleeding vessel.

(3) *Lower Jaw.* Place the middle portion of a narrow fold bandage under the chin and carry one end of the bandage over the top of the head, down to the other side

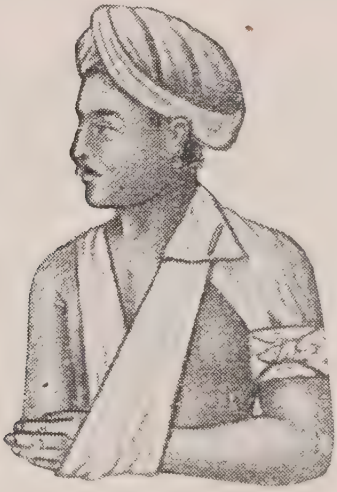


FIG. 11. Shoulder Bandage.

and then cross with the other end of the bandage at the angle of the jaw. Carry the long end across the chin and tie the ends together. (*Vide* Fig. 23, page 73).

(4) *Shoulder Bandage.* Put the forearm of the injured side on a small arm sling. Take a triangular bandage and fold the lower border of the bandage into a narrow hem, lay the centre of the bandage on top of shoulder, the point touching the neck of the patient, the lower hemmed border lying over middle of arm. Wind the ends round the arm, cross them behind and tie off on the front. Draw point of bandage under the arm sling, turn it down so that the knot of the sling gets covered, then keep fixed with a safety pin. (Fig. 11).

(5) *Elbow Bandage.* Bend elbow, raise it in the bent position with the back of the elbow towards you. Place centre of an open bandage over the back of the elbow, point upwards on the back of arm towards the shoulder. Place the centre of lower border over the back of forearm about three inches below the bend of the elbow—

fold it into a narrow hem. Pass ends round forearm crossing them in front of the elbow, pass ends up round arm, cross behind the arm and knot off in front of arm, tighten by drawing up the point which is then turned back. (Fig. 12). Put up the limb on a large arm sling.



FIG. 12. Elbow Bandage.

(6) *Hand Bandage.* The hand may be bandaged closed as a fist (Fig. 54, page 160) or with the fingers open (Fig. 13). In either case spread a triangular bandage to its full extent, and place the palm side of the hand on the

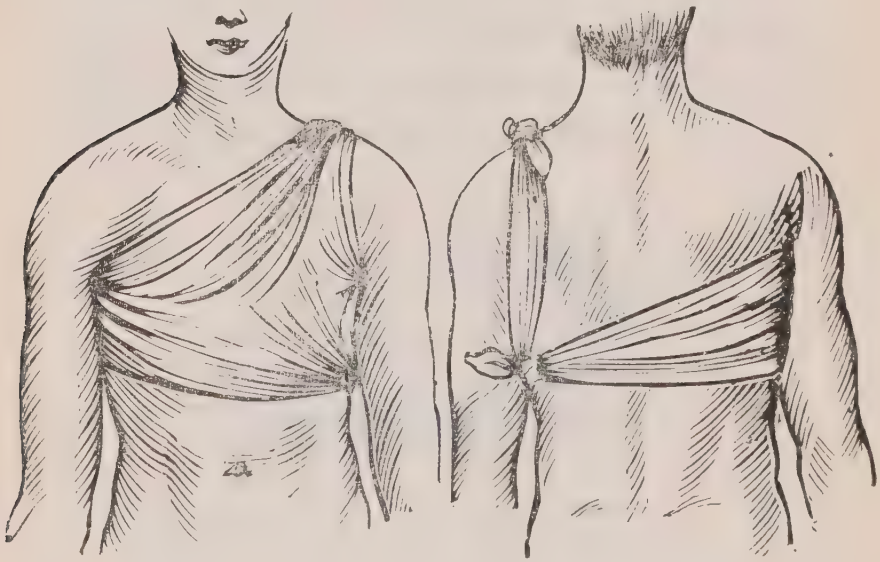
centre of the bandage, fingers towards the point. Turn back point over back of hand to reach the wrist, pass ends round wrist crossing them over the point. Pull and fold point towards fingers and cover it by another turn of bandage round wrist. Tie off ends in knot in front of or behind the wrist.



FIG. 13. Hand Bandage with Fingers Open.

(7) *Chest Bandage.* Place the centre of an open bandage on the dressings, the point over the shoulder of the same side over which dressings are put on. Take the ends round the body and tie them at one side of the back so that one end becomes

longer than the other. Pull the point over the shoulder to reach the back and tie it to the longer end. (Fig. 14).



Front View. Back View.

FIG. 14. Chest Bandage.

(8) *Hip Bandage.* Take a narrow fold bandage, put it round the waist and tie off so that the knot is on the injured side.

Take an open bandage, fold the lower border into a narrow hem, place the centre of the bandage over the dressing, bring the ends round the thigh and tie off on its



FIG. 15. Hip Bandage.

outer side. The point is carried up under the first bandage, pulled tight, turned down over the knot and pinned. (Fig. 15).

(9) *Knee Bandage.* Make a hem, place it two inches below knee, cover the knee with the bandage, the point of the bandage lying over the lower part of the thigh.



FIG. 16. Knee Bandage.

The ends are brought to the back of the knee, crossed and brought forward again above the knee joint and tied (Fig. 16).

The point is turned downwards and pinned.

(10) *Foot Bandage.* Open a triangular bandage and put it on the floor with the point away from the patient, placing foot on the bandage with toes towards point



FIG. 17. Foot Bandage.

and heel about 2½ inches from the lower border. Carry point over instep, lift up ends and cross over instep, then carry ends behind ankle, cross behind ankle and bring ends in front and tie off in front on ankle.

Pull point up and then bring it forward in front of ankle and fix it with safety pin. (Fig. 17).

CHAPTER VIII

Fractures: Causes, Varieties, Signs and Symptoms.

When a bone breaks, it is called a fracture. Fracture is defined as "a sudden solution of continuity in a bone usually resulting from external violence."

Causes.

(1) Direct violence, (2) Indirect violence, (3) Violent muscular action.

(1) By *Direct violence* is meant that the force causing the fracture falls directly on or over the bone which is broken. The bone breaks at the place where the force is applied. Examples:—The blow of a

"Lathi," the fall of a heavy stone, the passing of a wheel of a bullock cart or motor car over a limb or any part of the body—resulting in breakage of a bone.

(2) *Indirect violence* is that when a fracture occurs as a result of force or violence applied at some distance from the seat of fracture, *e.g.*, a person falls off from horseback and alights on his hands, the force transmitted through the fore-arm results in fracture of the collar bone. A man jumps down from a terrace and lands on his feet, the force transmitted results in fracture of his thigh bone or that of the bones of the base of the skull. A man gets caught in a big crowd during a "mela" and gets his chest pressed front aft and the ribs at the side break due to indirect violence.

Note.—A man falling on his hands or feet apart from the injuries caused by indirect violence may also get fracture of bones of his

forearm or leg or those of hands or feet as well, due to direct violence caused by the fall.

(3) *Muscular action.* A bone snaps in two by sudden and violent contraction of strong muscles attached in it, *e.g.*, the knee-cap bone snaps in attempting to jump across a *nullah* with a heavy load. Similarly the thigh bone may also break. In the same way sudden lifting up of a heavy load by a very muscular person may cause fracture of arm bone.

Varieties.

There are six different kinds of fracture —the two chief groups being Simple and Compound.

(1) A *simple* fracture is one in which the bone is broken but the skin over it remains unbroken. It does not matter whether the bone is broken in two or three or more fragments. As long as the

skin over the broken bone remains whole, it is called a simple fracture. (Fig. 18).

(2) A *compound* fracture is one in which not only is the bone broken but the skin over it is also divided either by the same violence which caused the fracture or by the broken ends of the fractured bone piercing it. (Fig. 19). Germs which cause disease are present in the soil and in the air and gain entrance through the wound in the skin into the broken bone and soft tissues around it and poison them giving rise to most serious consequences. Hence a compound fracture is a very serious accident.

(3) *Complicated* fracture is the name for that variety in which not only is the bone broken but in addition there is injury caused to some internal organ or blood vessel or nerve, *e.g.*, (*a*) a man is kicked on his side by a horse, it causes fracture

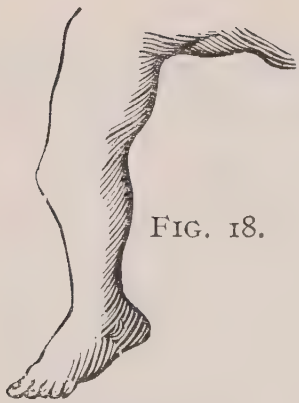


FIG. 18.

Simple fracture.



FIG. 19.

Compound fracture.



FIG. 20.
Comminuted
fracture.



FIG. 21.

Green stick
fracture.



FIG. 22.

Impacted
fracture.

of a rib by direct violence, the broken ends are driven in and injure the lung ; (*b*) an untrained person attempts to help a man who has been knocked down by a motor car and sustained fracture of his leg bones, he twists the broken limb in such a manner that the sharp end of the broken bone punctures an adjoining blood vessel and also lacerates the nerves lying close to it. A complicated fracture may be either *simple* or *compound*.

(4) *Comminuted*. In this the bone is broken into more than two pieces. A comminuted fracture may be simple, compound or complicated. (Fig. 20).

(5) A *green stick* fracture is one in which the bone does not break asunder into two or more pieces but is only bent, cracked and half broken like a doubled up green twig of a tree. This kind of fracture occurs in soft bones of young children. (Fig. 21).

(6) *Impacted.* It is that variety in which one end of a broken bone is driven into the other fragment by the force (impact) of the violence. (Fig. 22).

Signs and Symptoms of Fractures.

(1) The patient feels great *pain* over the broken part.

(2) The patient is unable to use the limb for its usual purpose. There is *loss of function and power ; e.g.,* when the arm is broken the hand cannot be raised and when the leg is broken the man cannot lift it up or stand on it or kick at any thing with it.

(3) There is *alteration in shape and contour.* The limb gets bent, twisted or shortened and assumes an unnatural shape —thus causing *deformity.*

(4) *Swelling* is observed due to effusion of blood around the seat of fracture

and also overriding of bones due to contraction of muscles.

(5) *Irregularity* (of surface) due to overriding of broken fragments felt and even seen easily when the broken bone is not deep under cover of muscles but has only thin tissues covered over by skin lying over it, *e.g.*, collar bone, shin bone, lower jaw. If the bone is placed deeply, one must have some experience to be able to feel irregularity. An ambulance man need not try to feel it lest he aggravate the injury and cause pain to the sufferer.

(6) *Unnatural mobility.* Bones give rigidity to the limb, which if broken gives way where the bone is broken ; this is unnatural.

(7) *Crepitus.* This is the grating sensation felt when one broken surface of bone rubs against another.

Note.—A First Aid man should not attempt to feel crepitus lest he aggravate the condition

and convert a simple fracture into a complicated or a compound one and it also causes much pain to the patient.

PRINCIPLES OF TREATMENT IN FRACTURES.

The main principles of treatment in fractures by a First Aid man are to (1) *reduce shock* caused by the injury and (2) *prevent aggravation* of the injury especially during transport. Make the patient comfortable, relieve pain and overcome shock.

CHAPTER IX

General Rules for the Treatment of Fractures.

(1) Attend to the fracture on the spot —no matter whether the place of accident is a crowded thoroughfare or a bazar, or a race course, or a battle field.

(2) If there is any bleeding, stop it first—the other injuries can be attended to afterwards. Save the life blood first and prevent further contamination of the wound. *See* page 5, and 6.

(3) Immediately support and keep the injured limb steady. If engaged in stopping bleeding see that some one else steadies the limb.

(4) While waiting for the doctor keep your patient comfortable and covered up so that he will feel warm. This reduces and overcomes shock caused by the injury.

(5) When in doubt, treat the injury as a fracture—you do no harm.

(6) Do not move the patient from the place of accident until after having attended to the fracture.

(7) Apply your improvised splints and bandages, so as to prevent movement not

only in the broken parts but also in the joints above and below the fracture.

(8) If you suspect a fracture of the head, spine, haunch or thigh bone, do not allow the patient to sit up. He must be carried lying down on a stretcher.

(9) Put up fractures of leg, keeping the foot at right angles to the leg—those of the forearm, with the hand open and the thumb up, pointing towards the chin. The hand in the sling should be kept higher than the elbow.

(10) Tie off your bandages and finish them up with the knot always lying on the splints and never on the broken part.

(11) Always pad your splints.

Improvise *splints* and *bandage* from any material available on the spot. Lathies, walking-sticks, umbrellas, police constable's baton, sword, bayonet, rifle, flat Indian pattern *Salim Shahi* shoes, pieces of card-board, stiff cover of a book,

straw and wire covers of medicine and wine bottles, folded map or magazine or news paper have all served the purpose of giving support and preventing movement of the broken part. If triangular bandages are not handy use instead handkerchiefs, neckties, pugries, jharans, gamchas, kumarbunds, belts, straps, strings, fibres and twigs of creepers. Improvise *padding* from cotton wool, tow, jute fibres, layers of cloth, triangular or any other bandage, in fact anything that will prevent the rough or hard surfaces of improvised splints from pressing over bony prominences and hurting the patient.

Note.—The surgeon "sets the fracture," *i.e.*, brings the broken ends in exact apposition to each other. The First Aider is likely to do more harm than good by trying to set a fracture or attempting to correct shortening or deformity in a broken limb; therefore a knowledge of what to do and what not to do is very important.

Nemonic tip for remembering treatment of broken bone is—

"FRACTURES."

F—Fix immediately.

R—Restrain movement.

A—Apply support (of your open hand, then splint).

C—Control hæmorrhage.

T—Traction—gently pull broken parts before applying splints.

U—Uncover—Inspect, and attend if there is free bleeding.

R—Remove—carefully.

E—Exercise extreme care in handling.

S—Shock, caused by the injury to be treated.

N.B.—*Remember* the tip is only for bearing in mind the different steps one should take in attending to a fracture and does not imply that a fracture may be attended in the order given in the tip. For instance control of hæmorrhage comes first in treatment and one does not uncover to see and attend to free bleeding after fixing splints.

Some Important Points to Remember in the Treatment of Fractures.

Note.—(1) If in doubt as to whether or not the splints are of suitable size, measure them from the sound limb. (2) Never put on a bandage over the splints or on the seat of injury, as this is painful. (3) Tie all your knots in a line. (4) Tie the upper end of splints first, the lower end next, then fasten a bandage on either side of (but not over) the injured part. (5) The conveyance on which you are removing your patient should travel slowly in order to avoid bumps on a rough road which would shake the fractured bones and cause pain. (6) While putting on an arm sling, always get an assistant to support the limb with his hands while you are fixing the sling in position. (7) Do not fix your splints and bandages too tight, they will give pain and if left longer than two hours, will cause gangrene. (8) *Choice of slings*:—For fractures anywhere from fingers to elbow joint, use a large arm sling. For fracture of the arm bone, use small arm sling. For collar bone, use St. John arm sling. (9) It is not necessary to take off clothes for rendering "First Aid" in simple fractures—they may act as pads.

CHAPTER X

Individual Fractures—Details of Treatment.

Fracture of the Skull. Cause: direct violence to the upper part and indirect violence to the base—they are both dangerous on account of the shaking (concussion) of the brain caused by the violence or pressure (compression) on the brain caused by a piece of broken bone driven in by the force of the violence.

In concussion unconsciousness comes on immediately. In compression it gradually grows worse and worse, it is associated with a fracture of the skull. *See* page 15 and 251.

Fracture of the Base.

It is almost invariably fatal. *Signs*: in addition to unconsciousness, there is discharge of blood from the mouth, nose,

eyes, ears. Part of the blood may be swallowed by the unconscious patient and then vomited out. A clear watery discharge is sometimes seen to come out through the ears in addition to blood.



Treatment of Fractured Lower Jaw.

FIG. 23.

This fluid is contained in the membraneous covering around the brain and its discharge through the ear shows that this covering is torn. *Treatment*: Confine

your efforts to overcoming and reducing shock, keep the patient lying down with head and shoulders slightly raised. Loosen clothes round the neck and over the chest. Keep him quiet, *do not give stimulants.* Wipe away the discharge but do not plug his ears as it will cause congestion. Arrange for transport lying down on a stretcher.

Fracture of Lower Jaw. Fairly common, always compound. *Special signs*: (1) Irregularity of teeth, (2) Bleeding from gums. *Treatment*: (1) Place the palm of your hand below the fractured bone and press lower jaw against the upper, (2) correct deformity, keep in this position and (3) apply a narrow bandage as directed on page 50.

Fracture of the Shoulder Blade. An uncommon accident. *Treatment*: (1) Place the centre of a broad fold bandage in the arm pit of the injured side, (2) pass over

sound shoulder cross, (3) go down and knot off in sound arm pit. (Fig. 24). (4) Apply St. John arm sling on injured side.

Fracture of Collar Bone. *Special signs*: (1) The arm on the side of broken collar



Treatment of fractured shoulder-blade.

N. B.—The illustration does not show the St. John sling.

FIG. 24.

bone lies helpless by the side, (2) the patient droops his shoulder on the injured side, (3) he supports the elbow of that side with the hand of the sound side.

Treatment. You need not remove the

patient's clothing. *When one collar bone is broken.* (1) Put a pad of the size of a small clenched fist in the armpit of the injured side, (2) bend the elbow and support the limb against the chest and (3) apply a St. John arm sling. (4) Fix elbow and lower part of arm of injured side by a broad fold bandage. (Fig. 25).



Treatment of fracture of one collar bone.

FIG. 25.

The shoulder of the injured side should be drawn back and the elbow raised by the application of bandage and sling so as to

correct the deformity caused by the over-riding of broken pieces of the collar bone. If this is done carefully and well, the patient feels remarkably relieved and comfortable.

Note.—Feel the pulse after tightening broad bandage. If you cannot feel it, your pad in the armpit is pressing too much on the blood vessel at the side of the arm, therefore loosen your broad bandage round the elbow and arm and readjust so that the circulation is re-established and you are able to feel the pulse.

When both collar bones are broken: (*a*) *By four narrow bandages.* (1) Take four narrow bandages or pieces of pugree cloth or kumarbands, (2) pass one under each armpit and tie loosely over the shoulder, forming a loose ring, (3) the third narrow bandage is passed within these rings from behind and the ends are pulled tightly and tied in the middle of back, (4) when both shoulders are thus drawn back in a suitable position—the

elbows are bent, the forearms placed on the chest with the hands crossing each other and the fourth bandage is put round the body passing above the elbows over the forearm and is tied off in front. (Fig. 26).



FIG. 26. Treatment of fracture of both collar bones with four narrow bandages.

(*b*) *By two narrow bandages.* (1) Tie one end of a narrow bandage round each arm close to the shoulder, (2) pass each

free end across the back over the opposite arm, (3) bring them round to the front and tie them together by a knot, (4) the forearm is kept raised and supported by the bandage. (Fig. 27).

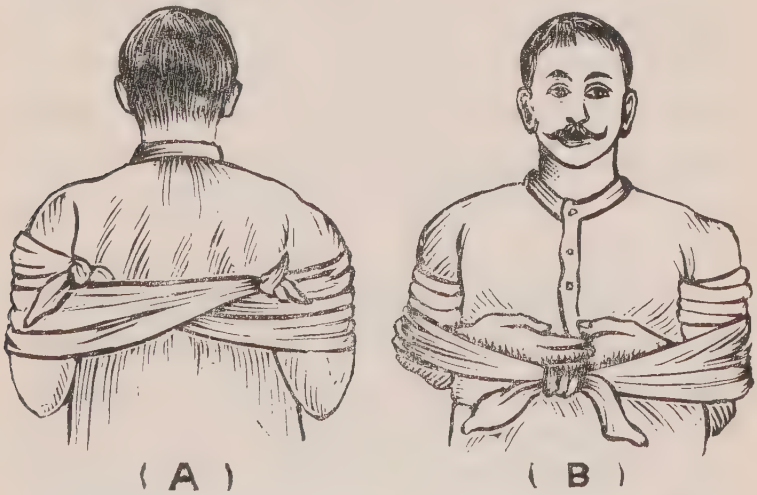


FIG. 27. Treatment of fracture of both collar bones with two narrow bandages.

Fracture of arm. This bone may be broken by direct, indirect or muscular violence and may break near the shoulder joint, near the elbow joint or in the middle

of the arm. Fracture in each region must be dealt with in a different way.

Fracture near the shoulder joint. (1) Apply a broad fold bandage placing its centre on upper and outer parts of the injured arm, (2) carry the bandage to the armpit of the opposite side (sound side), (3) cross there, (4) carry the ends above shoulder of sound side and (5) tie off, (6) put the fore-arm of injured side on a small arm sling.

Fracture near elbow joint. (1) Take two pieces of improvised splint—one piece long enough to extend from armpit to elbow and other long enough to reach from elbow to the fingers. Tie them together to form the letter L (Fig. 28). Pad well. (2) Bend elbow into a comfortable position and place it across chest. (3) Place the splint between the body and the injured limb, the elbow resting on it. (4) Secure splint by three narrow bandages one above

the fracture and one below the elbow joint and the third round the splint and the hand. (5) Apply a large arm sling.

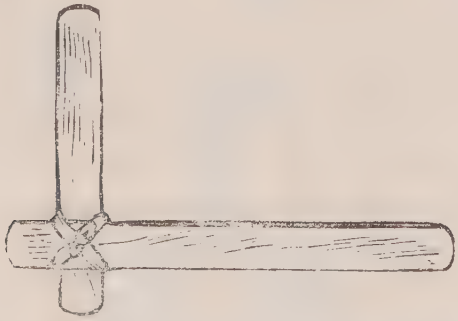


FIG. 28. Improvised angular splint.

Note.—When patient reaches home take off the splint, put elbow in a comfortable position on a pillow and put on cold water dressing and await arrival of doctor. In cases where this accident occurs indoors, this treatment is all that need be done by the First Aid man; he may not put up the limb in a splint except when the accident occurs out of doors or while travelling.

Fracture in middle of the arm. (1) Improvise four pieces of splint—three pieces

long enough to reach from the shoulder to the elbow, the fourth one about 2 inches shorter. (2) Apply the small splint on the inner side of the arm. (3) Of the three larger ones, one is placed in front of and one behind and one on the outer side of the arm. (4) Fix splints with two narrow fold bandages, one above and one below the seat of the fracture. (5) Put up fore-arm in a small sling. (Fig. 29).

Note.—Remember the essential splints are those on the outer and inner sides of the arm, and if only two pieces of splints can be improvised, apply them to these parts only. A pair of heelless Delhi-wal shoes have been improvised for use for this purpose, the soles facing the limb. If nothing else is available secure the upper arm to the side of the body with two broad bandages passing one above and one below the seat of fracture.

Fracture of fore-arm. This may happen to one or both bones, and may be near the wrist or away from it. When both

bones are broken, recognition is easy ; when only one bone is broken, look for loss of power, irregularity and other symptoms. Fracture near the wrist joint is often



FIG. 29. Treatment of fracture of middle of arm.

mistaken for a sprain. Treat a fracture of fore-arm as follows :—

(1) Bend fore-arm at right angles to

the upper arm, (2) place hand with palm towards chest and thumb pointing to the chin, (3) correct irregularity and get some one to hold the limb in this position, (4) take two splints slightly broader than the fore-arm and pad them well,—one splint should reach from inner side of



FIG. 30. Fracture of fore-arm.

elbow to the finger tips, the other from the outer side of the elbow to the back of finger tips, (5) secure by four narrow fold bandages, one at the upper end, one at the lower end of the splint and two on either side of the seat of fracture (Fig. 30), (6) apply a large arm sling.

Crushed hand is generally caused by the hand being badly "jammed" or a weight falling on it. There are lacerated wounds and compound fractures of some of the bones of the palm and fingers and possibly of the wrist also. *Treatment*: (1) Stop bleeding. (2) Prepare a well padded splint long enough to reach from



FIG. 31. Treatment of crushed hand.

above the wrist to beyond the tip of the fingers. (3) Place the whole hand with the fingers open, and the palm downward on the splint. (4) Secure the splint in position by a narrow fold bandage applied like a figure of 8 (Fig. 31). (5) Place the hand carefully in a large arm sling keeping the hand higher than the elbow.

Fracture of thigh bone. This bone may break at its upper end near the hip joint, or at its lower end near the knee joint, or anywhere between these two points.

Note.—In old people a trivial cause such as tripping over a carpet or missing a step in going downstairs may cause a fracture in the upper end (neck) of this bone.

Special signs. (1) The foot lies turned outward. (2) Marked shortening of the limb—as compared with the uninjured limb. (3) Marked inability on the part of the patient to move the limb.

Treatment. (1) Steady the limb by firmly but gently holding the ankle and foot.

(2) Gently pull the limb straight keeping the foot at right angles to the limb, and the leg in a line with its fellow. (Shortening and deformity of the limb will in this way be overcome.)

(3) While the leg is held in this position by an assistant, apply two well-padded splints, one on the outer side long enough to reach from the armpit to beyond the foot and another in the inner side to reach from the top of the thigh to the knee.

(4) Secure the splints by seven bandages, three broad and four narrow. (*i*) One broad fold bandage is put round the long splint and the chest just below the armpits. (No. 1, Fig. 32). (*ii*) The

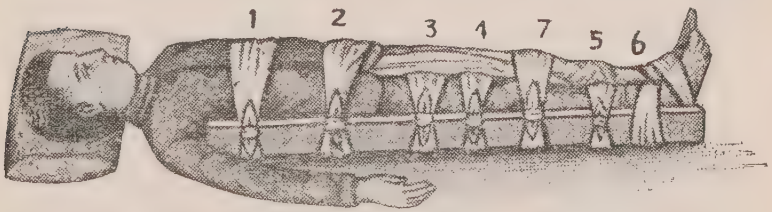


FIG. 32. Treatment of fracture of thigh in male.

second broad fold bandage is put round the splint at the hips. (No. 2, Fig. 32). (*iii*)

The third broad bandage is put on last, after all the others have been fixed ; it is to be passed round both knees, securing the injured limb to the other. (No. 7, Fig. 32).

The four narrow fold bandages are put on thus :—one above the seat of fracture, one below the seat of fracture, both bandages passing round the thigh and both the splints (Nos. 3 and 4, Fig. 32), one round the leg and the long splint (No. 5, Fig. 32), one round both ankles and feet and the end of the long splint and finished off by being tied below the feet (No. 6, Fig. 32).

Note.—See that the top end of the splint at the armpit is well-padded. When there are no assistants to help and also when the patient is a female, the inner splint is not to be applied, only the outer long one is to be put on. Pass the bandages round the outer splint and both the limbs, outside her clothing, so as to avoid exposure. Fig. 33. A long walking stick has been improvised as a long splint.

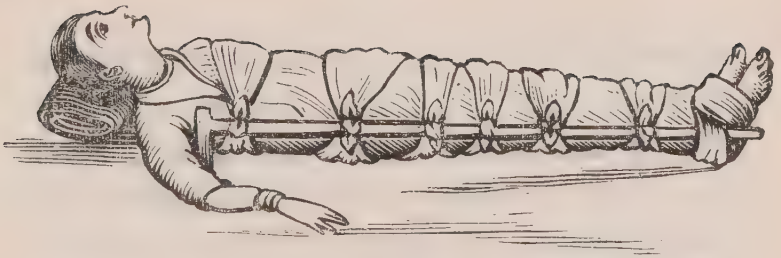


FIG. 33. Treatment of fracture of thigh in female.

Fracture of knee cap. This bone breaks owing to direct violence or violent muscular action.

Rapid swelling of the injured knee joint and pain and inability to use the limb are marked symptoms.

(1) Lay the patient in a half sitting position with his limb straight and the foot raised and keep in that position by an assistant or by placing it on a *mohndah,* a box or a log of wood. (2) Place a padded splint under the injured limb long enough to reach from the buttock to beyond the heel—fix by applying a narrow bandage

the centre of which lies immediately above the knee cap. (3) Carry the ends to the back of the splint and cross there, then bring them in front and tie off below the broken bone. (4) Apply a second narrow



FIG. 34. Treatment of fracture of knee cap.

bandage in a similar way, but begin below the broken bone in front, cross behind the splint and tie off just above the broken bone in front. In the diagram the position of this bandage is shown by dotted lines. (5) Secure the ends of the splint by means of narrow bandages passed around the thigh and leg. (Fig. 34). (6) Apply an ice bag or cold water and spirit dressing over the knee joint.

Fracture of leg. One or both bones may be broken. When one bone is broken, the deformity is not so apparent as when both bones are broken.

Treatment. (1) Support the ankle and foot and see that the foot is at right angles to the leg. (2) Gently pull the limb in the natural position and keep it steady. (3) Apply two well-padded splints, one on each side of the leg ; they must be long enough to reach from above the knee joint to beyond the foot. (4) Secure the splints by applying five bandages, four narrow and one broad. Of the four narrow ones apply one above and another below the seat of fracture (Nos. 1 and 2, Fig. 35), one just above the knee (No. 3, Fig. 35) (all these three bandages passing round both splints). (5) The fourth narrow bandage passes round both the splints and secures both ankles (No. 4, Fig. 35). (6) The fifth one is a broad fold bandage

and is passed round both knees. (No. 5, Fig. 35).

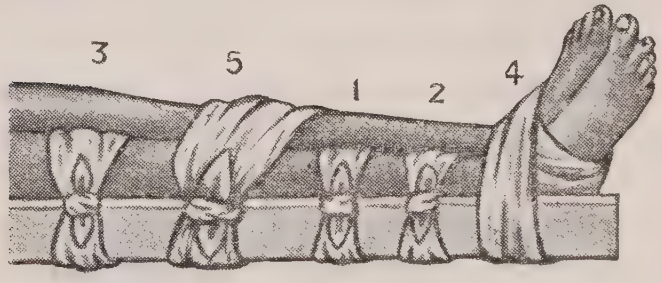


FIG. 35. Treatment of fracture of leg with outer and inner splints.

A very good method of dealing with this fracture is by putting up the leg in a box splint. (1) Take two pieces of thin, smooth board wider than the depth of the limb and reaching from above the knee to below the foot. (2) Fold a bath towel or a *purdah* to the length of the splints, and roll one splint at each side (of the cloth) until the box so formed just fits the sound leg. (3) Hold the broken leg in proper

position and gently put it in this box splint. (4) Fold two small face towels or *jharans* and place them over the front of the leg. (5) Tie bandages one below the

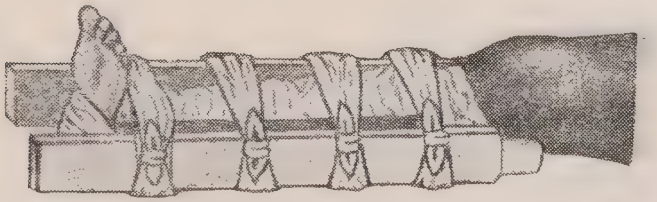


FIG. 36. Fractured leg in a box splint.

knee, one above and one below the fracture. (6) Fix the foot at right angles by means of the fourth bandage passed like a figure of eight 8. (Fig. 36).

Note.—If only one splint is available or if the patient is a woman use only the one splint on the outer side of the limb. Extend the limb and fix both feet together in the manner described already. Apply the broad bandages, passing them round both the limbs and splint.

(Fig. 37). When no splint is available tie the legs, feet and ankles together by means of bandages. The sound leg serves the purpose of

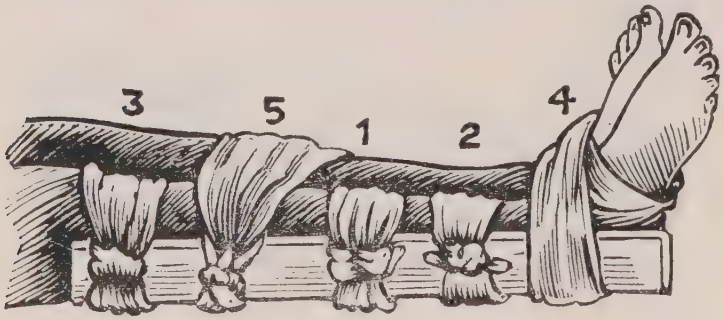


FIG. 37. Treatment of fractured leg with only outer splint.



FIG. 38. Treatment of fractured leg with no splint.

a splint. (Fig. 38). The bandage on the left leg is put over a dressing on an wound.

Crushed Foot. *Causes.* The same as those of a crushed hand. In the same way there are lacerated wounds and compound



FIG. 39. Treatment of crushed foot.

fracture of some of the bones in the foot, toes and possibly the ankle.

Treatment. (1) Gently remove the boot or shoe by slitting it down at the back —loosen or cut off the lace and cut off the socks, at the sides or at the back. (2) Stop bleeding. (3) Prepare a well-padded splint to reach from the heel to beyond the

toes. (4) Secure the splint to the sole of the foot with a figure of 8 bandage which passes round the ankle as well as the foot. (Fig. 39). Keep the foot elevated.

Fracture of the Spine. If a man falls on his back from a height or a heavy weight falls on his back or if he is hit hard on the back, a portion of his spine is liable to be broken owing to direct violence. A " broken back " consists of a fracture of one or more of the vertebræ with displacement of the fragments. The spinal cord contained in it and the nerves issuing from it may get torn or crushed, causing complete or partial paralysis of the part below the fracture, according to the nature and extent of the injuries received. The lower down the fracture occurs, the less is the immediate danger to life (*vide* page 18). *Railway spine* is a term applied to the condition resulting from jarring (concussion) of the spinal cord of the

patient in a railway collision. It causes temporary paralysis.

Treatment—If medical aid is expected within a short time, do not attempt to remove the patient. If not then follow the following rules of transport. (1) Spread out on the ground, with its centre in a straight line with the patient's head, a blanket or a small *durree,* gunny cloth or a strong *purdah* cloth of suitable size. (2) Let two persons one on each side gently draw in the cloth or the blanket under the head and the shoulders of the patient, and work it down without lifting him and gradually draw it under his back, beneath his hips and legs and feet. (3) See that the whole length of the cloth, *durree* or the blanket improvised to serve as a stretcher cloth has been passed underneath the patient and he is lying on it from head to foot. (4) Take two lathis a little longer than the length

of the cloth and strong enough to bear the weight of the patient and place one on each side of the edges of the cloth. (5) Roll the lathis round and round enclosing them in the stretcher cloth till the two lathis are quite close up to the patient's body. (6) Four bearers now gently lift up this improvised stretcher with the patient lying on it. (7) They move sideways to avoid jolting and convey him to the nearest place of shelter or hospital.

Fracture of the Pelvis. Usually the result of a crushing form of violence on or in the neighbourhood of the haunch bones such as the fall of a mass of coal, a log of timber or the passage of a wheel of a heavy cart. There may be no sign of damage to the lower limb, yet the patient will be unable to move the lower limbs, or can do so with great pain and difficulty. The organs lying inside the pelvis might be lacerated by the tearing asunder of the

bones or get perforated by their sharp fragments.

Treatment. (1) Overcome and reduce shock. (2) Restrain movements of broken fragments by applying a broad piece of cloth, a sheet or a towel as a binder around the hips—reaching from the waist above the hips to the upper part of the thigh. (3) Fix the knees together with a broad fold bandage. (4) Place the patient in whatever position he is most comfortable and keep his lower limbs drawn up with a pillow or a folded *durree* or an overcoat as a rest under his bent knees. (5) Remove him on a stretcher taking the same precautions as for a fractured spine.

If a piece of cloth or a binder of suitable width is not available, use three broad fold bandages, one round the hips, one above and one below the hips. Fix the knees.

Note.—In fracture of pelvis, pain is felt in the lower limbs as sensations are present. In

fractures of spine there is loss of sensation in lower limbs. By this symptom one may be distinguished from the other.

Fracture of the Ribs.

Causes: (*a*) Direct violence: A kick or a blow may result in fracture of any part of the rib hit upon.

(*b*) Indirect violence, such as pressure on the front of the chest with the back against some unyielding support commonly causes fracture of 5th, 6th, 7th, or 8th rib at the side.

In fractures caused by direct violence the broken ends are driven inwards and may puncture the lungs, liver or spleen and cause bleeding from the organ injured and in this way result in collapse. In fractures of the ribs due to indirect violence injury to an internal organ is very rare.

Recognition of a fracture of a rib is not

always easy: history of the injury and complain of severe pain felt on deep breathing or on coughing helps. If one can succeed in feeling a grating sensation by laying one's hand on the seat of pain and asking the patient to take a deep breath, the diagnosis is established beyond doubt. If a patient after having been hit or kicked over his ribs shows signs of internal hæmorrhage *see* page 165) or spits up frothy blood and gets collapsed, it is a case of fracture of a rib complicated with internal injury.

Treatment. (*a*) *In cases uncomplicated by internal injury.* (1) Apply firmly round the chest, two broad fold bandages, pugree cloth or any other suitable material. (2) See that the centre of one bandage is placed just above and the centre of another just below the seat of fracture. (3) The lower bandage should overlap the upper to half its extent. (4) Put up the



FIG. 40. Treatment of fractured ribs without internal injury.

Note it is not necessary to take off clothes.

forearm of the fractured side on a large arm sling. (Fig. 40).

If two bandages are not available, apply a towel or a *gudhree* or draw sheet folded

about eight inches wide, firmly round the chest and secure it with three or four safety pins.

(*b*) *In cases complicated by internal injury to the lungs.* (1) Loosen the clothing. (2) Lay patient on his back

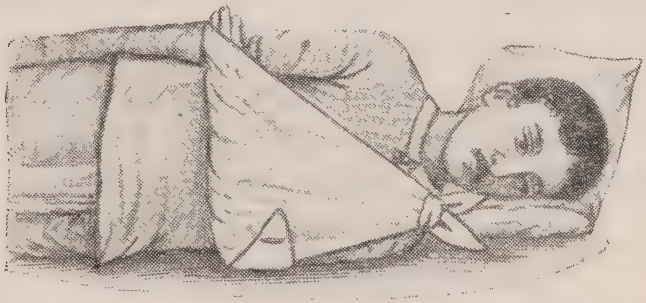


FIG. 41. Treatment of fractured ribs with internal injury.

inclined towards the injured side. (3) Give ice to suck, and apply ice bag or cold water dressings over the painful part. (4) Place the arm on the injured side in a large sling (Fig. 41). Do not apply bandages as the pressure of a bandage

would press the already depressed end of the broken rib into the lungs or any other internal organ injured by it. Keep the patient quiet. Give no stimulants, as it will aggravate bleeding and cause collapse. Seek medical aid.

Fracture of Breast Bone. A very rare injury and is caused by direct violence. *Treatment.* Loosen the clothing, put the person in an easy position. Apply cold dressings and send for the doctor.

CHAPTER XI

Dislocations, Strains and Sprains.

When as a result of an injury or violence such as a wrench, twist or jerk the ligaments of a joint and the muscles and tendons over it get overstretched, the patient is said to have *strained* himself or

to have got a strain, but when the strain is sufficiently violent to cause the ligaments to give way, the patient is said to be suffering from a *sprain*. But when as the result of violence or injury the ligaments of a joint are torn and one or more of the bones forming the joint slip out of their usual normal position and get displaced, a *dislocation* is said to occur. The joints that allow of free movement and which have got a loose capsule of ligaments, such as the shoulder joint, generally slip out of position, *i.e.*, get dislocated.

Signs and Symptoms of Dislocation.

(1) Sharp and sudden pain in and around the joint.

(2) The patient is unable to move the joint which becomes fixed (loss of function).

(3) The shape and contour of the joint is altered.

(4) There is swelling around the joint in a place where there is none on the sound side.

(5) There is a hollow in the joint where there is roundness on the sound side.

(6) Numbness and uselessness of the limb.

(7) Alteration in the length of the limb which becomes lengthened or shortened on account of the displacement upwards or downwards of the bone from its normal position. (Fig. 42).

Treatment—The First Aid man should not attempt to reduce a dislocation.

(1) Support the limb in a comfortable position by means of broad and medium fold triangular bandages and slings.

(2) Remove the patient as early as

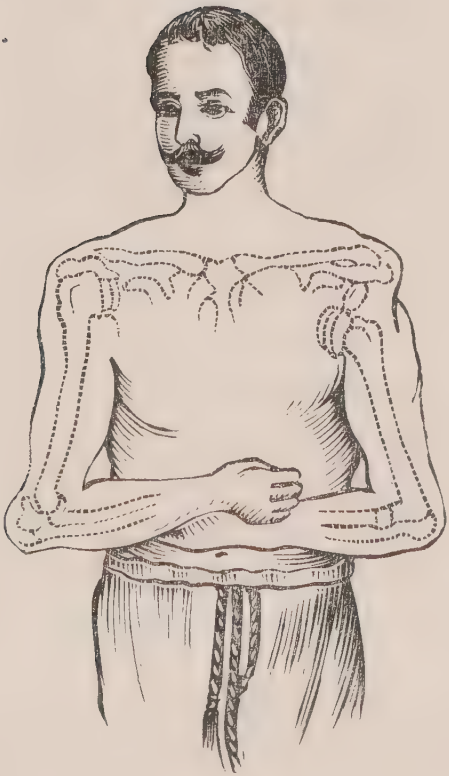


FIG. 42. Shows displacement of left shoulder joint and of right elbow joint.

possible from the scene of the accident.

(3) Put a cold water dressing or an ice bag on the dislocated joint.

(4) Keep the patient warm and take steps to reduce shock.

If the accident happens indoors, put the patient to bed ; take steps to reduce shock—let him lie down in a comfortable position—place the limb on a pillow. Apply cold water dressing or an ice bag and send for the doctor. If the patient complains of great pain and there is delay in the arrival of the doctor, apply hot fomentations to the joints to relieve pain.

Sprains.

Signs and Symptoms—(1) There is inability to move the joint, muscle, or tendon without severe pain.

(2) Swelling and discolouration of the

tissues of the skin over the joint—on account of effusion of blood from the torn tissues.

Treatment—In cases of a sprained ankle, which is of most common occurrence, do not remove the boot as it gives support to the joint.

When out of doors—(1) Apply a bandage over the boot with the sole resting on the centre of the bandage. (2) Bring the ends round to the front and cross them firmly just in front of the ankle. (3) Wind them round the ankles and tie them. (4) Lastly wet the bandage as it makes the bandage sit tightly and give more support. (5) Remove the patient to a place of shelter. The same treatment is adopted if the feet are bare.

Treatment indoors—(1) Remove the boot or shoe by slitting the back seam and promptly cutting the laces. (2) Place the patient in a comfortable position and raise

the foot. (3) Apply cold compresses or an ice bag and if these fail to give relief apply hot fomentations.

Treatment for Strains—(1) Rest to the affected part. (2) Application of cold compresses.

CHAPTER XII

The Heart.

The *Heart* is a pear shaped, hollow organ about the size of a closed fist composed of a special kind of involuntary muscles. It is not the seat of feelings and emotions as is commonly believed but is a most skilfully devised pumping machine. By its inherent power of contraction and relaxation known as the beats of the heart it acts like a force pump and causes the pure blood to flow in the arteries and circu-

late through the body. If the heart stops working death soon follows. It is situated in the space between the two lungs in the middle of the chest cavity lying obliquely with its base towards the neck and its apex inclining towards the left side. The beats of the heart can be felt under the left breast and the student should make himself familiar with it. The cavity of the heart is divided into four chambers, two occupying the right side and two the left side. The chambers in the right side are completely separated from those in the left side by means of a muscular wall between them. On each side there is an upper and a lower chamber communicating with each other by an opening guarded by thin muscular flaps (valves) which allow the blood to flow from the upper to the lower chamber but prevent its return flow in the opposite direction. The upper chambers of the heart are called *Auricles,*

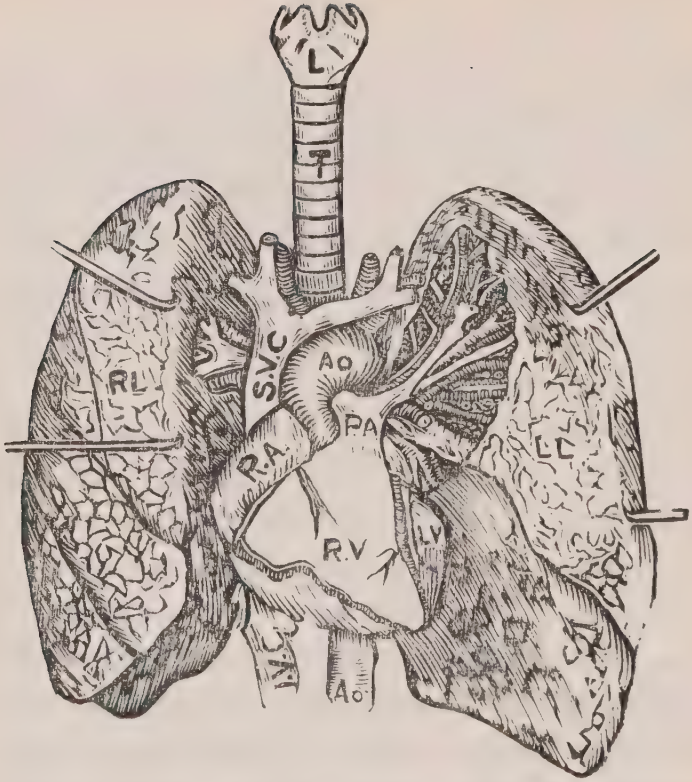


FIG. 43. Organs of Respiration and Circulation.

(L) Voice Box, (T) Wind-pipe, (R.L.) Right Lung, (L.L.) Left Lung, (R.A.) Right Auricle, (L.A.) Left Auricle, (R.V.) Right Ventricle, (L.V.) Left Ventricle, (P.A.) Blood-vessel carrying impure blood to the lungs for purification, (A.O.) Aorta.

(S.V.C., I.V.C.) The two large veins carrying impure blood from the upper and lower parts of the body respectively to the heart.

Note.—The lungs are drawn back by four hooks in order to expose the heart.

the lower ones the *Ventricles* of the heart ; thus we have a right auricle, and a right ventricle completely separated from a left auricle and a left ventricle. Both the right and left auricles have blood vessels opening into them by means of which they are filled with blood. Both the right and the left ventricles have blood vessels arising from them by means of which they empty themselves ; the openings of these blood vessels like those in the auricles are surrounded by valves which allow the blood to flow only in one direction. Thus in the case of auricles the blood flows from the blood vessels towards the inside of the auricles and in the case of ventricles the blood passes out from within the ventricles into the blood vessels.

Both the auricles and the ventricles relax and contract alternately and rythmically. During the period of relaxation of auricles, the valves around the openings of

those blood vessels which end in the auricles open and blood flows out from them into the auricles, which thus get filled with blood and then contract. During their contraction the valves in the blood vessels mentioned above close, but the auriculo-ventricular valves open and the auricles thereby empty themselves into the ventricles which simultaneously relax and get filled with blood. After the ventricles get filled with blood they contract, and during their contraction the auriculo-ventricular valves close, but the valves round the blood vessels which begin from the ventricles open and the blood from the ventricles flow into these blood vessels. This action of filling up and emptying goes on alternately with each beat of the heart and results in the maintenance of the flow of the blood in the blood vessels in the body. The left ventricle is the stronger of the two and pumps pure blood into the body. The

right ventricle drives the impure blood into the lungs, where it is purified.

Blood and Blood Vessels.

Blood is the fluid medium by means of which nourishment is carried to all parts of the body and waste products are carried away from it. Good or pure blood is of a scarlet red colour while blood which has absorbed the waste products of the body and become impure, is of a dark red or purple colour. The quantity of blood in the body is about 1/12th to 1/8th of the total weight of the body.

Blood flows through our body in tubes called *blood vessels*. There are three kinds of blood vessels, *viz.*, arteries, veins and capillaries. *Arteries* are those blood vessels through which pure blood is carried from the heart to the body. They have no valves except at their origin from the heart. Their walls are stronger and

more elastic than those of veins. (*Vide* the pulse, page 120).

Capillaries are a network of very fine sub-divisions of blood vessels as fine as the hair of one's head. They are situated between the ends of arteries and the beginning of veins.

These minute vessels reunite at their finest division into larger branches, and these into still larger ones, and thus form the veins through which the blood returns to the heart.

The blood by means of the capillaries reach every part of the tissues and carry nourishment to them. At the same time, the blood in these minute vessels absorb products of waste and repair of tissues.

The small sub-divisions of arteries serve as *inlet* pipes to one end of the network of capillaries and the small branches forming the veins at the other end act as their *outlet* pipes.

Veins are the blood vessels in which blood, which has become impure by absorbing waste products of the body, is carried back to the heart. The veins gradually increase in size as they approach the heart, they are not elastic like the arteries and do not pulsate like them. Throughout their course at certain intervals there are pouch like flaps (valves), which get filled with blood and join to close the opening inside the veins at that point and thus allows the blood to flow only towards the heart and prevents the flow of the blood towards the capillaries.

A *Varicose* vein is one in which the walls have become dilated and tortuous and thus the valves inside have become incompetent and do not function. As a result of this, blood can flow back in the reverse direction, and therefore varicose veins bleed from both ends.

It is to be remembered that a vein al-

ways follows the course of an artery and lies alongside of it. In some instances there are two veins one on each side of an artery. In accidents it is almost impossible to wound an artery without wounding the corresponding vein also. Occasionally bleeding is only arterial, as when a diseased artery (aneurysm) raptures, or only venous, as when a varicose vein bursts, or only capillary, as in the oozing from an abrasion, or from the sides of a wound.

CHAPTER XIII

The Circulation of Blood.

Blood passes through the heart and through the blood vessels in the whole body about twice in one minute.

By *circulation of blood* is meant the cycle of the flow of blood from the heart

to the body by means of the arteries and the capillaries, and back again by means of the veins to the heart. Let us follow the course of blood from the left ventricle of the heart :—

By the contraction of the left ventricle a volume of blood is pumped out into the main artery of the body called the Aorta. This pumping action of the heart sends volume after volume of blood which circulates all over the body through the branches of the Aorta and their sub-divisions which get smaller and smaller as they go farther and farther into the tissues till they finally merge into capillaries. which reach every part of the terms. After this the blood of the capillaries is collected into very small veins, which open into larger ones, till it is finally poured into two large veins which empty into the right auricle of the heart, which contracts as soon as it gets full of blood

and forces the blood into the right ventricle which similiarly contracts and through the blood vessels arising from it forces the blood into the lungs. In the tissues of the lungs the blood vessels break up into capillaries and here, in the capillaries of the lungs, the impure blood gets purified by absorbing oxygen from the air we breathe in. The purified blood from the lungs then passes out by means of blood vessels into the left auricle of the heart and from there to the left ventricle and thus the cycle is continued.

The Pulse is the term applied to the movement of expansion in the elastic walls of the arteries caused by the wave of blood passing through them with each beat or contraction of the heart. It is best felt in places where the arteries are near the surface of the body and are lying close to a bone. The pulse is usually felt at the wrist but the student should

accustom himself to feeling and counting the pulse at different convenient parts of the body such as wrist, temple, lower jaw, groin, the instep (back of the foot) etc.

The heart beats about 72 times per minute in the adult and 120 times in the child, therefore the normal pulse rate is also about 72 beats to the minute in the adult and 120 in the child.

It is a bad sign if the pulse is running rapidly in a sick person, or in one who has sustained an injury and has lost blood.

CHAPTER XIV

The Course, Distribution and Situation of the Main Arteries—Pressure Points.

The *Aorta* is the name of the largest artery in the body. It arises from the left ventricle of the heart and then arches

backwards and reaches the left side and front of the spine. The arch of the Aorta gives off different large branches—one set going upwards for supplying the head and neck, another set passing outwards to supply the upper limbs. The main trunk goes downwards and lies mainly along the left side and front of the spine supplying the trunk, the contents of the chest and abdomen and finally goes down to the lower limbs.

The arteries are named after the bones near which they lie (Sub-clavian, Radial), or the region through which they pass (Cervical, axillary) or the parts of the body which they go to supply (Cerebral, Brachial).

The branch supplying the neck and the head is called the *common carotid artery*. There are two carotid arteries, right and left. Each is divided into an *Internal carotid* artery supplying the brain and its

coverings, and the *External carotid* artery which supplies the superficial parts of the face and head. The branch of the external carotid passing along the region of the temple to the vault of the head is called the *Temporal artery*, and the branch passing over the lower jaw on to the face is called the *Facial artery* and another going inwards to supply the tongue is called the *Lingual artery*. The branch of the Aorta which supplies the upper limb is distributed in the following way. The portion of the artery passing under the collar bone or clavicle is called the *Sub-Clavian*, its continuation in the region of the arm pit is called the *Axillary*. Its further prolongation in front of the arm is called the *Brachial artery*. At the bend of the elbow the brachial artery divides into two branches, the one passing along the radius bone is called the *Radial artery*, and the other passing along the ulna is

named the *Ulnar*. These two reunite in the palm and form an arch of arteries which is named *Palmer Arch*. From the palmer arch are given off the *Digital* branches which supply the digits or the fingers.

The portion of the Aorta inside the chest is called *Thoracic Aorta* and that in the abdomen the *Abdominal Aorta*. At the level of the navel the abdominal aorta divides into two *Common Iliacs*, right and left. Each of these divides into an internal and external branch. The *Internal Iliac* on each side supplies blood to the bowels, and the bladder; and also the womb, etc., in the female. The *External Iliac* of each side comes out of the abdomen at the top of the groin and is continued down the front of the thigh bone as the *Femoral artery*. In the lower third of the thigh the femoral artery lies deep under the muscles, and through them passes to the back of the

knee where it again becomes superficial behind the knee joint and is then called the *Popliteal artery*. At the lower part of the knee the popliteal artery divides into the *Anterior* and *Posterior Tibials* which pass along the front and back of the leg. The *Anterior Tibial* is continued downwards on the instep under the name of the *Dorsalis Pedis* which at the base of the great toe sinks down to emerge on the sole of the foot and meet the continuation of the posterior tibial, to form an arch of blood vessels named the *Planter Arch*. From this arch arise the digital arteries of the foot which supply the toes.

Note.—It will be noticed that the general plan of bony framework of the upper and lower limbs and their blood supply are also similar. The student should make himself familiar with the course of the arteries giving special attention to the regions where they are near the surface, or are lying on a bone or close to it. The course of the Femoral artery is indicated by a

line drawn from the centre of the groin to the back of the inner side of the knee, the leg being kept bent at the knee and the thigh rolled out. The External Iliac crosses the brim of the pelvis and becomes the Femoral at the top of the groin lying midway between the highest point of the hip and just above the private parts. This blood vessel can be conveniently pressed against the brim of the pelvis to stop bleeding.

Pressure Point—

Is that spot in the course of an artery pressure over which stops the flow of blood in it beyond the point pressed upon. For each artery there is a special point at which pressure should be applied when the artery is wounded and bleeding. A skilful application of the knowledge of the position of pressure points is one of the most ready and effective means of stopping bleeding and should always be resorted to when direct pressure on the bleeding

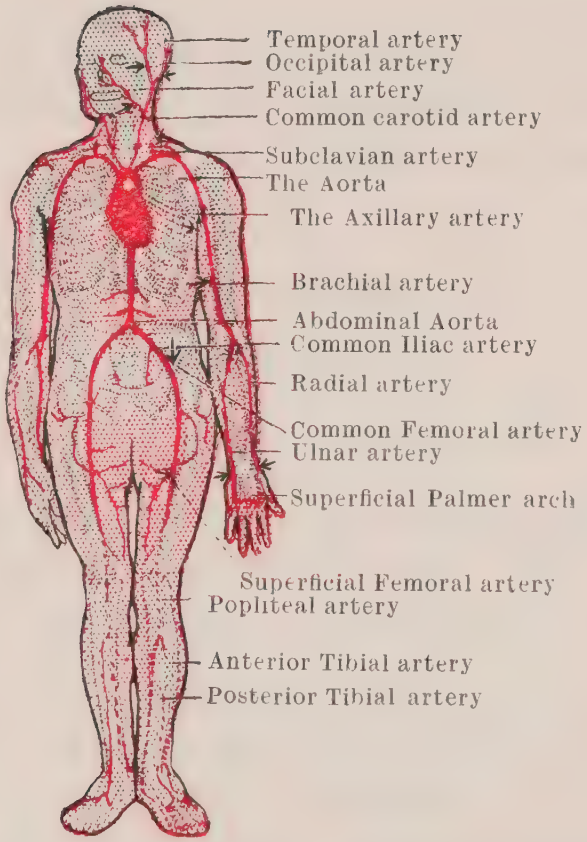


Fig. 44, showing the course of arteries. The arrows in the diagram show the "Pressure points" where pressure may be applied to stop bleeding from an artery from any part beyond that point. Compare figure in frontispiece.

wound or the artery is not advisable. See fig. in frontispiece on which 16 pressure points on different parts of the body is shown and also Fig. 44.

CHAPTER XV

Hæmorrhage, its varieties.

Hæmorrhage means bleeding or flowing out of blood from an injured blood vessel. There are two kinds of bleeding, (*i*) that which can be seen by the eye, *External hæmorrhage,* and (*ii*) that which cannot be seen by the eye but can be recognised by signs and symptoms, *Internal or concealed hæmorrhage.* Each of these may be of three varieties, (1) Arterial, (2) Veinous and (3) Capillary, each with special characteristic signs for recognition.

(1) Bleeding from a wounded artery or *arterial hæmorrhage.*

(*i*) It is bright red or scarlet red in colour. (*ii*) It flows out of the wound in jerky spurts or jets corresponding to the beats of the heart. (*iii*) It issues from that end of the wounded blood vessel which is *nearest* the heart. Thus if the carotid artery in the neck is wounded the blood flows out from the lower end of the wounded vessel, whereas if the axillary artery in the armpit is wounded the blood flows out from the upper end of the injured blood vessel, *i.e.*, in each case from the side *nearest* the heart.

(2) Bleeding from a wounded vein or *veinous hæmorrhage.*

(*i*) It is dark red or purple in colour. (*ii*) It flows out of the wounded blood vessel in a steady or continuous stream and not in jerky spurts. (*iii*) It issues from that end of the wounded blood vessel which is *farthest* from the heart. Thus when the jugular vein in the neck is

wounded the blood flows out from the upper end and when the axillary vein in the armpit is injured the blood issues from the lower end of the injured blood vessel, *i.e.*, in each case from the side which is *farthest* from the heart.

(3) *Bleeding from a Varicose Vein.*

(*i*) It bleeds very briskly. (*ii*) It flows from *both ends* of the wounded blood vessels, *viz.*, both the heart end and the end away from the heart. (*iii*) Like any other veinous blood the colour is dark red, and the flow is continuous, *i.e.*, without intermittent jerks.

(4) *Capillary hæmorrhage.*

(*i*) It may be light red or it may be pale pink in colour. (*ii*) It trickles out or slowly oozes out of a wound. (*iii*) It issues and wells up slowly from all parts and sides of the wound.

Note.—The reason why the blood from the veins does not spurt out in jets is because the

blood after passing through the capillaries no longer gets the pumping action of the heart to intermittently drive it forward; fresh volumes of blood, pressed through the capillaries, simply press the blood forward *en masse* in a continuous stream, the valves preventing their return in the reverse direction.

CHAPTER XVI

The General Means of Treatment for Hæmorrhage.

There is hardly anything of greater importance in First Aid treatment than the methods of stopping bleeding. In every wound blood vessels are injured, therefore every wound must bleed but the danger from bleeding varies with the size and kind of blood vessels involved *viz.*, arteries, varicose veins, ordinary veins or capillaries.

An accident associated with free bleeding is not only a most alarming sight to

the lay person but in reality is also most harmful to the sufferer as it rapidly produces weakness and fainting and if allowed to go on, in a short time proves fatal. We must therefore train the First Aid student to remain cool and to adopt measures to stop bleeding quickly and effectively.

The general means of arresting hæmorrhage is by (1) *position,* (2) application of *cold,* (3) application of *pressure.*

Position. Blood like all liquids finds difficulty in flowing upwards. Therefore we take advantage of *position* and *elevate the part that is bleeding.* Remember, however, that while you elevate a bleeding limb or a part of the body, a person bleeding from a wound is not to be made to stand or even to sit up. Lying down is the best position for him. In the lying down position the heart beats slowly and thus pumps out less blood through a wounded

artery; when sitting up or standing, the heart beats quicker, thus increasing the flow of blood through the blood vessels and its loss from the wound.

Cold. Nature has provided that blood, as soon as it is shed, that is, leaves the body, will settle into a jelly-like mass which is called a clot. The clots act as plugs on the bleeding ends of blood vessels and in this way stop further loss of blood. Similarly the mechanical effect of an injury makes the blood vessels to contract, with the result that less blood can flow through them and bleeding is thereby controlled. These are Nature's efforts to stop bleeding.

Ice, and cold applications, have also the same effects on blood and blood vessels and in the case of capillaries and small blood vessels cold may effectively stop bleeding, but in the case of big blood vessels, it is of no use as the force of the

pumping action of the heart is very much stronger than the power of stopping bleeding by the above means.

Pressure. Bleeding of all kinds can be stopped by *firm pressure.* The mechanical action of pressure also causes the ends of bleeding vessels to contract and promotes formation of clots. Therefore adopt one of the following six methods for stopping bleeding from arteries by means of *pressure,* and add to it the advantages derived from *position.*

Treatment of Bleeding from Arteries.

(1) *Pressure by means of fingers and thumb* is an ever ready, quickest, and most effective means of stopping bleeding. This can be applied: (*a*) *Directly* on the bleeding end of an artery or on the whole wound. (If handy take a piece of clean cloth or dressing between the fingers and the wound as it gives a better hold on the

slippery blood vessels and the surface of the wound). (*b*) By the fingers and thumb compressing the artery at its *pressure point* or at any other convenient spot towards the heart end of the wound where on account of the artery being near the surface and close to a bone, it can be effectively pressed against it (*vide* Page 126).

(2) *Pad and Bandage.* Pads may be improvised by folding a suitable piece of clean cloth, *e.g.*, from a *pugree,* handkerchief or a *chadder* or *kurta,* and kept in position by a narrow bandage. The bandage should be put on tightly over the pad of dressings and thus ensure continued pressure.

(3) *Graduated Pad and Bandage.* In hollow parts of the body such as the palm of the hand or the sole of the foot a cone shaped pad serves the purpose better than an ordinary flat pad. A number of cir-

cular pads of clean lint or cloth are made of similar shape but gradually diminishing in size, the smallest being of the size of a four anna piece. Put the smallest end of the pad on the bleeding surface. This cone of pads lies with its point over the bleeding surface and is kept in position by a narrow fold bandage.

Note.—Do not put pressure of any kind on the wound if foreign bodies such as pieces of broken glass or pieces of shell or an exploded bomb may be suspected to be still lying embedded in the wound as you are likely to press these into the wound and do more damage and cause pain. If the bone in and around the wound is shattered in fragments do not apply direct pressure of any kind for the same reasons. In such cases **stop bleeding by compressing artery** at its **pressure point** or at a convenient point on the heart side of the wound.

(4) *Forcible Bending of Joints on a Pad with Bandage.* At such situations as the elbow or the knee a pad is put in the fold of the joint and the limb kept forcibly

bent over it by means of a figure of 8 bandages round the limb. The flow of blood through the artery is thus prevented by compression and bleeding is stopped beyond the joint. (Fig. 45). Forcible bending of a joint as a means of stopping bleeding should only be used when there are no concomitant fractures of the bones forming the joint.

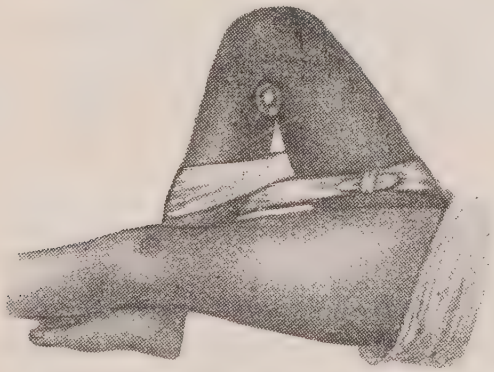


FIG. 45. Forced bending of knee with pad and bandage.

(5) *Compression of the Main Artery of Supply* (see Chapter XVII).

(6) *Tourniquet.*

Tourniquet is the name for an appliance for maintaining effective pressure over an artery and thereby stopping bleeding. It essentially consists of (*a*) a cord or a strap to go round the limb, (*b*) a small pad to be placed over the course of the artery which is intended to be compressed, and (*c*) a twister (which is a strong stick about 6″ in length) to twist round and round.

The chief kinds of tourniquets in use are (1) St. Johns, (2) the Screw, and (3) the Improvised. Use the one which is handy.

Improvised Tourniquets (Fig. 46).

Take a piece of rope, *pugree* cloth, *kamarbund*, or handkerchief to use as the cord of the tourniquet.

Place a suitably small pad on the course of an artery at its pressure point you intend to compress. In the absence of a suitable pad tie a knot in the middle



FIG. 46. Compression of main artery of thigh by an improvised tourniquet. A piece of paper is put under the knot to prevent the cord nipping the skin

of the improvised cord of the tourniquet. (The pad and the knot becomes firmer and more effective by enclosing in them a small piece of stone, wood, cork or coins). Place the knot over the artery you intend to compress. Pass the improvised cord round the limb and tie it loosely over the

limb with one knot, then place a twister improvised from a piece of stick, a pencil, a ruler or a door key of suitable size on this loose knot and tie a knot or two firmly on this (twister). Now twist the stick round and round until the artery is compressed and the bleeding stops. Keep the twister fixed in position by means of a narrow bandage.

It is essential that the pad or the knot should be placed exactly over the artery, and should press on the artery with each twist of the cord, otherwise by twisting the cord you would be compressing the whole limb and stopping its circulation and doing damage before stopping the bleeding. Therefore if bleeding continues after a few turns of the twister has caused moderate amount of pressure on the limb, you must conclude that the pad or the knot has not been placed correctly over the artery. Undo the tourniquet and put on the pad or the knot exactly on the line of the artery and compress as before by twisting and look for the desired result.

The St. John Tourniquet (Fig. 47).

This consists of:—(1) A. Pad, (2) B. Strap, (3) C. Twister, (4) D. Buckle, (5) E. String.

1. Place the rounded side of the pad (A) on the pressure point.

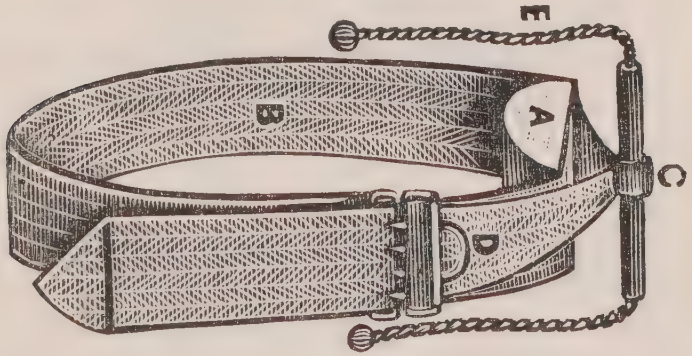


Fig. 47. St. John tourniquet.

2. Pass the strap (B) round the limb and buckle it.

3. After noting that the pad is in the correct position apply sufficient pressure by means of the twister (C) to stop

bleeding. The twister should be kept as near the centre of the pad as possible. Should moderate pressure fail to stop bleeding the pad is incorrectly placed.

4. Pass the appropriate end of the string (E) which runs freely within the cane, through the " D " attached to the buckle, and tie it. To ensure ease in relaxing pressure, the hitch or knot securing the string should be finished with a loop.

Note.—The twister may be quickly secured temporarily by passing the appropriate end of the string between the strap and the part of the buckle on which the points of the spikes rest. In this position it will be automatically gripped and held tight.

The Screw Tourniquet consists of a strap with buckle which goes round the limb, a pad to press on the artery and a screw arrangement to increase pressure of the pad on the artery. (Fig. 48.)

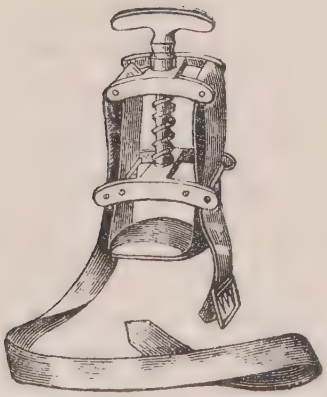


FIG. 48. Screw Tourniquet.

TREATMENT OF BLEEDING FROM VEINS AND CAPILLARIES.

Bleeding from a vein is stopped much more easily than that from arteries and therefore is not so serious. Bleeding from a varicose vein may be very profuse and is often dangerous.

Normal Veins. (1) Lay the patient down and keep the limb elevated if possible. (2) Apply pressure on the

bleeding point by fingers and thumb or by means of a clean piece of dressings or cotton wool held in position by fingers. (3) Apply a firm pad and bandage. If No. 2 is not sufficient and there is oozing and the dressings get soaked, do not remove the dressings, but (4) apply pressure by means of a pad and bandage near the wound on the side away from the heart. Cover the soaked dressings with fresh dry dressings.

Varicose Veins. (1) Same as for normal veins. (2) Should oozings occur, remove them but apply pressure by pad and bandage *both above and below* the bleeding point.

Treatment of Capillary Hæmorrhage. Least serious form of all varieties of external hæmorrhage.

(1) Raise the bleeding part. (2) Apply pressure by means of a clean piece of dry dressings, apply bandage firmly.

Practical Points to Remember in Stopping Bleeding.

1. Apply pressure directly to the bleeding point by means of your fingers or a piece of *clean* dry cloth pressed into the wound with the fingers. Keep pressure with finger and thumb whilst your assistant is preparing a pad to apply the pad and bandage method or the tourniquet.

2. Remember Note 2 on Page 5.

3. Remove all tight lacing and clothing in order to relieve congestion which promotes bleeding.

4. Elevate the wounded part in order to diminish the amount of blood flowing into it.

5. Do not attempt to wash off or otherwise remove or disturb clots of bloods as it will re-start bleeding.

6. Do not attempt to clean compound fractures or penetrating wounds with water

or lotion, wounds dressed dry do much better.

7. Avoid giving stimulants as they cause the heart to beat quickly and drive more blood through the injured part ; for the same reason do not let a bleeding man stand up or sit up, but keep him lying down. Stimulants are given after stopping bleeding if there is no head injury.

8. If you can avoid it, do not use elastic rubber cords as tourniquets, as they not only stop bleeding by compressing the main artery, but also cut off all circulation of blood in the limb and are likely to cause much harm.

In First Aid practice it is not always possible to have Esmarchs bandage handy, in spite of its value on the field of battle and in the hands of surgeons, First Aid students are not recommended to use it.

9. Never leave a tourniquet on for more than twenty minutes to half an hour.

It will cut off all circulation of blood, and cause death of the part beyond, by setting up strangulation gangrene. Gently relax the tourniquet and observe if bleeding starts again. If bleeding re-starts, tighten tourniquet again slightly shifting its position. If bleeding does not re-start then let it lie loosely in position round the limb. Do not remove it. Watch carefully for bleeding ; should it re-start, tighten the tourniquet.

CHAPTER XVII

Treatment of Bleeding by Compression of Arteries by Digital and Instrumental Pressure at their Pressure Points.

Bleeding from the face, lips, cheek, chin, side of the nose, is due to injury to blood vessels, the main artery of supply to

which is the facial artery. Press the facial artery against the lower jaw by putting your thumb over the hollow two fingers-breadth in front of the angle of the lower jaw (*see* No. 3, fig. in frontispiece ; all bleeding in branches of the artery beyond this point will be stopped by this means.

In the case of profuse bleeding from a cut in the lips or tongue, grasp the wound between the finger and the thumb with a piece of clean handkerchief or cloth to get a better hold.

Bleeding from the scalp. From regions behind the ear to the back of the head, put digital pressure on pressure point four fingers breadth behind the ear. (Fig. 49).

Bleeding in the region of the temple and the side of the head. Compress pressure point of the vessel in front of the lappet guarding the opening of the ear (Fig. 49).



FIG. 49. Digital compression of arteries of the side and back of head.

BLEEDING FROM THE NECK.

The two main arteries which supply the head and neck lie immediately one on each side of the wind pipe. There is also a large vein lying alongside and superficial to the arteries. Bleeding from one or both of these blood vessels is very dangerous and is met with in cases of cut

throat or stab and bullet wounds in the region of the neck.

A point in the neck about an inch above the top of the breast bone in the hollow just on the outer side of Adam's apple is the pressure point of the artery. (Fig. 50 and No. 4 of fig. in frontispiece). Press your thumb backwards and inwards towards the spine, this will stop bleeding ; take care not to press the wind pipe and strangle the patient. As the arteries on each side of the neck communicate freely with each other, bleeding from the upper portion of the wounded artery will start in a short time, therefore also compress the upper part of the wound. (Fig. 50). This will also stop bleeding from the big vein on the side of the neck (Jugular vein).

The chief vein of the neck is called the *Jugular vein* and because of its close neighbourhood to the main arteries of the neck (carotid arteries), it is almost

invariably wounded along with the artery and therefore in a case of cut throat the bleeding is from both the artery and the vein. It is therefore obvious that we should adopt means for stopping not only arterial, but also veinous hæmorrhage.



FIG. 50. Digital compression of main artery of the neck.

This will be attained by putting pressure also on the upper part of the cut throat wound. No tourniquet should be put around the neck nor firm pressure applied on bleeding blood vessels by means of a bandage tied around the neck. It is obvious in this way you will be strangling

the patient. Hæmorrhage should be kept under control by relays of fresh assistants who should put their fingers immediately over the fingers of the person he is relieving, who then gently withdraws his finger ; in this way the pressure on the artery is never relaxed. Do not waste time in finding out the *pressure point* if you do not know it. A pad of dressing of a suitable size, pressed on a stab wound or a cut in the neck and pressed firmly by means of one's fingers, will stop bleeding even if the actual *pressure point* is not compressed.

Bleeding from the artery in the Armpit. (1) Stop bleeding by digital pressure on the bleeding end of the artery. (2) Press upon the bleeding point on the side nearest the heart by means of a hard round pad about the size of a billiard ball. After pressing it home in the armpit put more dressings over it. (3) Keep them in position by a

narrow fold bandage. Pull the ends up firmly and then cross on the shoulder. Take one end across the chest, the other across the back and fasten them under the opposite armpit. (4) Bend the fore-arm and tie the limb tightly to the trunk



FIG. 51. Digital compression of Sub-clavian artery for stopping bleeding from main artery of armpit.

with a broad bandage applied on a level with the elbow. (Fig. 52).

The above method takes time. In order to quickly stop bleeding the artery at its origin can be compressed quickly at

its pressure point, and kept pressed either by means of relays of fingers or a well padded handle of a large size door key. Apply pressure in the hollow immediately above and behind the centre of the collar bone. Place the patient's arm against the body, depress the shoulder, make him incline his head towards the injured side ; these will relax the muscles around the artery. Press the thumb deeply downwards against the first rib, which lies beneath the collar bone at this spot. The artery passes below the collar bone and over the first rib and by the above means it is compressed against the first rib. (Fig. 51).

Bleeding from the main artery in the upper arm. The guide to the course of this blood vessel roughly is the inner seam of the sleeve of the coat or the jacket. To apply digital pressure, pass the fingers of your hand under the back of the

sufferer's arm over the seam of the coat or on the bare body over the groove which can be seen on the bare arm on the inside



FIG. 52. Compression of main artery in the armpit by pad and bandage.

of the biceps muscle. Press the pulp or the fleshy part of the fingers over the artery above the wound and press outwards and backwards against the arm bone. (Fig. 53).

A tourniquet might be applied on the arm over the pressure point to effectively stop bleeding on any part below it. (Fig. 54).

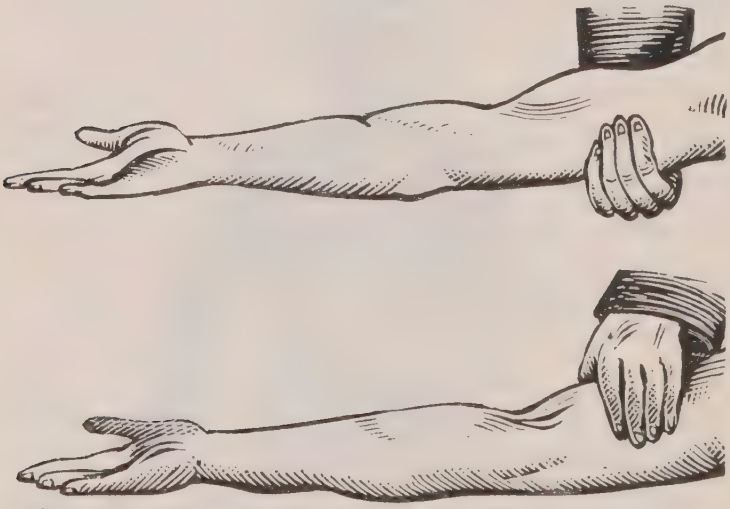


FIG. 53, showing compression of the main artery of the arm by digital pressure.

Bleeding from one of the main arteries in the forearm. Tourniquets are not applied anywhere along the forearm to stop bleeding, but they can be applied to

the upper arm to arrest bleeding in the forearm (Fig. 54). Pressure may be applied to the trunk of the artery at the fold of the elbow by forcible bending of the elbow joint on a pad kept in position by means



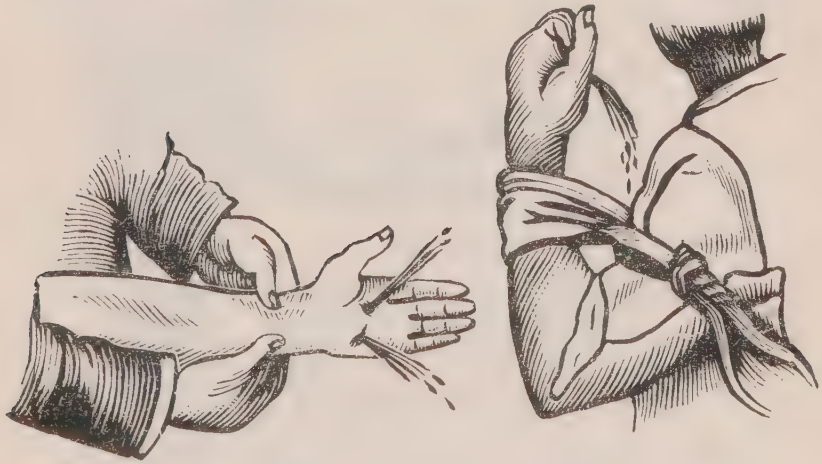
FIG. 54. Compression of main artery of arm by tourniquet.

of a narrow bandage passed round the arm and the forearm as a figure of 8 bandage. If no pad is available, the sleeve of the *kurta* or coat rolled and firmly folded up

will serve the purpose. (Fig. 55). A pad and bandage applied firmly directly over the wound will in many cases stop bleeding.

Bleeding from the palm. Forcible bending of the elbow joint or the application of tourniquet to the arm will stop bleeding from the palm. The principle underlying the use of tourniquet, however, is to apply it as near the bleeding point as possible, so as not to cut off blood supply from an unnecessarily large area. Therefore adopt other measures before putting on a tourniquet on the arm. Apply pressure by means of fingers on the two main arteries of the forearm about half an inch above the wrist and about half an inch from the outer and inner sides of the forearm where they can be easily felt pulsating (Fig. 55). Bleeding from the Palmer Arch of blood vessels may also be stopped by putting a suitable pad or piece

of split cork on each of the above mentioned arteries, and kept in position by means of a firmly applied narrow bandage. The usual and handy method of stopping



Compression of radial and ulnar arteries at the wrist.

Forced bending of elbow by pad and bandage.

FIG. 55. Treatment of bleeding from palm.

bleeding from a wound in the hollow of the palm is by means of a graduated compress or by dressings rolled up into the size

of a small ball placed in the palm, and the hand firmly closed on it and kept in this position by bandaging the closed hand, with a triangular bandage. (Fig. 56).



FIG. 56. Treatment of bleeding from the palm by graduated compress.

BLEEDING FROM THE THIGH.

The femoral or main artery of thigh comes out from the abdomen at the top of the groin and as it is very large, in a short time loss of blood would be considerable ; bleeding should therefore be arrested at once. Press the artery at the upper part of the wound. Keep up digital pressure at pressure point

by means of relays (Fig. 57) of First Aid men. Bleeding from the veins which lie



FIG. 57.
Digital compression of (Femoral) main artery of thigh.

alongside of the artery should also be stopped by pressure applied to the lower end of the wound. A suitably applied

pad and bandage will do the needful for this purpose, but digital pressure for the artery at its pressure point, must be continued by relays of trained First Aiders till the doctor's arrival. For bleeding from any other part of the femoral artery lower than the groin the application of a tourniquet is an effective means of stopping bleeding. (*See* Fig 46).

Bleeding from an artery in the leg. Digital pressure locally or the application of a tourniquet on the thigh will stop bleeding. Also forced bending of the knee joint on a pad will compress the artery at the joint and stop bleeding in any part of the leg and foot. (*Vide* Fig. 45).

Bleeding from the artery in the instep. Apply a firm pad of suitable size on the pressure point which is at the centre of the front of the ankle joint. For bleeding from any part of the sole and foot put one pad as for bleeding from the artery in the

instep and another pad immediately behind the inner protuberance of the ankle bone and keep the pads in position by a narrow bandage.

CHAPTER XVIII

Internal Hæmorrhage—Bleeding from Special Regions.

This is bleeding within a body cavity such as the chest, the abdomen, or the head. It cannot be seen but is recognised by certain definite signs and symptoms.

General Signs and Symptoms. 1. Rapid loss of strength. 2. Giddiness and feeling of fainting. 3. Shallow and hurried breathing. 4. Patient gasps for breath and attempts to loosen all tight

clothing around the neck and chest. 5. The patient yawns and sighs. 6. Face becomes pale and the lips blue. 7. The pulse becomes quick and thready. 8. A cold clammy sweat appears on the brow and the extremities get cold. 9. Patient becomes unconscious.

Treatment. (1) Send for the doctor. (2) Make the patient lie down with his head low and the feet slightly raised. If he is lying on a bed or *charpoy,* raise the foot end of the bed. (3) Undo all tight clothing about the neck and chest. (4) Allow free access of air to the patient. (5) Dash cold water on the face. (6) Give ice water to sip or small pieces of ice to suck. (7) Hold the limbs up and gently bandage them—beginning from the toes and from the fingers.

N.B.—It is most necessary to keep the patient very quiet, don't allow him to talk at all and don't give him stimulants. For treat-

ment of hæmorrhage in head injuries, *see* pages 73-74.

Bleeding from the Lungs. It might be entirely internal bleeding, and only recognised by signs and symptoms, or it might also be seen as when it is coughed up through the mouth. It is frothy due to the blood being mixed with air bubbles from the lungs, the colour is bright red. There is often pain in the chest and the patient looks very ill, there is a history of old disease of lungs like tuberculosis or dry pleurisy. It might also be due to injury as after fracture of a rib.

Treatment. Place the patient in an easy position and keep him perfectly quiet; do not allow any movement or talking, apply ice if it can be obtained or a piece of cloth or sponge wrung out of cold water over the region of the lungs. Give ice to suck or cold water to sip. Don't give any warm food or warm drink and don't give any

stimulants. Treat symptoms of collapse by methods of treatment for internal hæmorrhage.

Bleeding from the Stomach. This occurs in patients who had stomach trouble for some time and like bleeding from lungs is also a form of internal hæmorrhage, which may become apparent on being vomited up from the mouth. It is recognised by the blood being mixed with particles of food from the stomach and being dark red in colour, as the acid in the digestive juice in the stomach darkens it.

Treatment. The same as for all cases of internal hæmorrhage and for bleeding from the lungs. In the case of bleeding from the stomach, the ice bag or cold water compress is put over the pit of the stomach.

Bleeding from the Mouth. The blood may come out as a result of disease or local injury to the palate, the gums, the throat, the tongue or the socket of a tooth after

its extraction, or it may be coughed out from the lung or vomited up from the stomach.

If the blood is coming from a part of the tongue which can easily be got at, stop bleeding by holding the part firmly by means of a piece of clean lint or cloth held between the finger and thumb. If it is from the socket of a tooth, plug it with a piece of clean dressings—place on top of the plug a pad or a cork and ask the patient to bite on it. In cases of bleeding from the throat and tongue which cannot be dealt with by applying digital pressure locally, compress the main artery in the neck as for a cut throat case. For small bleeding surfaces from the palate, gums and other regions of the mouth, a solution of alum (two teaspoonfuls to a pint of cold water) held in the mouth causes the bleeding to stop.

Note.—Heat is more effective than cold for

stopping bleeding, provided that, by gradually increasing the temperature of the water, the mouth can bear sufficiently hot water to stop bleeding. *Warm* water promotes bleeding but if the water is *hot* it stops it. The liquid should be quietly held in the mouth as gargling or rinsing the mouth will disturb any clot that might be forming and cause more bleeding.

Bleeding from the Nose. It is not a condition to be always alarmed about because a moderate amount of bleeding from the nose is nature's cure for preventing congestion in the brain and can be left uninterfered with.

Treatment. 1. If indoors, sit the patient down in a comfortable position close to an open door or window. If outside, allow free access of air to the patient.

2. Throw back the patient's head to make the nose the highest point.

3. Raise the hands above the head and ask an assistant to hold them.

4. Undo all tight clothing around the neck, chest and stomach.

5. Apply ice bag or iced application such as a wet sponge or piece of rag to the nape of the neck and over the bridge of the nose.

6. Place the feet in hot water.

7. Instruct patient to *breathe through the mouth* and not the nose and so encourage him to keep his mouth open.

8. Do not allow the patient to blow his nose but wipe away the blood which flows out from the nose with a damp cloth.

Note.—Often the sight of blood causes a patient to faint. Do not let the patient see any blood stained cloth. In serious cases the nostrils should be gently syringed with a solution of alum in cold water (2 tea-spoonfuls to a pint of water). If this fails, it may be necessary to plug the nostrils with cotton wool dipped in the same solution.

CHAPTER XIX

Abrasions, Bruises, Cuts and Wounds.

Abrasion. It is an injury of the skin in which the thin layer at the top known as the *cuticle* is removed. It is not the same thing as a bruise although both often go together.

Bruises and *contusions* are injuries caused by falls or blows in which the skin remains unbroken but the flesh and the small blood vessels underneath it get crushed and lacerated.

The effusion of blood under the skin results in an immediate painful swelling and discoloration of the injured part. A *black eye* is a good example of what is meant.

The skin is at first reddened, then it assumes a dusky colour, then turns into

dark purple in a couple of days, and afterwards black. By the third day it turns violet, then greenish by the fifth or sixth day and finally it becomes yellow before it fades to the normal colour of the skin in about ten days to a fortnight's time. All these discolorations of the skin are due to the chemical changes in the colouring matter of the blood, which has been shed underneath the skin.

Treatment.—1. Rest to the injured part. 2. Cold application such as cold water compress or ice. 3. Application of evaporating lotion such as cloth wrung out of equal parts of cold water and tincture arnica, eau-de-cologne, liquid extract of witch hazel. The value of the time-honoured custom of school boys' applying raw meat on a black eye is due to its being a cold application.

Note.—Contusions and bruises over the region of the liver, spleen, lungs, bowels, head and

the back-bone should receive special attention, as important and vital organs underneath the bruised part may be injured, although the outward sign of injury might be trifling (*See* pages 35-36).

Wounds.

A *wound* is an injury in which the skin is cut or torn.

There are five distinct classes or varieties of wounds. (1) Incised; (2) Lacerated; (3) Contused; (4) Punctured; (5) Gunshot.

The danger of a wound depends on its number, size, depth and on the importance of the deeper parts involved.

General rules for treatment of wounds.

(1) Stop any brisk or severe bleeding. (2) Treat shock if present. (3) Guard against the wound becoming poisoned

Keep dry and apply Tincture Iodine. (4) Don't touch with dirty hands. (5) Afford rest to the wounded parts, supporting them with slings, splints and bandages.

Different kinds of wounds and their treatment.

(1) *Incised wounds* are caused by sharp instruments and have clean cut edges. The amount of bleeding depends on the part of the body injured and the depth and extent of the wound.

Treatment. (1) Treat as under general rules. (2) Dress the wound trying to bring the edges of the wound as close together as possible.

(2) *Lacerated wounds* are caused by machinery or the fangs and claws of animals. The wound has jagged, torn and uneven edges. More shock is present and

it heals up less quickly than an incised wound. It is more liable to get poisoned as the vitality of the parts is lowered by the tearing and mauling of the tissues. Treat as under general rules.

(3) *Contused wounds* are caused by a crushing blow received from a blunt but heavy weapon, such as a club, or a lathi. These also get easily poisoned and heal slowly. Follow general rules for treatment.

(4) *Punctured wounds* are caused by a stab from any sharp instrument such as a spear, a dagger or a bayonet.

Treatment. Remove weapon if present. Treat as under general rules.

(5) *Gunshot wounds* are caused by a shot or a bullet from a gun or any firearms.

Note.—In punctured and gunshot wounds, even if bleeding is apparently not severe, the shock to the system is great. These wounds

are usually far more dangerous than one would expect from their size, because deep and important organs and structure may be injured.

(*a*) *Wounds from buck shots*:

A person is often peppered with several small shots from a sporting gun.

Treatment. Don't attempt to remove any pellet. If they come away easily with the dressings, well and good, but never endeavour to extract shots embedded in the flesh; they cause no immediate danger. Treat as usual. Stop bleeding, dry the wound. Apply Tincture of Iodine and clean dry dressings. Treat shock which is always markedly present.

(*b*) *Wounds from bullets*:

As a rule there are two wounds, a small one caused by the entrance of the bullet, and a larger one caused by its exit. Wounds from hard nosed modern bullets

from a rifle however are practically of the same size at both the ends.

Treatment. Overcome shock which is markedly present and treat as in wounds from buck shots.

Rules for treatment of wounds in or near a joint.

If there is a gunshot wound or a punctured wound inside or near a joint—

(1) Stop bleeding.

(2) Dry the wound and apply Tincture of Iodine.

(3) Cover the part up with any dry clean dressings or cotton wool.

(4) Keep the injured limb at rest. (*a*) In the case of the upper limb, keep elbow in the bent position, supported by a large arm sling. (*b*) In the case of the lower limb, keep it in a straight position, supported by a splint.

If the joint gets an open wound or a stab wound, put on clean dry dressings and a firm bandage to stop bleeding and treat as under (3) and (4) above.

Poisoned Wounds.

Very often a member of a shikar party, out shooting in India, is mauled by a wild animal or accidentally gets peppered by buck shots or sustains an arrow or a bullet wound. Such wounds become easily poisoned as the fangs and teeth of wild animals contain poison or a living virus as in case of mad dogs and jackals, and also our hands, clothes, the dust, water, air have germs of disease in them to poison any open wound.

Treatment. (1) Treat shock. (2) Wash the wound with a strong solution of Permanganate of Potash—adding enough crystals to a pint of boiled water to make

the fluid bright red in colour. (3) Dry the wound and paint it with Tincture of Iodine or pure Carbolic Acid. (4) Wrap the wound with clean dry dressings.

If medical aid is not available and the wound becomes painful and swollen after the first day, wash it three times a day with a warm solution of Permanganate of Potash, or a lotion of Tincture Iodine or Carbolic Acid (two tea-spoonfuls to a pint of boiled water) and cover up with dressings wrung out of one of the antiseptic lotions mentioned above and cover it with a piece of water-proof material and bandage it up. Take steps to keep the part at rest by splints, bandage and slings.

Abdominal wounds, generally cuts and stabs. If the wound goes through the wall, some of the bowels come out of the wound. If the wound is transverse, raise the patient's head and shoulders, roll up a small *durree* or an overcoat and place

it as a support under the patient's bent knee. By this means the muscles of the belly are relaxed and the edges of the wound are brought together. If the wound is vertical, lay the patient down flat keeping his legs straight.

Dip a piece of clean cloth or lint in salt solution (2 tea-spoonfuls to a pint of boiled water), take out and squeeze it and apply it while warm over the wound. Keep it damp with the salt solution.

Practical points.—Avoid the use of sponges when cleaning wounds. They are difficult to clean and harbour germs. In the treatment of wounds always use boiled water which has been allowed to cool down to the required temperature. Water which has been kept on the boil for five to ten minutes is rendered free from all germs (is sterilised). Clean your hands with soap and water and use a nail brush which has been sterilised by boiling. Mechanical cleaning by scrubbing your hands and finger nails with soap and water is better than dipping them for a time in antiseptic lotions. Clean

your utensils by scrubbing them with soap and water and then wash by pouring boiling water on them. You may pour spirits of wine into them and set them on fire. This quickly kills all germs, then wash with boiled water. Do not forget the value of the application of Tr. Iodine and Pot. Permanganas in preventing the poisoning of wounds, and in overcoming the bad action of germs in poisoned wounds. Excepting as a treatment of poisoned wounds, do not wash wounds with the object of cleaning them as experience has shown that wounds cleaned even with lotions and sterilised water at the place of accident at the time of rendering First Aid have not done so well as those wounds which have been treated dry. Remember that river water and tank water are not safe, not even all well water. The only safe water is that which has been sterilised by boiling. Filtered water from a tap is fairly safe and can be used without boiling in an emergency for washing wounds and cleaning hands, so is water from a deep drinking water well. Remember always to have a sterilised nail brush to scrub your hands with soap and water. Boil your nail brush, as it may carry disease

germs from one's dirty hands to those of another. It is much better to wash your hands only with soap and water rather than with a nail brush which has not been boiled and is not clean, and thus is a source of infection.

(1) Avoid touching any wound with hands that have not been cleaned. Only when bleeding is so profuse and quick as to be dangerous to life, one may handle the wound at once in order to stop loss of blood. (2) Apply Tr. Iodine and cover up the wound with dry clean dressings. Use a clean piece of lint, cloth, or cotton wool and bandage it up. (3) If pieces of clean dressings or cloth are not handy, cover the wound with a piece of clean unprinted paper, such as the inside of an envelope or the unused inner fold of handkerchief. (4) Remember that a small poisoned wound is more serious than a very large but clean one. (5) Tincture of Iodine is a most valuable remedy. Apply it immediately to an open wound or an abrasion but remember that it smarts and if applied without warning the patient will suddenly withdraw the part touched with it, which may be very harmful, for instance, in cases of fractures with a wound;

therefore prepare the patient to expect this "pain." (6) Do not probe and unnecessarily interfere with punctured and bullet wounds. (7) Don't disturb clots or wash and clean them out of a wound.

CHAPTER XX

Burns and Scalds, Frost Bite, Snake Bite, Bites of Mad Dogs and other Animals, Stings.

Burns and Scalds.

A *Burn* is the destruction of tissues by *dry heat*, produced for instance by the flames of a fire, a live electric wire, thunder-bolt or lightning, and red hot metals, also by strong chemicals such as corrosive acids and alkalies.

A *Scald* is the destruction of tissues by *moist heat*, produced for instance by steam, boiling oil, boiling milk or water

or hot coal tar. The treatment is the same as for burns.

There are three degrees of burns. (1) When there is only reddening of the skin. (2) When the skin blisters. (3) When the skin and the tissues lying deeper than the skin are also injured—(*a*) muscles and even (*b*) the bones may get charred.

N.B.—A burn is dangerous according to the extent of the surface and the region involved and not according to the extent of the destruction of the deeper tissues. For instance, a small burn in the forearm charring the tissues down to the bone is not as dangerous as a burn causing only blisters but involving the whole arm and chest, because in the latter case the shock is far greater on account of the extensive surface involved in the burn. Burns over the region of the chest, abdomen and neck are always dangerous. *Burns and scalds in children are more dangerous than the same in adults.*

Treatment. (1) Treat shock (see page 241, para. 8 and 9). (2) Carefully remove clothing. (3) Do not break or prick blisters. (4) Lightly bandage after gently

covering up the burnt surfaces with clean dry dressings, which if not handy, then dry pieces of a *saree, dhoti* or a *chaddar* will do, provided they are clean from the wash. The object is to keep the burnt surfaces protected from exposure to air and to avoid risk of contamination by germs of disease. (5) Remember that application of carron oil or its substitute, such as a mixture of lime water with cocoanut oil or any other sweet bland oil ; boro-vaseline, cold creams and oily substances generally, are both soothing and protective, but they must not be used in the treatment of burns unless one is absolutely certain that they have been made free from germs of disease by having been thoroughly boiled, sterilised. After application of ointments and salves cover and lightly bandage as mentioned in No. 4 or else flies will sit and dust will settle on the oily surface and poison the

burnt surface with germs of disease. (6) Secure medical help as soon as possible.

When a child is extensively burned or scalded.

Immediately put the child in a warm bath; do not waste time by attempting to take off its clothing. The warm bath will overcome shock; soak off the adherent bits of clothing without causing any pain and keep the burnt surfaces protected from air and contamination by outside germs. If handy, add some baking soda to the bath (See page 186). The temperature of the bath must be kept warm, just above the normal body temperature. This is done by taking out a few mugfulls of bath water and in its place mixing some warm water to the bath. Carefully watch the breathing of the child and see that its nostrils and mouth are well out of the surface of the bath water. Keep the child

in the warm bath till the doctor arrives and takes over the treatment.

Note :—Hot water is not always handy in Indian homes. Until a hot bath is prepared keep the child warm and covered and adopt measures indicated on page 241 para 8—9.

Points to Remember.

Note.—(1) To (*a*) *overcome shock* (*b*) *exclude air* and (*c*) *prevent contamination of the burnt surfaces* are the main aims in the treatment of burns. Observe the same scrupulous cleanliness in handling burns as you would in the case of wounds—see page 179 and following.

Two tea-spoonfuls of baking soda (Soda Bicarb) added to every pint of warm water makes a very soothing lotion and is used to remove any adherent clothing as well as to cleanse the burnt surface.

(2) If the body of a person is burnt extensively and a bath tub is handy, fill it with warm water and put the patient in it. If you can add soda bicarb to make the bath alkaline so much the better. Immersion of the burnt part in warm water and in alkaline lotions is

soothing and admirably reduces shock and cleanses the burnt surface, and soaks off adherent clothing without pain.

(3) When the burnt surface is extensive, use several small pieces of dressings instead of one large piece to cover it, so that in subsequent dressings the wound can be dealt with piecemeal. This causes less pain and less shock.

(4) If the clothing is stuck to the body, do not drag it off the burnt part roughly but cut the cloth all round, and remove gently after soaking with warm alkaline lotion or soak it off by putting the limb or the whole body in such a bath if one can be arranged. A mixture of equal parts of peroxide of hydrogen and warm water poured on the dressings causes bubbles and quickly loosens adherent dressings and clothing. It is however expensive and not always handy for emergency treatment. If nothing else is handy loosen the adherent clothes by pouring any kind of warm sterilised sweet oil such as Cocoanut, Olive, Linseed Oil or Ghee.

If the face is burnt: Cut a mask out of a piece of lint or a clean piece of cloth leaving holes for the eyes, nose and mouth. Apply

sterilised boro-vaseline or sterilised sweet oils and put this mask over the face, and keep it in position.

Burns are unfortunately common in India, as Fire Works and illuminations by naked lights, candles and lamps are largely used in social and religious festivities and also as a rule open *chulas* are utilised for purposes of cooking food. Under the circumstances if by chance one is careless the loose oriental garments made of light materials easily catch fire. Thus the loose ends of *sarees, chaddars* and *dopattas*, the *kocha* and *kachotas* of *dhoties* easily catch fire. If one whose clothes catch fire has the presence of mind to throw oneself down and roll on the ground one can smother the flames in this way, but more often than not the sufferer does the very worse thing possible and runs out screaming for help. The quick movement causes a draught and fans the fire to leap into flames and to spread all over the body of the unfortunate person, who in this way soon becomes a moving pillar of flames. (*See* Fig. 58).

When a woman's dress catches fire the bystanders should immediately throw her down on the floor, flames uppermost. If



FIG. 58.

it is the front of the dress that is burning lay her down on her back. In this position the flames ascend away from the limbs, which thus escape injury. Throw a mat, a blanket, a durree or a coat over the burning part and smother the flames with it. If nothing is handy take your own coat off and hold it between the sufferer and yourself and quickly wrap it round the burning part and in this way smother the flames. Don't waste your time in looking for water as you can never get it

in sufficient quantity to absolutely drench the sufferer and put the flames out. You will only succeed in partially subduing the flames and they will smoulder and burn and scald the skin. Therefore when a person's clothes catch fire, try only to smother the flames in the dry manner described above. (*See* Fig. 58).

Scalded Throat.

Scalds if extensive are often fatal in children. We often hear of mothers and maid servants leaving utensils with boiling milk or hot water within reach of little children and they seizing it pour the contents over their little faces, necks, and other parts of body causing serious scalds which; if the child survives, on healing leave ugly scars and contractions which may or may not be cured even by surgical operations.

It often happens that a child tries to drink water out of the spout of a tea-pot full of boiling or hot water or hot milk and scalds its throat so severely that it may result in suffocation due to spasm of the throat or to the swelling of the throat preventing the passage of air into the

windpipe following the injury of the delicate tissues.

Treatment. (1) Reduce shock. (2) Apply a piece of woolen cloth or flannel or a sponge wrung out of hot water to the front of the throat. (3) Let it have a small drink of *warm* olive oil, and then later on, (4) sip cold water.

Frost Bite.

It is seen in cold countries and hills with frost and snow, never in the plains of India.

The fingers, toes, ears, nose and chin are the usual places to get affected, they become swollen, red and often get covered with blisters.

Treatment. By no means allow the patient to warm his hands or feet or face near a fire. If he is unconscious, do not bring him into a warm room. He will certainly die of shock caused by the

sudden change on account of the warm temperature of the room. Ice and snow if available should be rubbed over the affected parts, this will restore circulation of the blood. Cover lightly with cold coverings and sheets, then by degrees cover with warm clothes and increase the warmth of the room very gradually. When conscious, give the patient hot tea or coffee to drink.

Snake Bite.

Cobras, Kraitas, Vipers and other venomous snakes by means of their fine needle-like sharp fangs inject a deadly poison into the body of the unfortunate victim. These fangs have a central channel, which runs into the poison pouch situated at their bottom. The poison pouch collects the poison, secreted by poison glands situated at the back of their gums. In the act of biting the pouch gets

squeezed and the poison in it is squirted out through the hollow fangs into the wound. Two small punctures or bleeding points about one inch apart is all that is noticed. The venom, however, is so deadly and powerful and is so rapidly absorbed by being mixed in the blood that unless prompt and energetic action is taken at once, certainly not later than within five to ten minutes, the unfortunate victim dies very soon of heart failure or stoppage of breathing. A large number of lives are lost in India specially in mufassil towns and rural areas from snake bite. *Carry on First Aid treatment immediately and send for the doctor or quickly take the patient to a hospital or a doctor.*

First Aid Treatment.

1. Tightly tie one ligature round the limb between the wound and heart and

another a little higher than the first. For instance in the case of a bite on the foot or leg tie the second ligature above the knee and in that of the hand or the fore-arm, above the elbow.

2. Enlarge the punctured points caused by the fangs of the snake by making a cross shaped cut with a clean sharp knife. The wound thus enlarged will bleed and with the blood some of the poison will leave the body.

Note.—Do not waste time in sterilising by boiling or otherwise making your knife surgically clean. Every minute is so important that any knife or sharp tool or instrument will do. Inform the doctor of that you have done, and he will know what to do later on.

3. Rub crystals of permanganate of potash on the wound as it destroys the snake poison.

4. Keep the wounded part hanging down and wash it with warm water, these

methods encourage bleeding. If a warm lotion of pot. permanganate is used, in addition to the advantage of promoting bleeding it will also neutralise or kill the poison.

5. Give stimulants to overcome possible failure of the heart.

6. Perform artificial respiration if there is any sign of stopping breathing.

Note.—Tip to remember in the treatment of snake bite is—

"COBRAS."

C—Constriction between wound and heart by applying a ligature.

O—Obstruct further spread of poison by applying a second ligature higher than the first.

B—Burn (cauterise) by pot. permanganas.

R—Remove poison by making the wound bleed and washing it with warm lotion or warm water.

A—Artificial respiration if breathing fails.

S—Stimulants to prevent heart failure.

Bites of Animals.

Bites of animals cause lacerated wounds which do not heal easily, see pages 174 and 177. The general principles for the treatment of poisoned wounds should be adopted. Bites in the region of the head, neck, and face are particularly dangerous on account of the shortness of the distance for the poison or the virus to travel to the brain.

Bites from mad dogs and *jackals* are common in India. They are particularly dangerous on account of a living virus in their saliva the special poison of which causes Rabies and Hydrophobia which is a fatal disease. In India all dog bites should be regarded with suspicion. Find out if the dog has been showing *signs of Rabies, viz.*, has it bitten other people without provocation, has it been aimlessly running about with its head down and with frothy saliva dribbling from its

mouth? whether any difficulty in walking, such as a staggering gait, has been observed? whether the animal has shown signs of difficulty in swallowing food? or has it been off its food? whether it barks in an unnatural manner? and whether it is frightened of its own shadow and of water (Hydrophobia)?

Treatment. 1. Apply two constricting ligatures round the (bitten) limb as in the case of a snake bite. 2. Encourage free bleeding from the wound and wash it with warm water. 3. Dry the wound and burn it with pure carbolic acid, or crystals of pot. permanganate. If nothing else is handy burn it with an iron at black-heat. Remember lunar caustic is of no use and pure carbolic acid is the best as it penetrates into the tissues quickly and destroys the living virus. Be sure that all the edges and bottom of the wound is well cauterised with the disinfectant.

Note.—Immediately send all cases to the nearest Pasteur Institute. With every patient send a short history of the case and also the living dog, which should be muzzled and securely locked in a suitable box. If the dog is dead then send its brain and spinal cord preserved in spirits or packed in ice.

N.B.—If the dog is suspected of having rabies do not kill it but keep it under observation for ten days. If it dies within this period even without showing any of the signs of rabies enumerated above, and without any apparent natural causes of death, be sure to send the person bitten by this dog to the nearest Pasteur Institute for anti-rabic treatment, as almost certainly the dog was suffering from some variety of rabies.

Stings.

Stings of *scorpions*, bees and wasps cause burning sensation and swelling and if neglected sometimes festering sores. They inject, through their hollow sting, an irritating poison secreted in a gland near the tail. In some scorpions the poison is more virulent than others. The pain and

swelling from the sting of a scorpion is often very great and is accompanied by symptoms of collapse, vomiting and unconsciousness. In children they easily prove fatal, even grown up people stung by scorpions have been known to die. The *centipedes* do not sting but inflict a bite by means of the poison claws which are perforated like the fangs of a snake and the stings of a scorpion, and give exit to the venom which is secreted in their bases.

Treatment. The tiny and brittle stings of bees and wasps often break and can be seen embedded in the part stung. Remove the sting by squeezing the part between the nails of your finger and thumb, or by pressing the hollow part of the stem of a key over the sting. Application of scrubs ammonia, salvolatile, raw onion, juice of tobacco leaf, eau-de-cologne and other spirits are the usual remedies to relieve pain.

In virulent cases of scorpion bite the pain is very great and swelling of the parts comes on rapidly. Apply a ligature as for snake and dog bite and cauterise with pot. permanganate or pure carbolic acid which is an antiseptic and also deadens pain.

If the pain is severe apply hot fomentations of hot water with poppy head boiled in it or a little Tr. of Opium mixed with it. Continue every two hours till pain is relieved.

After the pain is relieved clean the part with soap and warm water, dry it and apply Tr. Iodine and then let it remain covered up with dry dressings. Always obtain medical aid as soon as you can.

There are some plants like stinging nettles and fish, like jelly fish, contact with which causes a stinging sensation. Treat by washing with ammonia or lime water or any other alkaline solution.

CHAPTER XXI

Foreign Bodies in the Eye, Nose, Ear, Throat and Stomach, Fish-hook in Skin.

Foreign bodies cause a varying amount of damage or discomfort according to the part concerned. For instance in the throat a foreign body may cause choking and thus endanger life. In the eye they cause a good deal of discomfort and may result in permanent damage to the delicate tissue of the eye-ball, whereas in the nose or the ear the discomfort is not much to speak of and its presence is not fraught with any immediate danger.

Eyes.

When beneath lower lid. (1) Gently pull the lid downwards. (2) With the tip of a clean rag or handkerchief remove the foreign body,

the chest cavity is enlarged, partly by the elevation of the ribs and breast bone by the action of the muscles attached to them, and mainly by the depression of the dome of the midriff which gets flattened during its contraction. The capacity of the chest is thus increased which creates a tendency to produce a vacuum, and causes the air to enter the elastic air cells of the lungs which expand and occupy the increased chest space.

In *expiration,* the capacity of the chest cavity is reduced partly by depression of the ribs and the breast bone and mainly by the relaxation of the midriff which makes its dome to come up. The lungs in this way get squeezed into smaller size and air is forced out.

There is a short pause in breathing between the end of expiration and beginning of inspiration.

CHAPTER XXIII

Asphyxia: *Causes of Suspension of Breathing*: *Artificial Respiration.*

Asphyxia is the name for that dangerous condition in which breathing is stopped, and the person becomes unconscious. Stoppage of breathing will deprive blood of its life-giving element, *viz.*, oxygen, and cause death.

Causes of suspension of breathing or Asphyxia.

1. Drowning. 2. Hanging. 3. Strangulation. 4. Suffocation by smoke, fumes and gases. 5. Electric Shock. 6. Choking. 7. Smothering and pressure on the chest.

Do not think the person is dead because he is unconscious and is not breathing (*See* note on page 218).

insect enters the ear, pour salad oil or any vegetable oil into the ear—and the insect will be floated out. Or you may saturate a piece of cotton with salt solution and place it in the ear and the insect will stick to it.

Choking.

Foreign bodies in the throat: While hurriedly swallowing food, a piece of meat, a bone or particles of solid food or even false teeth may go down by the wrong way towards the wind-pipe instead of the food pipe and choke the sufferer by obstructing the passage of air into the wind-pipe. In such an emergency (1) open the patient's mouth, and put a pencil or the handle of a spoon between the grinders to prevent his biting your fingers, (2) quickly pass your fore-finger right into the back of the throat and try to remove the foreign body which is causing the

choking. If the irritation caused by the fingers induces vomiting, so much the better. It will help to expel the foreign body.

The other method is to place the head down well forward and thump the back sharply between the shoulder blades and also to suddenly press the chest front and aft so as to cause a sudden expulsion of air from it.

Foreign bodies in the stomach. When a foreign body is swallowed, don't give a purgative nor a substance to cause vomiting, both may cause harm. The treatment should be to put the patient, who is generally a child, in bed and give bulky diet such as *dhall* and rice, mashed potatoes, *khichri,* thick *sujee,* bread and milk, *kanji,* and secure medical help.

Fish-hook in skin. (1) When a fish hook has entered any part of the skin, do not attempt to draw it out by the same

way it went in. Cut the dressing or line on the hook—so that only the metal part is left. Push the barbed part firmly and quickly through the skin and pull out the rest of the hook. (2) Dry the wound. (3) Apply tincture of iodine and dress with clean dry dressings.

CHAPTER XXII

The Organs and Mechanism of Respiration.

Respiration means the act of breathing. It consists of two parts, (1) the act of drawing in air into the lungs or *inspiration,* (2) the act of expelling air from the lungs or *expiration.* We draw in air by the nose and mouth and it passes through the windpipe and its branches called bronchial tubes into the lungs. In the lungs an interchange takes place ; the air gives up its oxygen which purifies the blood flowing in the capillaries of the lungs and from it

absorbs the waste product of our body called carbonic acid gas, and is then breathed out as bad air.

The organs concerned in respiration are the wind-pipe, its sub-divisions called the bronchial tubes and the lungs in which the minute sub-divisions of the bronchial tubes terminate.

A grown up person breathes about 15 to 18 times in a minute and a child about 20 to 22 times. With each inspiration we draw in about 30 cubic inches of air into the lungs and thus expand the lungs and enlarge the chest cavity.

The mechanism of respiration consists in the enlargement of the space in the chest cavity and the expansion of the lung by the air entering it, and the diminution of the space inside the chest cavity and consequent contraction of the lungs, and forcing out of air from the lungs.

During the act *inspiration* the space in

the chest cavity is enlarged, partly by the elevation of the ribs and breast bone by the action of the muscles attached to them, and mainly by the depression of the dome of the midriff which gets flattened during its contraction. The capacity of the chest is thus increased which creates a tendency to produce a vacuum, and causes the air to enter the elastic air cells of the lungs which expand and occupy the increased chest space.

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CHAPTER XXIII

Asphyxia: Causes of Suspension of Breathing: Artificial Respiration.

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Do not think the person is dead because he is unconscious and is not breathing (*See* note on page 218).

Artificial Respiration.

By this is meant the performance of certain recognised methods for causing air to enter and leave the lungs of a person whose breathing has temporarily stopped and thus restore the functions of the lungs (Respiration) and heart (circulation). Three chief methods are in use.

I. Schafer's Method.

1. Waste no time in loosening or removing clothes from neck and waist.

2. Lay the patient on his chest with his back upwards. No pad is placed under his chest.

3. Kneel astride or if convenient at one side of the sufferer, facing his head.

4. Clear the nostrils and mouth to ensure free entrance of air. Keep his head turned to one side.

5. Place palms of your open hands, one on either side of the lower part of the back of the sufferer, your fingers extend-



FIG. 59. Expiration, Schafer's method.

ing over his lowest ribs. Lean forwards and slowly let the weight of your body press straight downwards upon the lower part of the back and chest. This drives the air out of the chest. (Fig. 59). This movement should take three seconds.

6. Draw back somewhat rapidly and relax pressure without removing your

hands; this produces inspiration. (Fig. 60) This should occupy two seconds.

Note.—Immediately resort to silvester's method should any congestion be noticed in the patient's face.



Fig. 60. Inspiration, Schafer's method.

II. Silvester's Method.

1. Loosen and if possible quickly remove clothing from neck, chest and waist to ensure easy access of air. Unfasten belts, braces and upper buttons of trousers,

and corsets in women, and *izar bund, dhoty* or *saree* in those dressed in Indian fashion.

2. Lay the sufferer flat on his back with an improvised pad placed under the shoulder blades.

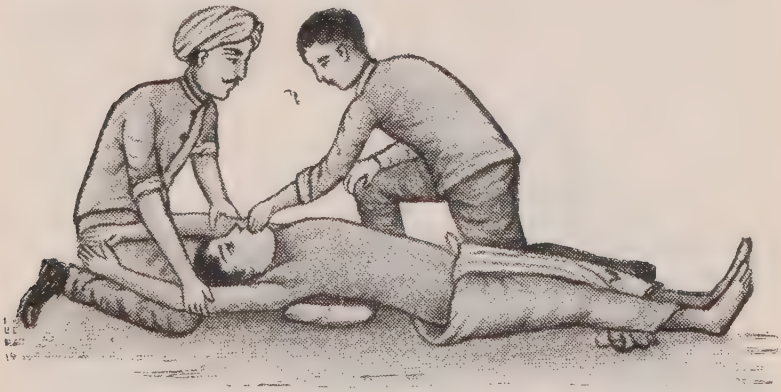


FIG. 61. Inspiration, Silvester's method.

3. Kneel at a convenient position behind the sufferer's head facing his chest.

4. Clear the nostrils, lips and mouth and ask an assistant (if available) to draw forward the tongue and hold it in that

position, by a handkerchief or cloth to avoid its slipping off.

5. Grasp the hand just below the elbow, draw the arms upward, outwards and towards you with a sweeping movement making the elbows touch the ground. This causes the chest to expand and draws air into the lungs (Inspiration). (Fig. 61).



Fig. 62. Expiration, Silvester's method.

6. Lift up the forearms bending the elbows and bring the bent arms forward, downward and inwards, press the arms and

elbows firmly on either side of the top of the chest. This drives out the air from the chest (Expiration). (Fig. 62).

III. Laborde's Method.

1. Loosen clothing round neck, chest, and waist.
2. Lay sufferer flat on his back or on one side.
3. Kneel at the side facing his head.
4. Clean nostrils, lips and mouth.
5. Grasp the tongue with a handkerchief between your fingers and pull it forward.
6. Hold the tongue in this position for two seconds and then let it go back into the mouth without releasing your hold on it.

N.B.—Each method has its own advantages. Silvester's method is most suitable for a case of drowning. Schafer's method can be very quickly applied and managed by only one person.

Laborde's method is useful when the asphyxiated person's ribs are broken or when the patient has got extensive burns on his body.

Be gentle and perform artificial respiration slowly. We breathe about 15 times per minute, therefore artificial respiration is to be performed about 15 times in the grown up and about 20 times in a child.

In the unconscious state the tongue is apt to fall back and block the passage of air down the wind-pipe (see page 247). For this reason in Schafer's method the head is kept turned to one side and in Silvester's method the tongue is kept drawn out by an assistant. Here also be gentle and do not pull the tongue out too much, nor violently press the chest as this may cause fracture of the ribs.

While artificial respiration is being performed help to start natural breathing by methods described on page 220.

CHAPTER XXIV

Drowning, Hanging, Strangulation, Smothering, Choking—Suffocation by Gases and Electric Shock.

Drowning.

Death by drowning as a method of committing suicide is not common in India but death due to accidental drowning is frequent.

In the usual variety, the unfortunate man struggles to keep himself afloat and to get out of water, till he gets exhausted and goes down. During his frantic efforts to save himself, he swallows a large quantity of water and also inhales water instead of air which results in suffocation and death.

In the less common variety a person, on falling into water, faints and goes down without a struggle. The entrance to the wind-pipe closes spasmodically, and

breathing stops. The lungs do not get water-logged but the man dies of suffocation as no air enters the lungs. If the action of the heart does not also stop with the act of breathing these cases are much more hopeful than the first variety.

There is a third variety in which both the action of the heart and the breathing stop due to the shock of sudden immersion in water and the man sinks without a struggle and dies. Plunge baths, specially in cold months, are therefore dangerous for persons with a weak heart.

The fourth variety is that in which strong swimmers sometimes get drowned on account of a sudden attack of cramps in their legs or in the heart.

Note.—Treat all drowned persons as not really but only apparently dead. Persons submerged in water for fully fifteen minutes and in some rare instances even after longer period have been restored to life. Therefore carry on

artificial respiration for resuscitation for two hours or till the doctor arrives and declares that life is extinct.

General Principles of Treatment in drowning.

1. Prompt removal of obstruction to breathing.

2. Restoration of the act of breathing.

3. Restoration of circulation of blood and body warmth.

Begin No. 1 and 2 at once *but not* No. 3 *till after breathing has been restored*, otherwise the patient will die from congestion of lungs and brain.

Special Treatment in Drowning.

(1) Do not wait until you have got the patient into a house or shelter but begin treatment on the spot at once. (2) Gently place the patient on the ground with the face downwards, and head turned to one side. (3) Fold one arm, lay it on the ground and place the head to rest upon it.

(4) Place an improvised pad under the stomach. Press on the back over where the pad is placed underneath. This will cause all fluid from the lungs and stomach to come out by the mouth. (5) Wipe mouth and nostrils. (6) Do not lose a moment but quickly turn the patient and place him on his back. (7) Loosen all clothing and lay bare the body from the neck to the waist, in men as well as in women. (8) Make a small cushion with some of the clothes and place it underneath the shoulders. (9) Perform Silvester's method of artificial respiration. If you have no assistants then begin artificial respiration in the prone position (Schafer's method). **Excite natural breathing** by holding smelling salts near the nostrils, tickling them with feather or snuff or a *falita*. Flick chest with damp towel.

Send for a doctor, and a conveyance: bullock cart, *palki, doly* or a *gharry* or

motor car, whichever is available for quick transport. Send to the nearest house for dry clothes, blankets, hot water bottles, hot bricks, clothes to wrap them up in, hot tea or coffee, hot milk, etc., and keep them ready for use after breathing is restored.

After breathing is restored promote circulation and warmth and when the power of swallowing has returned give hot milk, hot tea, hot coffee or, if nothing else is available, hot water. Keep him warm and watch for any signs of failing breathing and begin artificial respiration again, keeping as much of his body covered with *rezai* or blanket as you can. Apply hot water bottles, hot bricks, &c., to promote warmth.

Note.—An Indian method of getting water out of a drowned person's stomach and lungs is by placing a *mohnda* or a *dholak* under his stomach and gently pulling him backwards and forwards on it. The weight of the body presses

the water out of the stomach and lungs (Fig. 63).

Do not hold a person up by the heels or place him hanging across a railing to get the water out of him. Such rough handling extinguishes the feeble spark of life that might be lurking in the person.



FIG. 63. Getting water out of lungs and stomach of a drowned person

When you see a man in danger of getting drowned, be careful if you jump into water and swim out to his assistance. The man in his frantic efforts to save himself may unintentionally drown you. It is expedient to throw one

end of *pugree, puttee, kamarbund* or a rope to the man to hold and pull him out. A bamboo held out to him will also serve the purpose. A strong swimmer may get into the water and pull the victim ashore.

A person who does not know how to swim, will save himself from drowning (1) by lying on his back and throwing his head backwards and his mouth upwards; (2) by keeping his lungs well filled with air (by long inspirations and short expirations); and (3) by not raising his arms out of water. Shouting for help, and struggling to get out of water uselessly tires the drowning person and is the surest way for him to go down and hasten the end.

Hanging.

A common form of committing Suicide in India.

Treatment. 1. Call for help, but don't rush out looking for help and waste precious time.

2. Try and get at the victim with the least possible delay. Get up on a stool,

a *mohnda, ukli, machia* or anything which will raise you to the height sufficient to get at your patient.

3. As soon as you reach the patient pass one arm round his body and lift it up, so as to take the weight off the cord round the neck.

4. With the other hand, cut the cord. If there is a helper let him do it, and you hold the body with both your hands.

5. Lower the patient gently on the floor and prevent his body being injured by falling to the ground.

6. Cut and remove the cord from the neck, loosen all clothing about the neck and chest.

7. Attend to the tongue and begin artificial respiration. Excite natural breathing. (*See* page 220).

Strangulation by handkerchiefs and cords round the neck, throttling with fingers and also *smothering* with pillows

used to be a common means of murdering travellers in India by a class of robbers known as Thugs. It is still not a very uncommon form of crime. Treatment in strangulation, as in Nos. 6 and 7 for hanging. In smothering, as in suffocation see page 226.

CHOKING—(*Vide* page 204).

Asphyxia may be the results of being crushed under debris or in a crowd when pressure on the chest prevents breathing. Treatment as in suffocation.

SUFFOCATION BY GASES, FUMES AND SMOKE.

These are caused by irritating gases or fumes of chemicals, by smoke from a burning house or that from charcoal or an open fire in a closed room ; by accumulation of carbonic acid gas and wells in lime kilns, by choke damp after an explosion in a mine ; sewer gas, &c.

Treatment.

(1) Remove patient at once into fresh air. (2) Undo all clothing about the chest and neck. (3) Dash cold water on face and chest. (4) If this fails, resort to artificial respiration as in drowning.

Method of rescuing an unconscious person from a room full of irritating fumes, gas, or smoke :—

(1) Do not carry any light except electric torch.

(2) Soak a handkerchief in water or better still a mixture of vinegar and water (Strength 1 in 4). Fold the handkerchief into a triangle and tie it over your face, the bottom edge across the bridge of the nose and the point hanging down over the nose and mouth.

(3) Take a few good deep breaths in the fresh air before entering the room.

(4) Enter the room crawling on all

fours or bending low and quickly get to the patient's side. Smoke and most fumes and gases being lighter than air have a tendency to rise above ground. Fumes of petrol and some others like that from burning Sulphur, being heavier than air keep to the ground. In such cases one should not crawl but move in an upright position. It is useful to endeavour to obtain some idea of the nature of the fume or gas in order to know how to avoid it.

(5) Carry him out into fresh air by fireman's lift, page 275 and dash cold water on his face.

(6) Treat as for suffocation.

Note.—Open or break open doors and windows in order to establish currents of fresh air from outside which will clear the room of smoke or gas. Open them from inside on your way to the patient if they are not accessible from outside.

Electric Shock.

Shock from being struck by lightning or thunder bolt, or contact with live electric wires and cables are very dangerous and according to the strength of the electric current passing through the victim cause (1) severe shock, (2) insensibility, with stoppage of breathing or (3) death.

The shock of a powerful current deprives the person of the power of breathing and moving away from it and in order to give him the chance of life, he should be forthwith moved away from the live wire or the current should be switched off immediately and measures taken to restore breathing.

To quickly stop the current is the best but it is not always easy and delay is fatal. Therefore promptly remove the victim from the current without yourself

getting an electric shock. If you can prevent electricity from passing through your body, you will not get the shock. Therefore before trying to push off the victim from a live electric wire, protect yourself by substances through which electricity cannot readily pass. Such substances are dry wood, dry straw, dry glass, India-rubber articles, such as tobacco pouches, surgeon's gloves, India-rubber tubing or sheeting, and rubber mats, etc. Avoid all metallic substances, as they allow electricity to readily pass through them. For the same reason avoid water and moisture. Thus while dry wood and dry cloth, specially silk cloth, is safe to be used, they are dangerous and must be avoided as soon as they become damp or moist.

Treatment.

1. Quickly remove the victim from

electric contact by any of the following methods :—

(*a*) Take a *dry* bamboo or wooden pole and push him off the wire or cable.

(*b*) Push him away with your hand covered by an India-rubber article or if nothing else is handy by the sleeve of your coat if it is perfectly dry.

(*c*) Throw a loop made by a *dry* kumarbund, pugree, puttee or your coat held by the sleeves and let it go round the victim's body or his neck and drag him away from the contact.

2. *After removal from the live wire do not lose any time ;* dash cold water on his face and try to excite natural breathing. (*See* page 220).

3. *Perfrom artificial respiration* if breathing has stopped.

4. Treat Shock.

5. Dress burns.

CHAPTER XXV

The Digestive System.

This consists of the organs concerned in the eating and assimilation of food. They are the teeth and mouth by which we take food, the throat through which we swallow food, the stomach into which the food goes and in which it is digested, and the bowels (intestines) into which the digested food from the stomach passes out and is further digested and is then expelled as useless excreta after the absorption and assimilation of all the useful products of digestion.

The liver and sweetbread (pancreas) also secrete juices which are poured into the intestines to mix with the digestive juices there in order to further help digestion. This system is very interesting but not of much importance to the student of First Aid.

Rupture (Hernia).

Protrusion of a portion of the bowels under the skin is commonly called "rupture." It generally occurs at the groin in females, and just above it in males. It forms a soft rounded swelling under the skin which in many cases disappears on lying down. Generally it does not cause much discomfort or severe symptoms. If it comes on suddenly, a sensation of something giving way is felt by the sufferer. When from any cause the portion of the bowels in the rupture (hernia) gets nipped in the abdominal wall, it is said to get "strangulated." This is a very serious condition and causes great pain and discomfort to the sufferer and the following are the *Signs and Symptoms*. (1) Acute pain. (2) Swelling in the region of the groin which does not disappear on lying down. (3) Vomiting

and collapse. (4) Relief on lying down and keeping the knee and thigh of the affected side bent on the abdomen.

Treatment.

(1) Lay the patient down. (2) Bend the thighs and legs and raise the head and body as in a half sitting position. (3) Apply ice or cold water dressing over the hernia. (4) Send for a doctor. (5) (*a*) *Do not apply hot fomentations* or heat of any kind. (*b*) *Do not rub* or press down the hernia.

CHAPTER XXVI

The Nervous System.

The nervous system is the master tissue of the body and is found in a progressively better state of development as we ascend the scales of the animal kingdom, till in man it reaches a highly

specialised form. It is divided into three groups:—(1) *The central nervous system.* (2) *The peripheral nervous system.* (3) *The sympathetic nervous system.* The whole nervous system has been compared to a network of telegraph wires, with the brain as the central office, whilst the sub-offices and trunk lines are found in the spinal cord or marrow, the nerves representing the single wires. The messages come and the orders go, almost with lightning rapidity.

1. The central nervous system consists of the *brain* and the *spinal cord.*

The *Brain* is enclosed in the skull and controls all functions of the body and mind and is the seat of intellect, emotions and will power. It is oval in shape, its surface is thrown into tortuous folds and its structure is very complicated and it differs in weight and size according to the degree of intelligence of

the animal or the individual. Thus the brain of man is 1/40 of the weight of his body weight, while that of an elephant is 1/500th part and of a whale 1/3000th of its body weight. The brain of the Caucasian race to which the Indians and Europeans belong is heavier and bigger than those of the primitive races.

The *Spinal Cord* is the prolongation of the posterior part of the brain and occupies the spinal canal. It controls the functions of movement and of sensation in the body. It is a long cylindrical whitish grey cord made up of nerve cells and nerve fibres, which gives off thirty-one pairs of nerves from its side.

2. The *peripheral* nerves are white cords which arise from the sides of the spinal cord and divide into ever smaller and smaller branches which are distributed as fine pearly white filaments to all parts and reach the surface (periphery)

of the body and ramify under the skin. (*See* Fig. 64). The minutest filaments of nerves can only be seen through a micro-

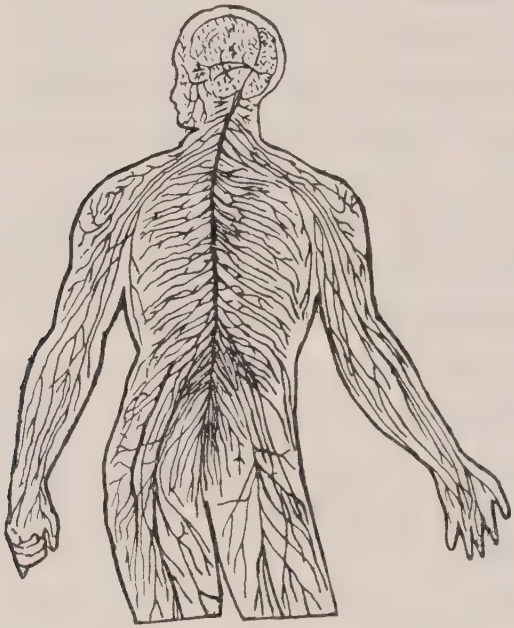


FIG. 64. Showing the brain, the spinal cord and the peripheral nerves.

scope, their function is to transmit will power for *voluntary action*, convey *sensa-*

tions of comfort and pain. They also serve to maintain *nutrition* of muscles and skin and control the *secretion* of the glands.

3. The *vegetative or sympathetic nervous system* is made up of a double chain of delicate nerves lying in front and on either side of the backbone. They send branches of infinitely fine filaments to the heart, lungs, stomach, bowels, kidneys and all other organs in the chest and abdominal cavities which are not controlled by the will. These nerves *control the automatic functions of organic life,* such as circulation, digestion, respiration, secretion and excretion and carry on their function even during sleep and insensibility and are not subservient to the will.

Injury to the brain or effusion of blood on its surface causes the following symptoms.

(1) Insensibility—either coming on suddenly or gradually deepening into complete unconsciousness. (2) Loss of the power of movement. (3) Loss of sensation. (4) Loss of the power of speech.

Injury to the spinal cord causes loss of movement and sensation (paralysis) of the part of the body below the site of injury or even instant death (*vide* page 18).

Injury to the perepheral nerves causes loss of movement and feeling, which is complete if the nerve gets cut across, *e.g.*, as a result of stabbing or cutting wound with a sharp instrument, or a gunshot injury.

Injury to the sympathetic system of nerves is rapidly fatal, even very slight injury causes severe symptoms. *See* page 36.

CHAPTER XXVII

Shock, Fainting or Syncope and Collapse.

In all the three conditions there is a general depression of the whole nervous system. Whether the result will be shock, fainting or total collapse is only a question of the degree of depression.

Causes.

1. *Physical conditions* such as profuse bleeding, severe injuries, burns, severe operations, injury on the abdomen, exhaustion from fatigue, hunger or exposure to extreme heat or cold, and being in a close and overcrowded room.

2. *Mental conditions* such as sudden grief, fright, suspense in anticipation of receiving pain or injury.

3. *Poisoning, e.g.,* Ptomain, Arsenic, Alcohol, Dhatura, Bhang, Opium and Narcotic poisons.

4. Constitutional—weak heart, hypersensitiveness, neurotic temperament.

Symptoms.

(1) Pale lips and face and a pinched appearance. (2) Cold and clammy skin. (3) Rapid and weak pulse. (4) A sick feeling or sometimes vomiting. (5) Unconsciousness is complete in fainting and collapse but not necessarily complete in all stages of shock.

Treatment for Fainting and Shock.

1. Arrange for free access of fresh air.

2. Lay the patient down with his head and shoulders low.

Stop bleeding and attend to fractures and crushing injuries should these be present.

3. Raise limbs in order to make the blood from the limbs to go to the head and abdomen.

4. Rub the extremities briskly to produce warmth by friction. Begin at the ends of the extremities and rub towards the heart.

5. Dash cold water on the face.

6. Hold a half open bottle of smelling salts near the nostrils.

7. Loosen all tight clothing about the neck ; chest and abdomen, see that there is no pressure on the chest specially in the region of the heart. If available lay the patient on a bed and raise the foot end of the bed.

8. Keep the patient warm by covering up with *rezais* and blankets. Apply warmth by hot water bottles or hot bricks to the body, feet, and hands.

9. If the patient is sufficiently conscious to be able to swallow, give hot sweet tea, hot milk with sugar, a teaspoonful of salvolatile mixed with about a chitak (a wineglassful) of water. Send for

a doctor and watch carefully. Ability to swallow in returning conciousness is tested by gently pouring about a teaspoonful of cold water at the angle of the mouth between the gums and the cheek and watching if it causes the person to swallow.

Note.—Hot water bottles or hot bricks should always be wrapped up with flannel or cloth to prevent their burning the skin. The damage might be considerable in a person unconscious or insensible in a state of shock.

CHAPTER XXVIII

Insensibility.

Insensibility means unconsciousness. It is called *coma* when unconsciousness is so deep that the person cannot be roused. It is called *stupor* when he can be roused by being shouted at. In stupor the pupils of the eyes respond to the action of light,

i.e., if the eyelids are drawn up and eyes exposed to light the pupils become smaller and if the eyelids are closed and the eyes turned away from light and then opened and examined the pupils dilate and appear bigger. But in a condition of coma the pupils remain fixed. In a condition of stupor if the eyeballs are touched the person quickly shuts the eye, and shows signs of getting hurt but in coma he lies listless and does not mind it.

Unconciousness may come on suddenly or its onset may be gradual.

Causes.

1. Asphyxia. 2. Injuries. 3. Effects of heat and cold. 4. Diseases and fits. 5. Drugs, Poisons and snake bite.

Classification of Conditions in Insensibility.

I. During which breathing stops (Asphyxia). The person can be saved

from death only by correct and prompt action.

Example :—Severe electric shock, struck by thunder bolt or lightning, hanging, drowning, choking, suffocation from smoke irritating fumes or poisonous gases ; smothering, getting crushed in a crowd, or under debris after accidents in mines and or collapse of a house, etc.

II. During which breathing is present and there are no convulsions or fits :—Two groups.

(*a*) **When the face is flushed or congested.** Example: Sun-stroke, Heat-stroke, Apoplexy; Head-injuries with compression of the brain; poisoning by alcohol or opium.

(*b*) **When the face and lips are pale.** Example :—Fainting, shock, profuse loss of blood, head injuries with shaking (concussion) of the brain; Heat-exhaustion.

III. During which breathing is present and is accompanied with convulsions or fits.

Example :—Epilepsy, Hysteria, Teething fever and stomach trouble in infants;—poisonings by strychnine or prussic acid.

Treatment.

I. General rules for treatment of insensibility.

II. Special treatment according to cause and effect. See page 249 and following.

General Rules for Treatment of insensibility.

1. Send for the Doctor.

2. Treat the unconsciousness and stoppage of breathing if present, immediately, do not waste time in attending to minor injuries.

3. Stop severe bleeding, should it be present, and examine the patient gently but quickly to find out if he has got a fracture or dislocation or a head-injury.

4. Do not jump to the conclusion that the patient is drunk because he smells of alcohol. It is quite likely that the patient might have been given alcoholic

stimulants just before he became unconscious. Coma due to abuse of alcohol is also a serious condition and should not be neglected.

4. Put the patient on his back stretched out at full length and in an easy position with his *head turned to one side*. So that should he get sick, the vomitted matter will flow out of the mouth and be not drawn into the lungs with the breath. Also to prevent the tongue falling back on the opening in the wind pipe and obstruct breathing.

5. Put the head low if the face is pale as is seen in faintness and insensibility after great loss of blood. Keep the head raised if the face is red, as in the case of heat apoplexy and sunstroke or if there is head injury with pressure on the brain.

6. Loosen tight clothing of all kinds.

7. Let him have plenty of fresh air.

8. Do not give him alcohol or stimulants of any kind.

9. Do not give him anything by the mouth.

10. Do not leave the patient till a responsible person takes over charge.

Note.—The opening in the wind pipe is guarded by a flap which prevents the entry of food and drink into it. During insensibility it does not act and therefore there is the danger of food and drink given during insensibility going down the wind pipe (the wrong way) and cause choking. In an insensible person the tongue often falls on the flap and obstructs breathing.

Treatment for special conditions in Insensibility.

See Chapter xxiv page 217 and the following pages.

Sun-stroke Heat-exhaustion, Heat Apoplexy.

During the summer months cases of sun-stroke, heat-stroke, and heat-exhaustion

commonly called (*Looh Lagjana*) are of frequent occurrence. Engine drivers, firemen, and those who work in the stoke-holes of steamers and in smithies, workshops and factories easily fall a victim unless sufficient precautions are taken. In the same way coolies and labourers in the fields and soldiers on the march specially with the load of their kit on them are particularly liable to be affected, often passengers in crowded trains suffer from heat-stroke.

One should avoid working for long hours in a hot or crowded place, and not expose oneself to the direct rays of the sun for a long time, specially when doing some hard physical work. Iced drinks and cold acid drinks, like lime or tamarind *sherbet,* and green mango *panna* are useful. Also the use of a solar hat, umbrella and glare glasses are a great protection for those who have to be out in the

sun. The steamy heat when there is much moisture (humidity) in the air is more trying and dangerous for susceptible persons than only the rise of temperature by itself.

Sun-stroke.

Causes :—Prolonged exposure to direct rays of a strong sun. *Signs and Symptoms.* The onset is more or less sudden. The patient feels sickness, giddiness and thirst and falls down. The skin appears dry, the face is flushed and both the pupils of the eyes, become contracted. The pulse is strong and jumpy and breathing is hurried and snoring. The temperature is generally high and it may rise up to 108°F. and the person lies insensible in a stupor which passes on to coma.

Treatment.

First Stage. (1) If the patient is sick turn his face to one side till vomiting is over. (2) Remove the patient to a cool

and shady place. (3) Lay him down with the head and shoulders well-raised in order to prevent the blood from rushing towards the brain. (4) Fan him vigorously. (5) Take off his clothing up to the waist and loosen the cloth round the waist. (6) Apply ice or cold water to the top and back of his head and neck. (7) Give the patient a cold sponging or wrap him up in a wet sheet dipped in ice-cold water. (8) As soon as the patient is able to swallow, he should be given cold drinks and if available green mango *panna* or tamarind *sherbet.*

Later Stage. When the temperature drops to 102°F. Remove the wet sheet, stop cold sponging and the application of ice to the head and neck. Then wrap the patient up in quilts or blankets and watch him carefully as there is fear of collapse due to sudden fall of the temperature. If necessary, the patient should

now be given hot milk and hot tea and hot water bottle and friction should be applied to the limbs and the soles of the feet and palms of the hands.

Heat Exhaustion.

Causes:—It is caused by prolonged physical exertion in a high temperature. *Signs and Symptoms.* The onset is gradual. The patient's expression becomes dull, and the face is first flushed then turns pale, and he may feel sick of stomach. The pulse is quick and feeble and the temperature is low. There is sighing, respiration and the person eventually falls in a stupor.

Treatment. Remove the patient to a cool place, but keep him warm with his head low. Apply friction to his limbs and hot water bottles to the flanks. If the patient is able to swallow, let him drink hot tea, coffee or hot milk.

Note the difference in the onset and treatment of sun-stroke or heat-stroke and heat exhaustion.

Heat Apoplexy.

Cause:—Rupture of a diseased blood vessel in the brain—the result of excitement caused by anger, grief, use of alcohol and drugs, exposure to hot sun or being in a crowded and hot room. Straining in the water closet due to constipation. The sufferer is generally an elderly person or one with high blood pressure or diseased arteries. There is no history or mark of injury to the head. *Signs and Symptoms.* Sudden unconsciousness which is deep, (Coma). Pupils, unequal and fixed. Skin, hot, Face flushed. Breathing, and pulse, slow, strong and full. One side of body limp (paralysed).

Treatment.

(1) Keep the head slightly raised.

(2) Apply ice bag on the head and nape of the neck. (3) Keep body and limbs covered. Carry out general rules for treatment of insensibility and of head injury with compression of brain. .

Head Injury (Compression).

Injuries causing pressure on brain. *Signs and Symptoms.* Skull often shows signs of injury. Onset—gradual and progressive. Deep unconsciousness ; cannot be roused. Pupils, unequal—first contracted, later on dilated. Fixed, not affected by light. Breathing slow, but snoring in character. Pulse, slow and strong. Skin, hot.

Treatment. 1. Dress wound if present. 2. Keep head slightly raised and put ice bag on head. 3. Hot water bottles to feet. 4. Do not give *stimulants.* Remember that more blood rushes to the brain if cold is applied only for a

short time and then removed. We dash cold water on the head and face of a fainting person to stimulate circulation and revive him but ice bag is put on the head and nape of neck to reduce the circulation of blood in the brain and therefore must be continued without break.

Head Injury (Concussion).

Injuries causing shaking of brain. *Signs and Symptoms.* Skull usually shows no signs of fracture, the person is simply stunned by the blow. Onset of insensibility is sudden ; can be roused by shouting in most cases. Pupils, equally dilated, sluggish reaction to light. Breathing, slow and feeble, sometimes sighing. Pulse, slow and weak. Skin, cool, face pale. *Treatment.* 1. Dress wound if present. 2. Ensure absolute rest and quietness. 3. Keep warm by covering with quilts and blankets and apply hot water bottles. 4. Put ice

bag on the head. 5. Avoid stimulants. If a patient has been unconscious even for a short time warn him to consult a doctor and not to resume mental or physical exertion.

Fainting.

Cause. Over-exertion, fatigue, hunger, overcrowding, *e.g.*, in stuffy rooms and railway carriages, loss of blood, heart disease. Physical or mental shock. *Signs and Symptoms.* Onset—gradual, pale face, white lips ; complete unconsciousness, pupils—normal. Breathing—silent and quick. Pulse—quick and weak. Skin—cold sweat on forehead and extremities.

Treatment. 1. Give smelling salts. 2. Give teaspoonful of Salvolatile in a wine glass of water to drink. 3. Keep head low and body and limbs warm. (*See* treatment for fainting, page 240).

Epilepsy.

Hereditary predisposition. *Signs and Symptoms.* Onset—sudden, without warnings. Complete unconsciousness. Patient suddenly falls down. Tongue is bitten, bloody froth from mouth. Eyes fixed, insensitive. Breathing—at first almost stops during the fit, when muscles stiffen. When fit passes off breathing returns and becomes snoring. Pulse—quick. Skin—normal.

Treatment. 1. Place patient so that he cannot injure himself. 2. Prevent further injury to tongue by placing a well-padded piece of wood or pencil between the teeth. 3. Dash cold water on the face. 4. After fit is over, encourage natural sleep.

Hysteria.

Cause. People with very neurotic temperaments. *Signs and Symptoms.* Onset—

gradual, jerky movement of body ; patient always conscious. Eyelids closed but patient resists opening them. Pupils unequal. Breathing—quick. Pulse—quick. Skin—normal.

Treatment. 1. Cold douche to head. 2. Flick face and neck with corner of wet towel or handkerchief. 3. Keep away sympathetic friends. 4. Apply mustard plaster to the back of neck. 5. Smelling salts, and snuff to the nostrils.

Infantile Convulsions.

Cause. Teething, fever, with bowel troubles (constipation or diarrhœa). *Signs and Symptoms.* Onset—gradual body stiffens with twitching of face muscles. Complete unconsciousness. Eyelids closed. Breathing—very quick. Pulse—quick but weak. Skin—cold and wet with perspiration.

Treatment. Relieve constipation by

a soap-water enema. 2. Give cold douche (in the bowels) if temperature is very high. 3. Put the child in hot bath from neck downwards for about 15 to 20 minutes and apply ice to head and continue after the bath. 4. After the bath, dry up the body, put on dry light clothing and place the child in bed covering very lightly and keep the ice bag on his head till temperature falls to 101°F. *N.B.*—Temperature of bath should be the same as that of the child's body or else it will get shock in the heart.

Malaria.

(That kind which affects the brain). *Signs and Symptoms.* Onset—gradual, headache, fever and restlessness preceding unconsciousness. Breathing—sighing, pulse—full and fast. Skin—hot and dry.

Treatment. Same as for heat stroke till medical aid is obtained.

Poisons (by Drugs).

Opium, Bhang, Majun, Dhatura, alcohol, etc. Insensibility varies with the amount of poison taken and the stage in which it is seen. See Chapter on poisoning.

Poison (from Urine.)

Cause. Scanty or stoppage of urine due to chronic kidney disease. *Signs and Symptoms.* Onset—gradually deepening unconsciousness, when complete, cannot be roused. Pupils—normal or dilated. Breathing—*Sighing.* Pulse—slow, full and bounding. Skin—normal.

Treatment. 1. Apply hot sponging, keep covered up and *encourage sweating.* 2. Give enema to open bowels. 3. If patient can swallow, give a purgative. 4. *Don't give alcohol* or *animal food* or soups.

CHAPTER XXIX

Poisons and their Treatment.

A poison is defined as "a substance which, if taken by the mouth or otherwise absorbed into the blood, affects one's health seriously or destroys life"; generally a poison is taken by the mouth, although it is possible to get poisons injected under one's skin or mixed up with the blood, as in snake bite and bites of mad and wild animals.

Cases of poisoning both homicidal and suicidal are unfortunately very common in India.

In order to avoid detection a poison whose colour and taste can be easily masked or the symptoms caused resemble those of a common tropical disease are generally chosen and it is mixed with ordinary articles of food like milk, *sherbets*, sweets, tea and coffee. Suspect

poisoning if a person in good health without apparent cause suddenly feels ill and shows alarming symptoms, which rapidly progress from bad to worse and are accompanied with marked exhaustion. Remember, however, that without criminal intention on the part of any one, accidental poisoning may occur from poisons, bad tinned provisions and also from several articles of food which rapidly go bad in our climate and turn into an irritant poison called Ptomaine. Similarly food cooked in copper utensils, which are not properly tinned, act as an irritant poison. See page 268.

General Rules to be Observed in Investigating Unknown Poison Cases.

(1) Quickly look round the room and the vicinity and try to detect a bottle, a pill box or a powder wrapper, which may

give a clue to the poison which has been taken.

(2) Throw nothing whatsoever away, vomited matter specially must be preserved for the doctor. This is most important.

(3) Go close to the patient and look for burns or stains on the mouth or clothes, so frequent in corrosive and acid poisonings.

(4) Notice whether the breath has the odour of alcohol, opium or any other substance familiar to you.

(5) Notice the presence or absence of drowsiness.

(6) Notice whether the patient is boisterous and excited or delirious.

(7) Note the state of the pupils of the eyes. Are they dilated or contracted?

General Rules for the Treatment of Poisoning Cases.

(1) Neutralise the poison, give antidote,

(2) Dilute the poison and prevent its absorption.

(3) Give an emetic (*see* page 264) to get rid of the poison except in poisonings which stain the lips and mouth.

Stains or marks of burns in the mouth or lips show that the delicate tissues of the stomach and throat have been injured by the action of the poison, and may get torn as a result of the strain caused by vomiting.

(4) Protect the injured internal surfaces by giving demulcent drinks, such as sweet vegetable oils, *e.g.*, olive oil, cocoanut oil, milk, barley water, white of eggs, mucilage made by boiling linseed (Tisi) sesamum (Til), *isafgul,* Gum Arabic (*Gond*), etc.

(5) Look after patient's general condition and overcome shock and exhaustion.

Definitions: An *Antidote* is a substance which is given to neutralise the bad and deadly effects of a poison, *e.g.*, white

of egg in corrosive sublimate, pot. permanganate in opium poisoning.

An *Emetic* is a substance which causes vomiting.

The *common method of inducing vomiting* is to make the person drink.—

(1) A tablespoonful of common salt in half a tumbler of warm water.

(2) A teaspoonful of mustard powder in half a tumbler of warm water.

To induce rapid vomiting, by methods (1) and (2), ask the patient to tickle his throat.

Classification of Poisons.

1. *Corrosives.* Those which corrode or destroy tissues and stain the mouth and lips, *e.g.*, strong mineral acids, carbolic acid, strong alkalies, etc.

2. *Irritants.* Those which cause vomiting, purging and rapid collapse, *e.g.*,

arsenic, phosphorus, ptomaines, kerosine oil, turpentine oil, powdered glass, mercury (corrosive sublimate), copper.

3. *Narcotics.* Those which produce deep sleep, *e.g.*, opium, morphia, laudanum, paregoric, etc.

4. *Delirients.* Those which cause excitement of imagination or delirium, *e.g.*, *Dhatura, Bhang, Ganja,* Alcohol, Cocaine.

5. *Convulsives.* Those in which convulsion and contracture of muscles is a prominent symptom, *e.g.*, Strychnine, Prussic Acid.

Poisons.	Signs and Symptoms.	Treatment.
CORROSIVES : *Acids* :—	(1) Extreme exhaustion, Rapid collapse. (2) Characteristic stain on mouth and lips, Black (Sulphuric Acid) Yellow (Nitric Acid) White (Hydrochloric Acid) Carbolic Acid (white eschar) Oxalic Acid (Dirty white). (3) Burning pain in the mouth, throat and stomach. (4) Dfficulty in swallowing. (5) Intense thirst. (6) Vomit may be mixed with blood,	*Avoid emetics.* In case of acids water mixed with chalk, bicarbonate of soda, Scrapings of *chunam* from a pucca wall or ceiling act as an antidote. In case of alkalies acid drinks made from vinegar, lemon or lime-juice or tamarind. In carbolic acid give ½ oz. of Epsom salts (Mag. Sulph) in ½ pint of water, whisky or brandy. Demulcent, drinks, *e.g.*, Olive oil, milk, etc.
Sulphuric Acid (Oil of Vitriol) Nitric Acid (Aqua Fortis) Hydrochloric Acid (Spirit of Salts)		
Oxalic Acid (contained in Brasso and other metal polish).		
Carbolic Acid (Phenol)	On account of the benumbing action of Carbolic acid, vomiting may not be present Characteristic smell.	Don't give oil in Carbolic acid as it is soluble in it and helps absorption of the poison, but give white of egg and milk freely. Treat for shock. Warmth to the feet.
Alkalies :— Caustic Soda, Caustic Potash, Strong Ammonia.	Slimy patches of dark chocolate or black colour.	

IRRITANTS.		
Arsenic (Sankhya. Sam-ulfar Senko) is largely used for criminal purposes. Accidental poisoning occurs from Fly-killers, Rough on Rats.	(1) Dryness and heat in the throat and burning pain in the stomach. (2) Continual vomiting of blood-stained matter, bile & mucus which does not relieve pain nor the feeling of nausea. (3) Frequent purging accompanied with much straining blood and mucus. (4) Painful cramps in legs. (5) Marked shock & exhaustion.	(1) Induce rapid Vomiting. (See page 231.) (2) Give special attention to overcome shock with application of warmth to body and hot fomentations to the belly, Artificial respiration. (3) Give milk, ghee, olive oil, egg, albumin,
Phosphorus Accidental from vermin pastes, rat poisons and matches.	Symptoms resemble Cholera. (1) Garlic taste in mouth and garlic smell from breath. (2) Vomited matter luminous in the dark.	As for other irritant poisons but don't give oils which dissolve the poison and make its action stronger.
Mercury, corrosive sublimate. (Ras-Kapoor.) Largely used in antiseptic lotions and dressings. *Turpentine & Kerosine Oil.*	Those of irritant poisons. (1) Metalic taste in mouth. (2) Tongue whitish with dribbling of Saliva, (3) Vomiting and purging. (4) Collapse. Vomiting shows oily substance and characteristic smell of Kerosine oil or turpentine. Thirst and collapse.	(1) Give emetics. (2) Give drinks of permanganate of potash 10 grs. in a pint of water. (3) Stimulants.
	(1) White of egg is an antidote for this poison. (2) Give ghee, flour and water.	Treat as for irritant poisons, *e.g.* emetics, demulcent drinks and apply warmth to body and feet.

Poisons.	Signs and Symptoms.	Treatment.
Powdered Glass.	Gritty taste in food and drink, powdered glass can be seen with a magnifying glass in the vomit. Other symptoms of irritant poisons, *e.g.*, pain in stomach diarrhœa and blood-stained stools and vomiting may be present.	Give large quantities of bread, rice, mashed potatoes and other bulky food. These get mixed with the powdered glass particles in the stomach and protect its walls against damage. After this give emetics, not before.
Iodine, tincture of Iodine.	Symptoms of irritant poison. Vomiting yellow or blue.	General treatment for irritant poisons. Antidote, rice, sago, arrowroot and other forms of cooked starch. Give plenty by mouth.
Ptomaines, Meat and fish that has gone bad, mushrooms.	As in other irritant poisons special symptoms. (1) feeling of uneasiness in the stomach. (2) Fever. (3) Brown furred tongue. (4) Marked exhaustion. (5) Vomiting and purging is mistaken for cholera but in cholera the stools are not offensive and there is no fever.	(1) Emetics as for other irritant poisons. (2) Give two ounces of castor oil to quickly get rid of the cause of irritation in the bowels. Overcome shock.
Copper Sulphate. (Tutia Nirmala) Food cooked in untinned copper utensils.	As for other irritant poisons Green vomit.	As for other irritant poisons.

NARCOTICS. *Opium.* Largely used for committing suicide in India specially by women. Accidental Poison from Laudanum, Morphia, Chlorodyne, Paregoric, Baby soothers often happens.	(1) Breath and mouth smell of Opium. (2) Drowsiness passing on to complete unconsiousness. (3) Pupils get small, become small like a pin point in later stage. (4) Slow and shallow breathing. (5) Skin is hot and dry in earlier stage, becomes cold and clammy and covered with perspiration in later stages. *N.B.*—In chlorodyne poisoning the pupils are not contracted because it contains Prussic Acid which dilates the pupils.	(1) Induce vomiting by Emetics (if the patient can swallow). (2) Give permanganate of potash (Condy's crystals) 10 grs. in 1 pint of water and tickle the throat to produce vomiting. Repeat this till the vomit comes out unchanged in colour. (3) Keep patient awake, flick his chest and legs with wet towel. Pinch him and shout at him. If the patient is not exhausted make him walk up and down. (4) Perform artificial respiration if breathing fails. If the patient is unable to swallow inject into the rectum with ordinary syringe about half a pint of strong tea or coffee every hour. If he can swallow give it by the mouth.

Poisons.	Signs and Symptoms.	Treatment.
Cocaine.	(1) Dryness of the skin. (2) Patient pale and faint (3) Skin dry and hot. (4) Pulse week and rapid (5) Excitement of imagination followed by depression. (6) Tremor of muscles. Staring expression. (7) Feeling of numbness. (8) Cemplete insensibility.	Keep the patient lying down. (1) Induce vomiting by common salt and tickling the throat, it may not act in the later stages as all the nerves are numbed with the poison (2) Overcome shock. (3) Perform artificial respiration if necessary.
CONVULSIVES. *Strychnine* (Kuchla) contained in many vermin killers and bug killers.	(1) Feeling of suffocation and difficulty in breathing (2) Jerking of head and limbs with stiffening of the whole body. (3) Violent convulsion with arching of the back, patient lies on his head and heels. (4) Eyeballs staring. Pupils dilated (5) Pulse feeble and rapid. (6) Difficulty of opening the mouth (Lock jaw).	(1) Emetics. (2) Give patient 10 grs. of permanganate of potash in a pint to drink then make him vomit. Repeat this every half hour (3) Strong tea and coffee, (4) A Doctor will give chloroform and perform artificial respiration which is possible only under chloroform. Avoid noise and strong light in the room.

DELIRIENTS.		
Stramonium *Dhatura* (Belladonna) Frequently a passenger in a waiting room, or a sarai is drugged through sweets and fruits and robbed.	(1) Staggering gait. (2) Dryness of mouth and throat, great thirst. (3) Difficulty in swallowing. (4) *Flushed face.* (5) Pupils widely dilated, sees double. (6) Gets easily frightened, grasps at imaginary objects, picks at his clothes and draws imaginary threads in the air.	Induce vomiting by salt and water. Pour water on the head. Rouse by flicking with wet cloth. In the later stages take measures to overcome shock. Give hot drinks, apply friction to the limbs. Perform artificial respiration if necessary.
Cannabis Indica or Indian Hemp. Ganja, Bhang, Majum, Charas. Bhang is generally given with sweets, sharbets and siddhi.	(1) Patient is boisterous, merry, sings and laughs and shows great excitement. (2) Pupils dilated. (3) In the later stages he becomes drowsy and then unconscious.	
Alcohol, Whisky, brandy and country liquor.	(1) Breath smells of alcohol. (2) Face flushed with red eyes, excitement with confusion of ideas, indistinct speech. (3) Staggering gait. (4) Vomits and is drowsy. (5) In later stages face becomes pale, breathing becomes snoring, drowsiness passes on to unconsciousness and the patient gets collapsed.	(1) Dash cold water on the face and wash head with cold water. Give him a bath except in the stage of collapse. (2) Induce vomiting. (3) Hold smelling salts to the nostrils. (4) Give hot tea and milk. (5) Perform artificial respiration if necessary, (6) Warmth to the feet by friction and hot water bottles.

Poisons.	Signs and Symptoms.	Treatment.
Prussic Acid.		
If taken in any appreciable quantity death is instantaneous. Ambulance man can do very little. In cases of overdose of a medicine containing this drug or of taking minute quantity from bitter almonds which contain prussic acid prompt action may save a patient. *Send for the doctor immediately.*	(1) Symptoms appear immediately and urgent. (2) Patient falls down unconscious. (3) Pupils dilated, eyes shining. (4) Breathing, gasping and difficult. (5) Convulsions and collapse. (6) Smell of bitter almonds from the breath.	(1) Don't waste time in giving emetics, if possible quickly carry him out into the open air. (2) Dash cold water on head and face. (3) Hold smelling salts to the nostrils. (4) Perform artificial respiration. (5) Give strong hot tea if patient can swallow.

CHAPTER XXX

Improvised Methods of Lifting and Carrying the Sick or Injured.

The methods vary according to the number of persons available for lifting and carrying the patient.

I. *Carrying by one person.*

(1) The friends grip. (2) Pick-a-back. (3) The back lift. (4) The fireman's lift.

The Friend's Grip. Useful only when a person has sustained an injury like sprained ankle or crushed foot on one limb only and when there is no fracture. The helper stands close to the sufferer and on the same side of the injury and (1) brings patient's arm round his neck, (2) holds the patient's wrist with one hand, (3) passes the other arm round the patient's waist, (4) the patient bends the knee of the injured side and keeps the

injured limb off the ground and hops along putting the weight on the helper, who



FIG. 65. Friend's Grip.

walks with broken steps as in a three-legged race. (Fig. 65).

Pick-a-back does not require any description. The injured person is carried on the helper's back, if the nature of the injury permits.

The Back Lift. (1) Let patient stand with his back to yours. (2) Stoop down a little. (3) Place your hands over your shoulders. (4) Grasp the patient's arms under his arm-pits. (5) Bring his weight well up on the small of your back. (6) Carry him, standing up erect or in a slightly stooping position whichever is convenient to you.

To lower the patient to the ground, get down on your left knee and place the patient in a sitting position.

The Fireman's Lift. The patient is lifted on to the shoulder and carried in this position. This method is specially useful when the patient is insensible.

1. Roll the patient over on his face and place his arm by his sides.

2. Kneel down straddling near the patient's head. Pass your hands under his chest and arm-pits and push them on till your hands reach the patient's back, interlock your fingers in this position. Fig. 66(1).



(1) (2)

FIG. 66. Fireman's Lift (initial stages).

3. In this position draw him forward

as high up towards your chest as you can manage. Fig. 66(2).

4. Loosen your grasp at the back and quickly move your hands down and encircle the patient's waist and interlock your fingers in this position.

6. Now get up lifting and raising the patient to an upright position with you, holding him firmly to your right side. Fig. 66(2).

7. Now grasp the patient's left wrist with your right hand, stoop down and draw his left arm round the back of your neck keeping the head bent. Stoop down till your head is opposite his left groin. While in this position pass your left forearm behind his thighs and grasp his left thigh firmly with your left hand. Fig. 67(3).

8. Bring the weight of the patient evenly on your right shoulder. When

the patient is evenly balanced upon your left shoulder, let go your right hand and hold his left wrist with your left hand



FIG. 67. Fireman's Lift (final stages).

and assume the upright position. Carry him out into fresh air. Your right hand is free for use. Fig. 67(4).

To lower the patient:—

1. Hold the patient's left hand by your right hand, left thigh with your left hand. Now take your left hand away from the patient's left thigh and support his buttocks on your left forearm.



FIG. 68. Lowering patient (Fireman's Lift).

2. Stoop down and gradually lower him until his feet touch the ground. Fig. 68(5).

3. Kneel on the left knee allowing his arm to drop from your back.

4. Interlock your fingers behind his back and gently lay him at full length on the ground. Fig. 68(6).

II. *Methods of lifting and carrying by two persons.*

(1) The fore and aft method. (2) Improvised stretchers. (3) Hand seats. (4) Chair seats.

The Fore and Aft Method. This is a rough and ready method for removing a person, who is not seriously injured and who has not to be carried a long distance. The method is as follows:—

1. One man stands behind the injured person and passes his hands and forearms under the patient's arm-pit and interlocks his fingers on the patient's chest.

2. The second person turns his back

to the patient and gets between his legs. He then stoops down and grasps the patient's knees from the outside.

3. Both helpers now stand up lifting the patient and carry him. This method is useful when space does not permit of a hand seat. An improvised chair-stretcher is however always preferable.

Hand Seats. This method of carrying is suited for moderate injuries, where stretchers are not available.

There are three varieties—(1) two-handed seats, (2) three-handed seats, (3) four-handed seats.

The two-handed seats.—Two methods—

1. *The clasp-hand.* The two helpers face each other. The fingers of the right hand of the one are interlocked with the fingers of the left hand of the other keeping the palms upwards. (Fig. 69). The palms of the hands should be kept together to prevent strain on the fingers.

2. *The hook grip.* Each helper bends his fingers at the second joint. One helper keeps the back of his hand upper-



FIG. 69. Clasp-hand seat.

most while the other keeps the back of the hand downwards. The bent fingers are hooked together. (Fig. 70). A hand-



FIG. 70. Hook-grip seat.

kerchief or a piece of cloth kept between the fingers relieves strain on the fingers and nails.

In order to remove a patient by two-handed seat the two helpers face each other and stoop one on each side of the patient. Each passes his forearm nearest to the patient's head under his back and

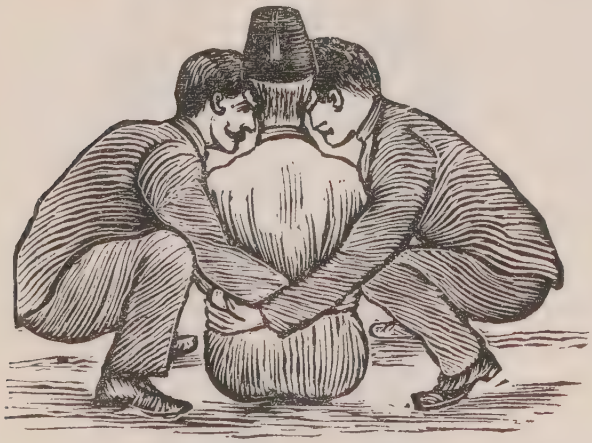


FIG. 71. Lifting on two-handed seat.

if possible grips his clothing. (Fig. 71). They slightly raise the patient's back and pass their other forearms under the middle of the patient's thighs and clasp their

hands by one of the methods mentioned above. The helpers rise together and step off. (Fig. 72).



FIG. 72. Carrying on two-handed seat.

The three-handed seat. This can be

utilised if the patient is able to use his upper limbs. One of the helpers also has one hand free to support an injured lower limb of the patient.

(1) The two helpers stand facing each other. (2) One helper grasps his own left wrist with his right hand and also the right wrist of the second with his left hand. (3) The second helper grasps the right wrist of the first with his right hand. This leaves his (the second man's) left hand free to support the patient's left leg. (Fig. 73). (4) The patient sits on the seat thus made and is raised from the ground by the helpers. The patient passes one arm round the neck of each helper and thus gets an additional support. (5) The bearers rise together and march with broken steps.

When the patient is unable to give any assistance due to injury to his arms, the second helper places his free hand on the

opposite shoulder of the first helper and asks the patient to lean on it.

For patient's right leg, the method is the same excepting that the first helper holds his right wrist with his left hand and the left wrist of the second helper



FIG. 73. Grip for the three-handed seat.

with his right hand, the second helper holds the first helper's left wrist with his left hand so as to leave his right hand free to support the patient's right leg.

The four-handed seats. Each bearer grasps his own left wrist with his right

hand and then grasps the wrist of the other with his left hand. (Fig. 74). The patient sits on the seat and supports himself by placing his arms round the neck of each bearer.



FIG. 74. Grip for four-handed seat.

Chair seats. Tie one pole on each side beneath the seat of the chair and convert it into a comfortable portable seat. (*See* Fig. 75).

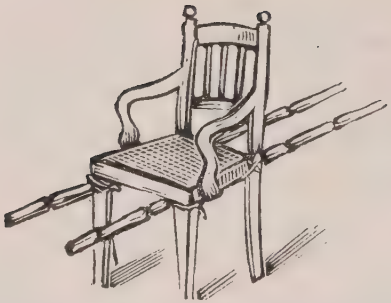


FIG. 75. Improvised Chair Seat

The Human stretcher. It is a variety of hand seat and is used for removing a patient in the recumbent or semi-recumbent position.

1. The two First Aiders face each other and stoop, one on each side of the patient. They pass their left hands beneath the patient's hips and hold them as in shaking hands. 2. The First Aider on the patient's left side passes his right hand and forearm under the patient's head, neck and shoulders. 3. The First Aider on the patient's left side passes his right hand

and forearm under the patient's legs. 4. The two rise together and carry the



FIG. 76. Carrying on a human stretcher.

patient, feet fore-most, taking short side-paces. (Fig. 76).

III. *Methods of lifting and carrying by three or more persons.*

These methods are used in removing a patient in a *doli,* a *charpoy* or a stretcher. Always use a stretcher for choice for the following advantages:—

1. It is light but strong. 2. It has a rigid bed which prevents jolting during transport. 3. It is not too wide, and can be placed in an Ambulance carriage or a wagon without difficulty, and can pass through narrow alleys and narrow flights of steps.

CHAPTER XXXI.

The Stretcher and how to use it.

Stretchers are narrow portable beds with a strong but light frame-work of wood and metal and a piece of canvas stretched between them. There are two chief

varieties in use, (1) the Army Service Stretcher which weighs about 34 lbs., (2) the St. John's Ambulance pattern which is lighter by twelve pounds (weighs 22 lbs.). A *Stretcher* (Fig. 78) consists of the following parts:—

1. Two poles, each measuring 7 feet 9 inches in length, with telescopic handles which when closed reduce their length to only 6 feet.

2. Two cross bars, or traverses. These are two metal bars with a joint in the middle by means of which they can be doubled up, when the stretcher is folded. The traverses remain under the canvas bed and when open keep the poles apart and also keep the canvas bed stretched.

3. Four rollers, or small wheels on which the stretcher can be moved along the floor of a wagon or an ambulance car with the minimum amount of friction and jolting.

4. The stretcher bed, a piece of strong canvas about 22″ in width and 6 ft. in length.

5. The pillow sack, a hollow case sewn on to the head-end of the stretcher bed. It is filled in with straw or any other



FIG. 77.
St. John Ambulance Pattern Stretcher—Open.



FIG. 78.
Ordinary Stretcher—Closed.

material handy to make a pillow, *e.g.*, doubled up coat.

6. Two slings, made of tan webbing with a loop at each end to go over the

ends of the handles of the stretcher. They are provided with buckles or grip-plates by which the length of the slings can be adjusted. At opposite ends is a cross piece of leather strap provided with a buckle. This is used to get around the stretcher when closed.

Improvised Stretcher. It is not always possible to have a standard type of stretcher available and one has to exercise one's resourcefulness to improvise a stretcher from the material available on the spot :—

A. (1) Take a material like a blanket, a piece of canvas cloth, *durree* or *purdah* of a suitable size and spread it on the ground. (2) Lay two poles over the edges of the cloth and roll it round the poles several times till about 22″ of bed is left. Make holes in the blanket and pass thin rope or twine through them, so as to fix the blanket or *durree* on to the poles.

(3) Lash a piece of split bamboo or stick of suitable size at each end to act as traverses.

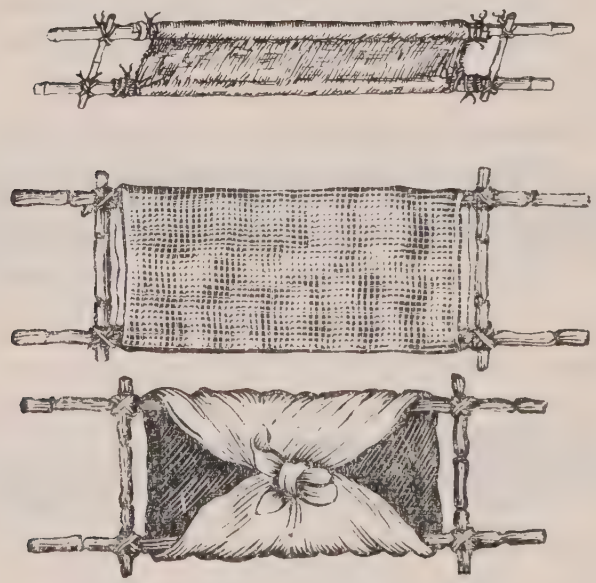


FIG. 79. Improvised Stretchers.

B. (1) Get a strong coat. (2) Turn the sleeves inside out. (3) Pass the poles

through the sleeves on each side. (4) Button the coat and place it button side downward. (5) Improvise two cross bars to keep the stretchers apart and lash them on to the poles at both ends of the bed formed by the coat.

N.B.—Two or more coats may be used according to the length of the stretcher required.

C. Two strong gunny bags placed end to end, with holes made at the bottom corners of each, might be used. Fig. 79(B). A blanket or a *chadar* may also be tied to form a stretcher bed. *Vide* Fig. 79(C).

D. Door and window shutter frames, a plank of wood, the common bamboo *sirhi* or *moi* might be utilised for carrying a patient.

E. A narrow country *charpoy* or *khatia* might be improvised as a stretcher but it jolts a lot which is a disadvantage.

Note.—Always test the strength of an improvised stretcher before use.

Rules for carrying patients in stretcher.

1. Carry a patient on a stretcher feet foremost.

2. When going upstairs or up-hill, carry a patient whose lower limbs are injured with his head foremost.

3. When going down hill with a patient whose lower limbs are injured carry him head foremost.

4. In order to prevent fainting in patients who have lost a considerable amount of blood, carry head side of the stretcher at a lower level than the foot side.

5. Avoid crossing *nullahs* and ditches or going over walls and bunkers as these cause uncomfortable jolting to the patient.

Note.—The stretcher should be carried with the hands or suspended by straps over the bearer's shoulder. The stretcher should never be carried on shoulders, because then the bearer cannot watch the patient and he may die unobserved or might fall off the stretcher.

Rules for carrying stretcher over an obstacle.

To cross a ditch or a nullah. (*a*) *When there are four bearers* :—

1. The stretcher is placed on the ground with its front end at the edge of the *nullah*. 2. The front bearer crosses over to the opposite bank. 3. The two side bearers enter the *nullah* and with the help of the bearer at the rear end of the stretcher, advance the stretcher across the *nullah*. 4. When the front handles of the stretcher reach the bearer on the opposite bank, they are grasped by him and he now helps the bearers in the ditch to advance the stretcher further. 5. After the stretcher has reached the opposite bank, all the bearers cross over and carry away the patient in the usual manner.

(*b*) *When there are three bearers* :—

1. The stretcher is placed on the

ground with its front end at the edge of the *nullah*. 2. The first two bearers then descend into the *nullah*. 3. With the help of bearer No. 3 at the rear end of the stretcher, it is advanced supported by the two bearers in the ditch. 4. When only the rear handles of the stretcher remain on the bank the last bearer also enters the ditch and all carry the stretcher to the opposite side. 5. The front end is now made to rest on the edge of the opposite bank and the rear end is supported by the 3rd bearer. 6. Of the first two bearers one climbs out of the ditch to the opposite bank and one remains in the ditch to help the bearer who is holding the rear end of the stretcher. 7. All the three bearers help to advance the stretcher till it rests on the opposite bank ; then the remaining two bearers also climb out of the ditch and all the three lift and carry the patient away in the usual manner.

To cross walls and bunkers.

1. On arriving at the edge of the wall one bearer stands at each corner of the stretcher. 2. The stretcher is raised and its front part placed on the top of the wall. 3. Two front bearers climb over the wall and all the four lift the stretcher from the wall and place it on the ground. It is then carried away in the usual manner.

Adopt the same method with three bearers.

Conveyance of the Sick or Injured by Rail or in Country Carts.

Transport by a train is the quickest and the least irksome method for the patient. Unfortunately, in India railway communication is inadequate and country-carts drawn by bullocks are very often utilised for the conveyance of the sick and the injured. The progress is slow and the

jolting, both on account of our bad country roads and springless carts, is very great. Such a means of transport should be always avoided. Even a *Khatia* or a *Charpai* is better. The *Doli* and *Palki* carried by professional bearers is much more comfortable and a fairly quick means of transport. A case of multiple injuries was safely brought to me one morning in a mofassil hospital having covered 30 miles in about six hours by relays of bearers.

A patient lying on a stretcher can be carried by rail in a wagon. Care must be taken to see that the stretcher does not slip out and therefore it should be securely lashed on to the sides of the wagon. The *method of loading a wagon* is as follows. The stretcher is brought close to the wagon and put on the platform or the ground about one pace distant from the edge of the wagon. The stretcher is then lifted and carried towards the wagon till its front

wheels rest on the floor of the wagon. One of the bearers enters the wagon and keeps this end of the stretcher steady. The other end is held in position by two bearers outside the wagon and they gently push the stretcher into the wagon in proper position and it is then tied on to the wagon. While unloading the same method is adopted in the reverse direction and the stretcher taken out of the wagon and placed on the ground or the railway platform about one pace distant from the wagon.

CHAPTER XXXII

Stretcher Exercise.

It requires training and practice to lift and carry a patient on a stretcher without causing him pain. Three different methods are generally adopted.

A.—With four bearers.

Choose 4 bearers of nearly the same size and mark them Nos. 1, 2 and 3. Should one man be taller and stronger

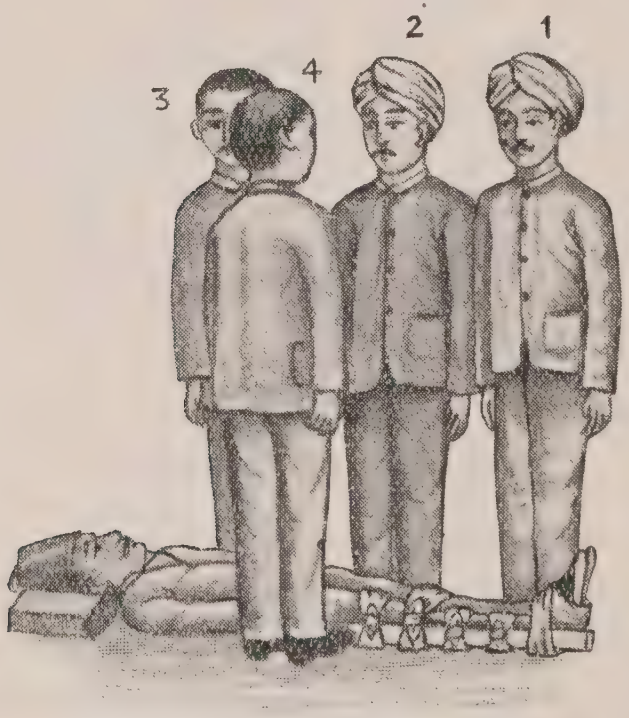


FIG. 80. Stretcher exercise—"Fall in."

than the rest, mark him as No. 3 and let him carry the heaviest part of the body, *viz.*, the shoulders and head ; No. 4 is the captain who gives the orders.

(1) At the word of command, "fall in," the bearers take their respective positions to the left of the patient and stand facing him—No. 1 near the patient's feet, No. 2 at his hips and No. 3 at the shoulders. No. 4 places the stretcher ready for use on the ground by the right side of the patient and about two paces away from him. After this he takes his position opposite No. 2 standing to the right of the patient. (Fig. 80).

(2) At the word of command "*ready,*" all drop down on their left knees and No. 1 passes his hands and forearms beneath the patient's legs, keeping his hands wide apart. No. 2 and No. 4 pass their forearms beneath the patient's hips and loins and grasp each other's hands. No. 3

either passes both his hands and forearms beneath the shoulders of the patient, or his left hand across the patient's chest and



FIG. 81.

Stretcher exercise—"Ready" (for lifting).

under the right shoulder and his right hand under the left shoulder. (Fig. 81).

(3) On the order "*lift—load stretcher,*" the bearers carefully raise the patient and let him securely rest on the knees of Nos. 1, 2 and 3. (Fig. 82). No. 4 now disengages his hands, runs round the head

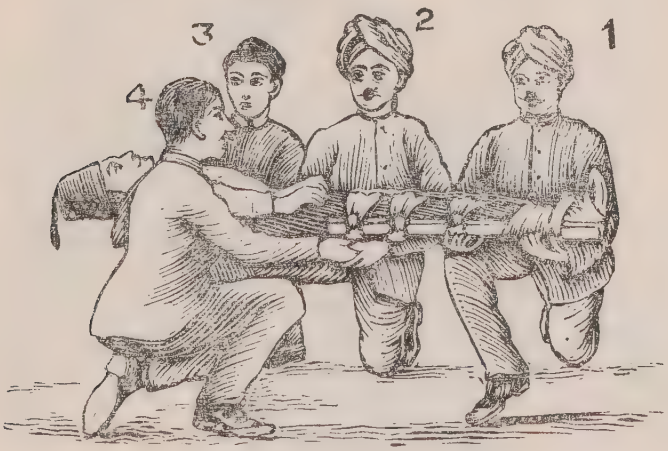


FIG. 82. "Lift—load stretcher."

of the stretcher and places it under the patient, pushing it close to the feet of the three bearers and taking care to see that the pillow of the stretcher is put under the

head of the patient. (Fig. 80). When this is done No. 4 kneels down on his left knee and grasps the hands of No. 2 as before.

(4) On the word "*lower—load*



FIG. 83. Placing stretcher in position.

stretcher," the patient is placed on the stretcher and the bearers take their hands away. (Fig. 83).

(5) At the command "*stand to stretcher,*" No. 1 marches to the foot of the stretcher and stands between the handles with his back to the patient,

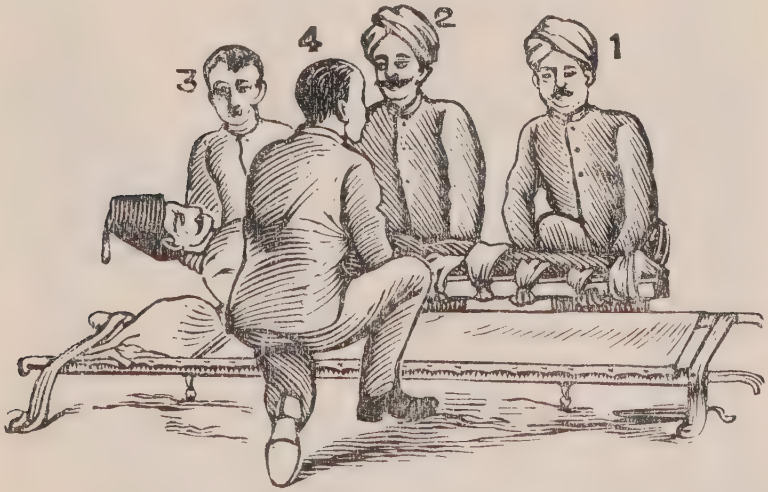


FIG. 84.
Placing patient. "Lower—Load Stretcher."

No. 3 to the head of the stretcher with his face to the patient, Nos. 2 and 4 stand on the left and right side of the stretcher.

(6) At the word "*ready,*" Nos. 1 and

3 pick up the slings, place them over their shoulders, stoop and pass the loops of the slings on to the handles of the stretcher which they then grasp.

(7) Now the command "*lift stretcher*" is given. Nos. 1 and 3 rise together lifting the stretcher evenly with them as they stand up. Nos. 2 and 4 now walk up to Nos. 1 and 3 and adjust the slings. See that they are below the level of the collars of the bearers and lie flat in front of the shoulder. By means of the buckles the length of the slings are lowered or raised to suit the height of the bearers or the needs of the patients. (Fig. 85).

(8) At the word "*march*" the bearers move off taking short steps of 20 inches with knees bent and with no spring from the front part of their foot.

Nos. 1, 2 and 4 step off with their left foot and No. 3 with his right foot. (Fig. 85).

(6) At the word "*halt,*" the bearers stop and remain steady.

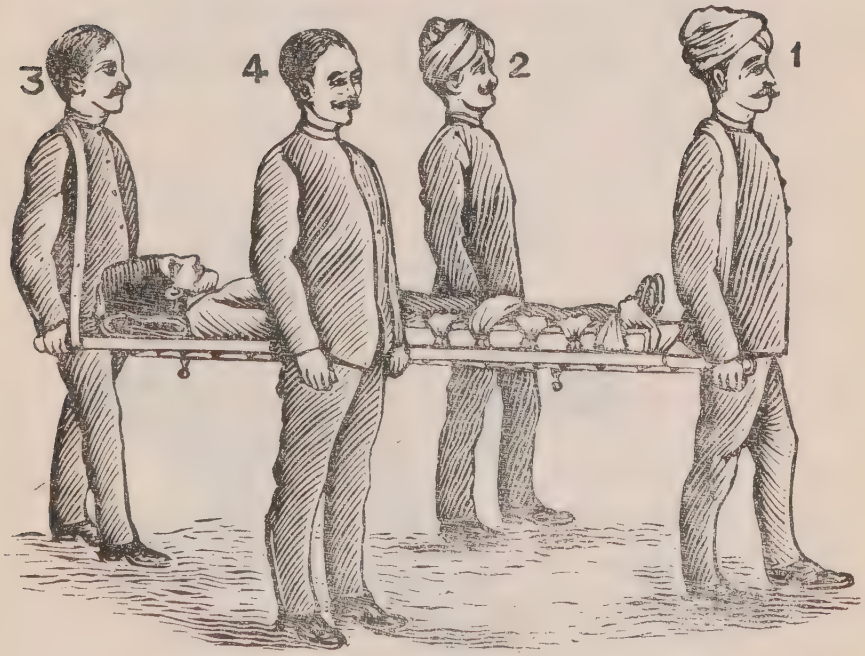


FIG. 85. "Lift—stretcher: March."

(10) "*Lower stretcher*"—The bearers stoop down and gently place the stretcher on the ground, remove the loops from

the handles and take off the slings from off their shoulders and then stand up.

(11) At the word "*unload stretcher—ready*" the bearers take up position as

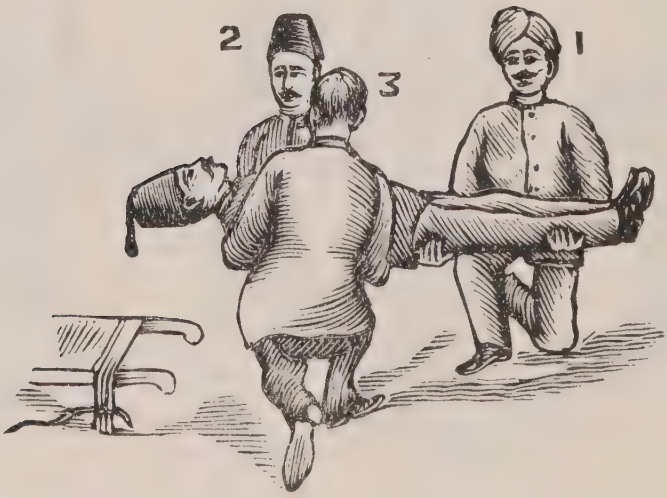


FIG. 86. Ready for lifting patient.

at the word "*fall in*" and "*ready.*" (Commands Nos. 1 and 2).

(12) At the word *"lift"* the bearers lift the patient as they did before. (*See* order No. 3).

No. 4 in this case after disengaging his hands from No. 2 removes the stretcher and returns to his former position. The bearers now steadily and evenly rise together with the patient and carry him to the bed prepared for him and at the word "lower" carefully put him on the bed.

B.—With three bearers.

No. 2 gives the command. The stretcher is placed in line with the patient's body quite close to his head. Nos. 1 and 2 stand at the left side of the patient, and No. 3 at the right. The patient is lifted by the three bearers and carried over the foot of the stretcher until his head is in line with the pillow. He is then carefully lowered on to the

stretcher. Nos. 1 and 3 lift and carry the stretcher, No. 2 adjusts slings and looks after the patient. (Figs. 86 and 87).

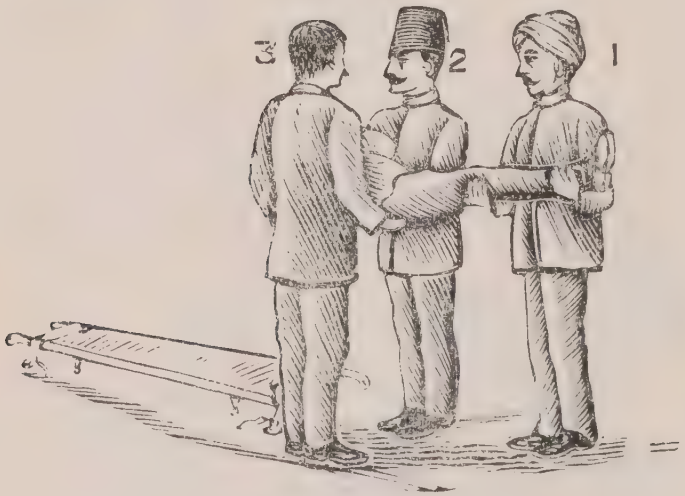


FIG. 87. Patient lifted.

Note.—If for want of room or any other reason, the stretcher must be placed beside the patient, then lift the patient as before but move in a kneeling position towards the stretcher and lower him on to it.

C.—With two bearers.

No. 1 gives the command. Place the stretcher in line with the patient's body close to his head and lift the patient straddling over him—No. 1 holding the

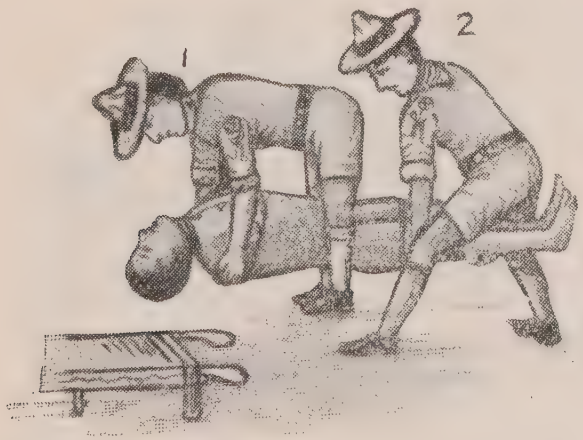


FIG. 88. Two Boy Scouts lifting a patient.

patient at the shoulders with both his hands over the back, No. 2 passes his right hand under the calves and his left hand under the hips. No. 1 who lifts the upper

part of the body moves forward by very short steps. No. 2 bends his body forward without moving his feet. No. 2 orders No. 1 to halt and remain steady, whilst he (No. 2) advances his right foot to his

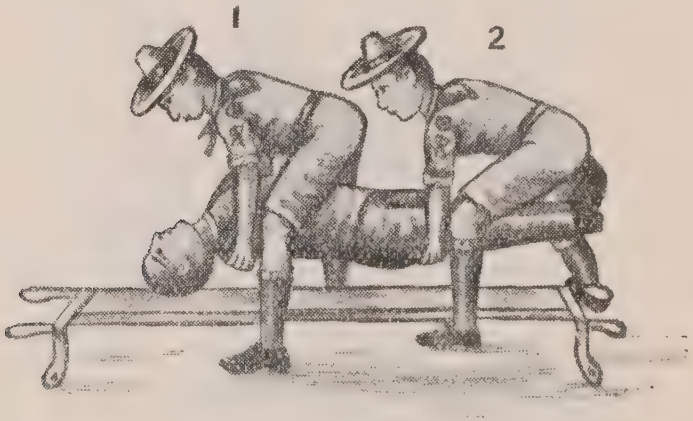


FIG. 89. Two Boy Scouts lowering a patient on stretcher.

left and then his left until his toe touches the heel of No. 1. These movements are repeated until the patient's head is on a level with the pillow of the stretcher, when

he is gently lowered and carried. (Figs. 88 and 89).

Note.—This method is useful in narrow places such as the shaft of a mine.

Ambulance Squads.

Ambulance squads during camps of training for competition should adopt the following method, (*a*) to render First Aid, (*b*) to prepare stretchers, (*c*) to close stretcher.

To Render First Aid.

No. 4 is the captain and gives orders.

1. At the word of command the squad pick up a stretcher and double up to the injured person by the shortest route and halt about one pace from his head.

2. On the word "*lower stretcher—prepare stretcher,*" Nos. 1 and 3 put the stretcher down on the ground and prepare it to receive the patient, No. 4 gets to the

right and No. 2 to the left of the patient and render First Aid to the patient.

Note.—If necessary No. 4 may call for the services of Nos. 1 and 3 also to assist in rendering First Aid, normally however they remain at the stretcher.

3. After rendering First Aid No. 4 gives the command "*load stretcher*," when the bearers take up their position as in method A, 3 and 4 of stretcher exercise already described. Similarly on the words "*lift, march, halt, lower, unload—lift lower*," the requisite movements described in the stretcher exercise are performed by the bearers.

On the word of command "*prepare stretcher*" and "*close stretcher*" the following method is adopted:—

"*Prepare Stretcher.*" Two bearers Nos. 1 and 3 kneel on their left knee, unbuckle the transverse straps and drop

the slings upon the ground, separate the poles, pull the telescope handles out and straighten the cross-bars or traverses to make the canvas or the stretcher bed even and taut. Lastly each picks up a sling, folds it in two and keeping the grip-plate (buckle) uppermost passes it on the handle nearest to him.

"*Close Stretcher.*" Two bearers Nos. 1 and 3 kneel on their left knee, remove the slings and place them on the ground beside them, fold the cross bar by pushing them in, raise the canvas of the stretcher bed and bring the poles together. They then rise lifting the stretcher and face one another, place the handles of the poles between their thighs and roll the canvas tightly over the poles to the right of the stretcher. Next each of them takes up a sling and passes the buckle or grip-plate end to each other. Each holds in his left hand the grip-plate end given him, threads

the transverse strap through the loop of the other sling and buckles lightly close to the roller keeping the slings on top ; then holding both handles in their right hands they rise and turn to their left together.

CHAPTER XXXIII.

Preparation at Home for the Reception of Accidents and for the Surgeon.

1. *Selection of the Room.* Whenever possible choose a room which can be reached easily such as one on the ground floor or the 1st floor. Be sure that it is large, bright, sunny and airy. Avoid a small, dark or damp room. Remove all unnecessary furniture, carpets and old mattings, as the one interferes with ventilation, and the other accumulates dust and

harbours germs of disease. A small table or a teapoy and two good chairs are all that need be left in the room.

2. *Selection and preparation of the bed.* Always choose a narrow and firm bed. A bed that sags or has a feather mattress is unhealthy. The Indian *takht* with a clean coir mattress and *toshak* will do for most cases. Pour boiling water in all cracks and joints of the *takht* or bedstead and after drying put petrol or kerosine oil into it and rub dry. In this way you will be sure of killing bugs and bacteria. A mattress of clean dry straw is quite good and is recommended for those who cannot afford anything better.

The best bed is the English single iron bed with a strong wire mattress. For cases of fractures however these must be made rigid by putting a board or plank of wood over the spring and beneath the coir mattress. The advantage of a narrow bed

is that the patient can be easily got at from the sides which makes nursing easy. For the same reason do not place a sick bed with one side against the wall. The head of the bed might be placed near a wall but the sides must be free. No boxes, or any other article should be kept under the bed as they obstruct free passage of air under the bed. If possible secure two narrow beds so that the patient can be put from one into the other. A change is always pleasant specially after extensive dressings.

Pull the bed sheet evenly over the bed all round and tuck it tightly under the mattress.

A piece of water-proof sheeting is laid over the bed sheet and a draw sheet of the same size is spread over it. It helps to keep the bed clean and saves frequent changes of the bed sheet. The draw sheet is spread across the middle portion of the

bed and tucked under the sides of the mattress on either side or fixed to it by safety pins. In accident cases, where soaking is expected an additional draw-sheet is spread underneath the injured part. When the draw sheet gets soiled or the patient complains of the bed feeling hot, instead of changing the whole bed the draw sheet only is changed. One end of the draw sheet is unfixed and rolled up and drawn away from under the patient, another roll of clean draw sheet is spread in its place. Arrange to secure or improvise *bed cradles* to take off the weight of bed clothes from an injured limb. A low tripod stool will do well or a small packing case or a kerosine oil tin could be utilised for this purpose after one side and the two narrow ends have been removed. Two pieces of split bamboo tied in the form of a semi-circle to the sides of the bed will also answer the purpose.

3. *Carrying of the Patient.* Before attempting to carry a patient, secure a sufficient number of bearers and see that they are, as far as possible, of the same height. If they are not trained, they should be made to thoroughly understand that they have to handle the patient gently, and lift him together and evenly without jerk. Clear the passage to the sick room of all possible obstacles. In carrying a stretcher upstairs or up-hill, keep the head end first. A helper should assist at the lower end so as to raise the stretcher and keep it horizontal. If a chair has been improvised as a stretcher or a seat, support it in the same way and prevent the injured person from slipping out. Having reached the bed, place the stretcher beside it. All the helpers should stand at that side of the stretcher which is farthest from the bed. The helpers pass their hands and forearms under the patient and support him beneath

the shoulders, the middle of the back, the hips, the knees and the ankles. An injured limb should receive special support if possible from an additional helper. Thus supported the patient is lifted off the stretcher. The stretcher is then drawn away by a spare helper, and the patient is gently carried to the side of the bed on which he is laid as far as possible in the middle. In the case of a very narrow bed, the stretcher with the patient may be placed in a line with it, the head of the stretcher at the foot of the bed. The patient is then lifted over the foot of the bed and placed on it.

Note.—St. John's Ambulance rules do not require females to be given instructions in stretcher exercise. Red Cross Training Manual however says :—" Women ought to learn stretcher drill. Should a woman be in a country district with no other help but untrained country folk, it is expedient to be able to direct how a

wounded man should be lifted, how an improvised stretcher can be put together, and how the stretcher should be carried. Women should not be allowed to carry wounded men on stretchers, unless there are six women bearers, one at each end of the stretcher, and four to support the sides of the stretcher."

4. *Removal of Clothes, etc.* Extreme care and gentleness is necessary in all cases, specially those suffering from painful injuries and shock. Sacrifice the clothes rather than run the risk of aggravating the injury and giving pain to the patient.

In removing a *kurta* or a coat or shirt from a patient with fracture and injuries in the upper limb, draw out the uninjured arm first and then gently remove the sleeve from the injured arm. In putting on a shirt or *kurta,* the sleeve is gently put on the injured arm first and then on the sound arm.

In removing a *pajama* or trousers, cut

it at the outer side seam and remove it gently.

If boots are to be removed, cut the back seam and the lace in front and let it be taken off.

For method of removal of adherent clothing in cases of burns and scalds, *vide* page 185.

Note.—In the cold season warm the clothes over a fire before putting them on a patient.

5. *Home Preparation for the Surgeon.*

(1) If the room is provided with a fire-place—and it is cold—have a fire in the room. This is not often necessary in India where rooms must be kept cold. (2) Keep plenty of hot water and boiled water which has been allowed to cool without removing the lid of the *deckchie* or kettle in which it was boiled. (3) If sterilised dressings are not handy, boil some pieces of clean *saree, kurta, dopatta* or some soft cloth of

latha or *mulmul* in a *deckchie* of water and then allow it to cool with its lid on. (4) Keep two or three clean towels, a cake of clean soap and a nail brush which has been washed and then boiled (sterilised). (5) Keep a slop pail, a *balti,* or a *chilamchi,* to throw waste water in. (6) Also keep a packet of safety pins, a clean pair of scissors. (7) About 4 ozs. of tincture iodine, two bottles of carbolic lotion (strength two teaspoonfuls to one pint), 2 ozs. of boric acid powder, a few bandages and some sterile or clean cotton wool and dressings—all these must be kept arranged neatly on a small table covered over by a clean cloth. (8) Keep two or three iron enamelled cups or small bowls made free from all grease by scrubbing with soap and water and sterilized by being kept in boiling water for ten minutes.

For cases of drowning—spread over the bed sheet a blanket which has been

warmed before a fire. Also keep several warmed blankets ready for use. Keep several hot water bottles or hot bricks ready wrapped up in flannel or cloth. Hot milk, hot tea or coffee, salvolatile, smelling salts should also be provided.

For cases of burns—keep baking soda about 1 lb., sweet oil or cocoanut oil or vaseline which has been boiled and cooled, several packets of dry sterilised cotton wool and dressings.

Note.—For all cases of injuries accompanied with shock, keep hot water bottles, warmed blankets and keep the bed and bed clothes warm with them.

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